Faculty of Commerce

Accounting Department

A Proposed Multidimensional Model for Measuring Earnings Management and Its Impact on Firm Value (An Empirical Study)

A Thesis Submitted to Obtain the Doctor of Philosophy Degree of Commercial Sciences (Accounting)

Prepared by:

Heba Allah Gaber El Sawaf
Assistant Lecturer in Accounting Department
Menoufia University

Supervised by

Prof. Mohamed Zedan Ibrahim
Professor of Financial Accounting
Faculty of Commerce
Menoufia University

Prof. Mohamed Saber Hamouda Elsayed
Professor of Financial Accounting and Vice Dean for Community Service and Environmental Development Affairs
Faculty of Commerce
Menoufia University

Dr. Tarek Ibrahim Saleh Saadah
Assistant Professor of Accounting
Faculty of Commerce
Menoufia University

2024

بِسْمِ اللَّـهِ الرَّحْمَـٰنِ الرَّحِيمِ

﴿قَالُواْ سُبْحَٰنَكَ لَا عِلْمَ لَنَآ إِلَّا مَا عَلَّمْتَنَآۖ إِنَّكَ أَنتَ ٱلْعَلِيمُ ٱلْحَكِيمُ ﴾

صدق اللَّهُ العظيم

[البقرة: ٣٢]

Discussion & Judging Committee

Prof. Mohamed Zedan Ibrahim **(Supervisor &Chairman)**

Professor of Financial Accounting, Menoufia University

Prof. Farid Moharam Farid Ibrahim **(Member)**

Professor of Financial Accounting and Vice Dean for Education and Student Affairs, Ain Shams University

Prof. Mohamed Aly Wahdan **(Member)**

Professor of Auditing and Dean of faculty of commerce, Menoufia University

Prof. Mohamed Saber Hamouda Elsayed **(Supervisor & Member)**

Professor of Financial Accounting and Vice Dean for Community Service and Environmental Development Affairs, Menoufia University

Dr. Tarek Ibrahim Saleh Saadah **(Supervisor & Member)**

Assistant Professor of Accounting, Menoufia University

Acknowledgements

In the name of Allah... First and foremost, I would like to thank Allah for giving me the health and mind to live and learn, particularly for granting me the ability, strength and faith to face all the hard times to complete this thesis.

I am deeply grateful to my thesis supervision committee. I certainly would like to direct my extreme appreciation to ***Prof. Mohamed Zedan Ibrahim*** for his kind help, sustained and unlimited support and continuous encouragement to me during my work. Without his guidance and persistent help, this thesis would not have been possible. My appreciation also extends to ***Prof. Mohamed Saber Hamouda Elsayed*** for his never-ending support, patient guidance and insightful comments that helped me to improve this thesis. I would like also to thank ***Dr. Tarek Ibrahim Saleh Saadah*** for his continuous encouragement and support.

Special thanks to ***Prof. Farid Moharam Farid Ibrahim and Prof. Mohamed Aly Wahdan*** for their acceptance to be members of my examination committee and dedicating their precious time to provide constructive criticism for improving my thesis. I consider myself fortunate to have the chance to benefit from their knowledge.

Last but not least, I take this opportunity to express my deep thanks to all the faculty members of the accounting department for their help and advice. I am also grateful to all my colleagues and professors in other departments for their encouragement. I am deeply thankful for them.

Dedication

To my lovely parents, for their endless love and encouragement

To my husband, for his support, understanding and companionship

To my precious children, for their love and patience with my busy time

To my beloved brothers, for their help and support

To in loving memory of my sister, whose spirit continue to inspire and guide me

Table of Contents

Discussion & Judging Committee…… 3
Acknowledgement…… 4
Dedication…… 5
Table of Contents…… 6
List of Tables…… 10
List of Figures…… 11
List of Abbreviations…… 11
Abstract…… 12
Chapter 1…… 13
General Framework of the Study…… 13
1.1 Introduction…… 14
1.2 The Study Problem…… 16
1.3 The Study Objectives…… 18
1.4 The Study Significance…… 18
1.5 The Study Limitation…… 19
1.6 Literature Review and Hypotheses Development…… 19
1.6.1 Studies on Classification Shifting as an Earnings Management Tool…… 19
1.6.2 Studies on the Impact of Earnings management practices on Firm Value…… 22
1.7 The Study outline…… 33
Chapter 2…… 34
Firm Value and Earnings Management: A Conceptual Framework…… 34
2.1 Introduction…… 35
2.2 Firm Value Definition…… 35
2.3 Firm Value Determinants…… 36
2.3.1 Firm Size…… 37
2.3.2 Firm Age…… 38
2.3.3 Audit Quality…… 38
2.3.4 Profitability…… 38
2.3.5 Liquidity…… 39
2.3.6 Leverage…… 39
2.3.7 Dividend policy…… 40
2.3.8 Investment Opportunity…… 40
2.3.9 Capital Structure…… 41
2.3.10 Firm Growth…… 41
2.4 Firm Value Approaches…… 42
2.4.1 Asset Approach…… 42
2.4.2 Income Approach…… 42
2.4.3 Market Approach…… 43
2.5 Earnings Management Definition…… 43

2.6 Earnings Management Motivations…….. 45
2.6.1 Capital Market Expectation Motivation…….. 45
2.6.2 Contractual Motivation…… 45
2.6.3 Regulatory Motivation…… 46
2.7 Earnings Management Determinants……..…….. 46
2.7.1 Accounting standards ……..…….. 46
2.7.2 Internal Corporate Governance…… 47
2.7.3 Audit quality ……..... 49
2.7.4 Firm life cycle……... 50
2.7.5 Firm size……..…….... 52
2.7.6 Financial leverage……. 53
2.7.7 Human Characteristics…….. 54
2.8 Earnings Management Approaches……..…..... 56
2.8.1 Accounting Earnings Management Approach……..……..…… 56
2.8.1.1 The Flexibility of Accounting Standards…….……. 56
2.8.1.2 Accrual based Earnings Management……..……... 57
2.8.1.3 Classification Shifting……..... 58
2.8.2 Real Earnings Management Approach……..……..…. 61
2.8.2.1 Sales Manipulation……..……..…. 61
2.8.2.2 Discretionary Expenditure Manipulation…….……...… 62
2.8.2.3 Overproduction…… 62
2.8.2.4 The Timing of Asset Sale……...……. 62
2.8.2.5 Stock Repurchases……... 63
2.8.2.6 Securitization…….……..….... 63
2.8.2.7 Financial Derivatives……... 63
2.9 Chapter Summary…….. 64
Chapter 3……...……... 65
Earnings Management Tools and their Models……...……. 65
3.1 Introduction…….……..…… 66
3.2 The relationship among Earnings Management Tools ……..……...……. 66
3.3 Measurement of Earnings Management Tools……..……...…….. 74
3.3.1 Accrual based Earnings Management Models……...…….. 74
3.3.1.1 The Healy (1985) Model…….……...……..... 75
3.3.1.2 The DeAnglo (1986) Model…….……..…..... 76
3.3.1.3 The Jones (1991) Model…….……...……. 77
3.3.1.4 The Modified Jones Model (Dechow et al. 1995)…….……..…….. 78
3.3.1.5 The Kotharie et al. (2005) Model…….……...……... 79
3.3.1.6 The Dechow and Dichev (2002) Model……...…….…… 80
3.3.2 Classification shifting Models……..……..…….. 81
3.3.2.1 The Mc Vay (2006) Model…….……...……. 81
3.3.2.2 The Poonawala and Nagar (2019) Model……... 83

3.3.2.3 The Malikov, et al. (2019) Model…….. 85
3.3.2.4 The Nagar and San (2014) Model……..…….... 86
3.3.3 Real Earnings Management Models…… 87
3.3.3.1 The Sales Manipulation Model……..…... 87
3.3.3.2 The Over Production Model…….……..…... 88
3.3.3.3 The Discretionary Expense Model……...…..…. 88
3.3.3.4 The Selling of Long-Term Assets Model…… 89
3.4 A Proposed Multidimensional Model for Measuring Earnings Management and Its Impact on Firm Value……..……..…….. 90
3.5 Chapter Summary…….…… 92
Chapter 4…….…..... 93
The Empirical Study……..…. 93
4.1 Introduction……..…….... 94
4.2 The Study Methodology…….....…… 94
4.2.1 Sample Selection…….……..……... 94
4.2.2 Data Collection Method……...…….. 95
4.2.3 Variables Description and Measurement……..……..…… 95
4.2.3.1 Dependent Variable…….……..…… 95
4.2.3.2 Independent Variables…….……..…… 95
4.2.3.3 Control Variables…….……..……... 101
4.2.4 Model Specification…….……..…….. 103
4.2.5 Statistical Techniques…….……..……... 104
4.3 Empirical Results and Testing Hypothesis…….……..……..…… 105
4.3.1 The Normal Distribution Test……..……..……... 105
4.3.2 Descriptive Statistics…….……..……..…… 106
4.3.3 Validation Tests of the OLS Models…….……...……..…… 107
4.3.3.1 The Normality Test…….……..…… 107
4.3.3.2 The Multicollinearity Test……...…….. 108
4.3.3.3 The Autocorrelation Test…….……..… 109
4.3.3.4 The Heteroscedasticity Test…….…….... 109
4.3.3 Statistical Analysis for Testing Hypotheses……...…….. 110
4.3.3.1 Investigating the Impact of Accrual based Earnings Management on Firm Value……..…….…….... 110
4.3.3.2 Investigating the Impact of Real Earnings Management on Firm Value…….…….... 113
4.3.3.3 Investigating the Impact of Income Classification Shifting and Cash Flow Classification Shifting on Firm value…… 114
4.3.3.4 Investigating the Impact of Aggregate Earnings Management on Firm Value……...……. 116
4.3.3.4 Investigating the differences among Sectors regarding the Study Variables 117
4.4 Chapter summary……..... 122

Chapter 5……..……... 123
Summary, Conclusion, and Recommendations…….…….. 123
5.1 Summary……..……. 124
5.2 Conclusions ……..…..... 125
5.3 Recommendations…….……..…….. 126
5.4 Future Research…….…….. 126

List of Tables

Table (1.1) Summary of the Studies for the Impact of Earnings Management practices on Firm Value…… 29
Table (2.1) Summary of the Nature of Classification Shifting Practices……. 60
Table (3.1) A Comparison among Earnings Management Tools……..…….. 68
Table (4.1) The Study Sample…….. 94
Table (4.2) Description and Measurement of Variables……..…… 101
Table (4.3) Results of the Normal Distribution Test……..……. 105
Table (4.4) Descriptive Statistics Results……..……. 106
Table (4.5) The Normality Test Results……... 108
Table (4.6) The Multicollinearity Test Results……. 108
Table (4.7) The Wooldridge Test Results……. 109
Table (4.8) The Heteroscedasticity Test Results…….. 109
Table (4.9) GLS Regression Analysis Results for the Impact of AEM on Firm Value……. 110
Table (4.10) GLS Regression Analysis Results for the Impact of REM on Firm Value……. 113
Table (4.11) GLS Regression Analysis Results for the Impact of ICS and CFCS on Firm Value …….. 115
Table (4.12) GLS Regression Analysis Results for the Impact of AGG_EM on Firm Value……..…….. 116
Table (4.13) Kruskal-Wallis Test Results for Investigating the differences among Sectors regarding MTB…… 117
Table (4.14) Kruskal-Wallis Test Results for Investigating the differences among Sectors regarding ROA……...……. 118
Table (4.15) Kruskal-Wallis Test Results for Investigating the differences among Sectors regarding AEM……..……. 119
Table (4.16) Kruskal-Wallis Results for Investigating the differences among Egyptian Exchange Sectors regarding REM……. 119
Table (4.17) Kruskal-Wallis Test Results for Investigating the differences among Sectors regarding ICS…….……... 120
Table (4.18) Kruskal-Wallis Test Results for Investigating the differences among Sectors regarding CFCS……..… 120
Table (4.19) Kruskal-Wallis Test Results for Investigating the differences among Sectors regarding AGG_EM …….. 121

List of Figures

Figure 2.1: Factors Affecting Firm Value……..... 37
Figure 3.1: The Choice among Earnings Management Tools……. 68
Figure 3.2: Factors Affecting the Choice of an Earnings Management Tool….. 74
Figure 3.3: A Proposed Multidimensional Model for Measuring Earnings Management and Its Impact on Firm Value…… 91
Figure 4.1 The Relationship among the Study Variables…… 103

List of Abbreviations

EM	Earnings Management
AEM	Accrual based Earnings Management
REM	Real Earnings Management
CS	Classification Shifting
ICS	Income Classification Shifting
CFCS	Cash Flow Classification Shifting
AGG_EM	Aggregate Earning Management
MTB	Market-to-Book ratio
ROA	Return on Assets
DA	Discretionary Accruals
CFO	Cash Flow from Operations
PROD	Production Cost
DISEXP	Discretionary Expenditure
SG&A	Selling, general, and administrative Expense
COGS	Cost of goods sold
R&D	Research and Development Expense
IFRS	International Financial Reporting Standards
OLS	Ordinary Least Squares
GLS	Generalized Least Squares
BIG4	Big four auditing firms
Age	Firm Age
Size	Firm size
LEV	leverage
Growth	Firm Growth

Abstract

Purpose: This study seeks to propose a multidimensional model for measuring earnings management (EM) and its impact on firm value. The current study empirically investigates the impact of EM practices namely, accrual-based earnings management (AEM), real earnings management (REM), income classification shifting (ICS), cash flow classification shifting (CFCS), and aggregate earnings management (AGG_EM) on firm value. Two alternative measures were used to proxy for firm value; market-to-Book ratio (MTB) and return on assets (ROA). AEM was proxied by discretionary accruals, REM was measured using an aggregate REM proxy as computed by the aggregation of abnormal values of operating cash flow, production cost, and discretionary expenditure. ICS was measured by unexpected gross profit whereas CFCS was measured by unexpected cash flow from operations. AGG_EM was captured by averaging the ranks of AEM, REM, ICS, and CFCS.

Design/methodology: The study was conducted on a sample consisting of 48 firms listed in the Egyptian Exchange during the period from 2017 to 2021. Generalized Least squares (GLS) regression test and Kruskal Wallis test were employed to test the research hypotheses.

Findings: The results found a significant positive impact of AEM, CFCS, and AGG_EM on MTB, but ICS was negatively significant. On the other hand, REM indicated an insignificant impact on MTB. As for the second measure ROA, the findings revealed that ICS, CFCS, and AGG_EM had a significant positive impact on ROA. whilst AEM and REM were negatively significant. Further, the results reported the existence of significant differences among sectors in the Egyptian Exchange regarding MTB, ROA, and AGG_EM, but no significant differences concerning AEM, REM, ICS, and CFCS.

Originality/value: This study contributes to the existing literature by presenting a multidimensional model for measuring EM and its impact on firm value. Most of the prior literature mainly focused on AEM and REM practices. The study adds to the existing literature by providing novel evidence of CS practices in the Egyptian context. In addition, this study integrates EM tools including AEM, REM, and CS, offering empirical evidence of how these tools collectively influence firm value. Finally, the study findings are valuable for standards setters, investors, analysts, regulators, and auditors.

Keywords: Firm Value, Earnings Management, Accrual-based Earnings Management, Real Earnings Management, Income Classification Shifting, Cash Flow Classification Shifting, Aggregate Earnings Management.

Chapter 1
General Framework of The Study

1.1 Introduction

The general purpose of financial reporting is to provide financial information about the reporting entity (IASB, 2018). Earnings are one of the most crucial accounting numbers in financial statements which attract the interest of many stakeholders like investors, creditors, and analysts, since earnings provide information about the firm's current performance and its prospects for the future (Orieby, 2017; Al Madbouly, 2021). The increasing use of earnings induces managers to exercise their discretion over financial reporting to manage earnings to achieve specific targets, leading to a significant impact on market valuation and ultimately the value of the firm (Darmawan, et al., 2019).

Essentially, firm value is considered as one of the most fundamental indicators that investors rely on when evaluating the firm's performance and making investment decisions. Moreover, maximizing the firm value is a strategic goal for management as it reflects its efficiency in managing the firm resources (Hussien, 2023). Firm value is the investor's perception towards the success of a firm. The increase in the value of the firm will attract investor interest and foster the trust of stakeholders (Sukesti et al., 2019; Darmawan et al., 2019).

Earnings management (EM) is a widely spread phenomenon. Two broad approaches to identify and investigate EM have been employed in accounting literature. The first approach is accounting earnings management whereby managers intervene in the financial reporting to manipulate reported earnings without changing the underlying cash flows to maximize the firm value or fulfill their opportunistic needs, including accrual-based earnings management (AEM) and classification shifting (CS). An alternative EM approach is real earnings management (REM), which occurs when manager's decisions affect cash flow level as well as the reported earnings (Mohamed, 2017; Abdelwahed, 2018).

Under AEM managers may borrow earnings from future periods to improve current period earnings or the opposite pushing earnings from the current period to future period earnings, through the manipulation of revenues and expenses recognition (i.e., acceleration or deceleration) (Al Maleeh, 2019; Bansal et al., 2023). In this sense, such practices are more visible than other EM tools due to scrutiny from auditors and regulators as well as having the potential for reversal in future periods (Ezat, 2015).

A second approach of EM is REM which is characterized as deviating from normal operational activities to mask firm's true economic performance (Roychowdury, 2006). This is undertaken through taking various decisions affecting operating, investing, and financing activities (Chang et al., 2018). In such a case, REM activities imply that firms often abandon long term growth to attain short-term earnings which in turn impairs firm value in the long term and imposes costs on shareholders (Ge and Kim, 2014; Chung et al., 2021).

One recently established form of EM is CS. Beginning with Mc Vay (2006) a sizable literature adduced additional empirical evidence for such practices. Mc Vay (2006) introduced CS as a viable EM tool whereby managers intentionally move their recurring items (revenues and expense) within the income statement as non-recurring items to artificially inflate their core earnings without impacting bottom line net income. Core earnings are usually used by investors as a base amount for their portfolio valuation since it excludes the effects of nonrecurring items (Bansal et al., 2021). Further Poonawala and Nagar (2019) denoted that managers are more inclined to boost gross profit than core earnings potentially arising from shifting costs of goods sold to operating expenses. In addition to income classification shifting (ICS), CS practices may take the form of cash flow classification shifting (CFCS) where managers misclassify items within the cash flow statement to record increased operating cash flows, holding the cash flow at the end with no change (Nagar and Sen, 2016; Mahmoud, 2023). Such manipulation might be explained by the investor's tendency to perceive that higher operating cash flows provide a positive signal concerning firm valuation and future cash flows (Nagar and Raithatha, 2016). Likewise, managers can exercise CS practices over the balance sheet statement. For instance, they are likely to misclassify short-term liabilities as long-term liabilities (tree et al., 2019).

Consequently, the current study aims to propose a multidimensional model for measuring EM practices and its impact on the value of firms listed in the Egyptian Exchange. This chapter presents the general framework of the study and is divided into seven sections. Section 1.2 highlights the study problem. Section 1.3 points to the study objectives. While section 1.4 presents the study significance. Section 1.5 addresses the study limitations. Literature review and hypothesis development are presented in section 1.6. Section 1.7 points to the study plan.

1.2 The Study Problem

Extant literature has documented three main tools of EM, namely AEM, REM, and CS. Arguably, CS is considered a relatively less costly tool to manage earnings than AEM and REM as it elicits lower scrutiny by auditors, in addition, it does not result in accrual reversals in the following periods as AEM, nor forgone benefits or increased cost as REM. Moreover, it is less probable to negatively influence firm value than AEM and REM (Zalata and Roberts, 2017; Hwang et al., 2022; Elsayed et al., 2023; Usman et al 2024). However, such misclassification could mislead investors, especially less sophisticated investors as they are likely to be fooled by abnormally overstated earnings, implying higher market and earnings expectations for the future periods (Alfonso et al., 2015).

Notably, managers can employ such tools complementary, jointly with a combined EM strategy to fulfill their greatest desired earnings targets as no complete substitution relation among the three EM forms employed to manipulate earnings. Alternatively, they can make cost/benefit trade-off decisions (Zang, 2012; Ismail 2017; Elkala, 2021; Al-Absy et al., 2021) i.e., they may switch to the one which is more difficult to be detected by auditors, relatively less costly and realize the desired outcomes of the firm (El Diri, 2018). In this regard, AEM, REM, and CS tools are perceived as substitutes.

Significantly, the directional impact of EM practices on firm value is an empirical question. Conflicting evidence on the implications of EM practices on firm value exists. On the one hand, the market might negatively perceive manager's opportunistic behavior leading to a decrease in firm value (Tulcanaza-Prieto and Lee, 2022). On the other hand, a positive impact of EM on firm value could occur when managers adopt specific accounting policies and methods to communicate the firm's specific information and convey their expectations of the firm's future flows (Garanina, 2024). In this sense, managers can use EM practices efficiently (Abbas and Ayub, 2019). They are likely to manage earnings to meet the expectations of analysts and stakeholders and show how their firms are more stable in the market to enhance its reputation which in turn reflects on the share price positively and subsequently increases the value of the firm in the short term. Nevertheless, these prices are unreal and don't reflect the firm's realistic performance, leading to a negative impact on firm value in the future. (Abo Taleb, 2022). Furthermore, detecting such opportunistic practices may pose challenges to the users of financial statements (Habib et al., 2022). More importantly, EM via REM activities is capable of conveying

positive signals concerning firm's growth and their market share (Alsharairi et al., 2020). Nevertheless, such activities focus on short-term business decisions rather than long-term ones, this could influence the firm's ability to generate income in the future; managers may abandon some economic decisions that are necessary for firm's long-term performance, such as reducing necessary expenses and/or postponing investments, regardless of their positive future value to the firm (Al Madbouly, 2021). Likewise, under CS practices, even if managers attempt to make earnings more informative, it may provide invalid information for the prediction of performance for the following periods (Anagnostopoulou, et al., 2021). Moreover, such mispricing would cause investors to incorrectly evaluate these earnings and lose excess returns (WU, 2016).

Given the above-mentioned discussion, the research gap can be determined by the following aspects:

Firstly: Since 2006, Mc Vay has opened up new avenues to conduct extensive studies related to measuring EM by CS to provide more evidence about this behavior, analyze its motivation, constraints, and measurements. Therefore, the current study seeks to investigate the existence of such practice in the Egyptian context besides other EM practices, AEM and REM to fully analyze the different ways allowed for managers to manipulate earnings and explain their integrated effect. In addition to that, understanding and realizing the relationship among the three EM tools and how managers use them complementary and jointly or substitutional is very considerable.

***Secondly*:** Prior studies independently investigated the impact of each EM tool on firm value and mixed evidence has been found. To illustrate, Sunardi (2018); Abbas and Ayub (2019); Lattanzio and Thomas (2019); AlSoboa (2020) provided a positive impact of EM on firm value. On the contrary, a negative impact of EM on firm value was found. This result is supported by the findings of Gill et al. (2013); Yuniarti (2017); Handayani and Ibrani (2020); Tulcanaza-Prieto and Lee (2022); Osman et al. (2022); Hussein (2023). Given the conflicting evidence, the current study attempts to fill this gap in the literature by empirically examining the impact of EM practices collectively including AEM, REM, CS, and their aggregate measure AGG_EM on firm value in the Egyptian setting.

Accordingly, the research questions can be written as follows:

Q1: To what extent does AEM have an impact on the value of firms listed in the Egyptian Exchange?

Q2: To what extent does REM have an impact on the value of firms listed in the Egyptian Exchange?

Q3: To what extent do ICS and CFCS have an impact on the value of firms listed in the Egyptian Exchange?

***Q4*:** To what extent does AGG_EM have an impact on the value of firms listed in the Egyptian Exchange?

***Q5*:** To what extent are there significant differences among Sectors in the Egyptian Exchange regarding the study variables?

1.3 The Study Objectives

The main objective of the current study is to propose a multidimensional model for measuring EM practices and testing the impact of such practices on the value of firms listed in the Egyptian Exchange. ***To achieve this, the main objective is divided into the following five sub-objectives***:

1. Examining the impact of AEM on firm value.
2. Determining the impact of REM on firm value.
3. Finding out the impact of ICS and CFCS on firm value.
4. Investigating the impact of AGG_EM on firm value.
5. Exploring if there are differences among sectors in the Egyptian Exchange regarding the study variables.

1.4 The Study Significance

This study contributes to the existing literature by presenting a multidimensional model for measuring EM practices including AEM, REM, and CS. Prior research have extensively focused on AEM and REM while there has been limited research on the pervasiveness of CS practices in Arabian nations especially Egypt. In this regard, this study investigates the relatively new EM tool namely, CS within the Egyptian context besides other EM tools; AEM and REM. Moreover, this study extends prior literature by providing empirical evidence on the impact of EM tools; AEM, REM, and CS collectively on the value of firms listed in the Egyptian Exchange. Furthermore, the study findings are anticipated to have several practical implications for standards setters, investors, analysts, and auditors. For standards setters, the current study turns their attention toward setting enough standards and regulations that limit the

manager's opportunistic behavior to exercise their discretion in financial reporting which in turn protect users of accounting information and help investors making better investment decisions. The findings are of particular importance to auditors to increase their awareness on the seriousness of EM practices to search for a relevant technique to limit and constrain such behavior in firms.

1.5 The Study Limitation

Only non-financial firms listed in the Egyptian Exchange with available data for all study variables are included to cover the study period from 2017 to 2021. Banks and financial services firms have been excluded from the analysis because of the unique nature of their activities. Moreover, the current study is limited to using modified jones (1995) model to proxy for AEM. REM was measured by an aggregate measure including abnormal cash flow, abnormal production cost, and abnormal discretionary expenses. Furthermore, to investigate the existence of CS practices the study depended on ICS and CFCS models. When measuring ICS, the study concentrated on gross profit manipulation models rather than core earnings models.

1.6 Literature Review and Hypothesis Development

The prior studies which are related to the current study can be classified into two main categories. The first focuses on CS as an earnings management tool. The second demonstrates the impact of EM practices on firm value.

1.6.1 Studies on Classification Shifting as an Earnings Management Tool

EM is one of the most studied areas in accounting research. Literature has documented three main tools of EM which are AEM, REM, and CS. The latter has been the main concern of several research due to the long time focus of AEM and REM. CS was first introduced by Mc Vay (2006) as deliberate misclassification of items within the income statement thereby inflating core earnings while bottom line income remains unchanged. Majorly, Mc Vay developed a methodology to identify the evidence of CS practices among US firms. By using a sample of 76901 firm - year observation covering the period 1988 to 2003, she found that managers opportunistically shift core expense (cost of goods sold and selling, general and administrative expense) to income decreasing special items to inflate their core earnings. Beginning with Mc Vay (2006), initial empirical study, a stream of studies has begun to empirically investigate the practice of CS and provide additional evidence. For instance, by

using quarterly data, Fan et al. (2010), concluded that managers engage in CS by shifting core expenses to income decreasing special items, such practice is more prevalent during the fourth quarter than in the interim quarter. While Barua, et al. (2010) undertook a study aimed at investigating whether managers use CS to manage earnings by allocating operating expense to discontinued operations in order to meet or beat analysts' forecasts. Based on a sample of 79643 firm - year observations through the period from 1988 to 2006, the results reported a positive relationship between income decreasing discontinued operations and unexpected core earnings. In the same vein, using a sample of 1230 Korean firms in 2011, Noh et al. (2017) argued that when managers wish to inflate their reported operating income and therefore influence the markets' perceptions, they use non-operating income as well as special expense items for expense misclassification in the IFRS adoption period. While Wu (2016) investigated the pervasiveness of CS practices among European firms based on a sample of 1125 firm - year observations during the period 2005-2015. The results found a significant positive relationship between unexpected core earnings and non-recurring items. In the UK context, Zalata and Robert (2017) investigated the misclassification of recurring items as non-recurring items in the post era adoption. Based on a sample of 1552 firm - year observation over the period 2008 to 2010, the results denoted that managers tend to engage in CS practices following the IFRS adoption showing a significant positive relationship between non-recurring expense and unexpected core earnings.

On the other hand, another group of researchers shed light on gross profit manipulation rather than core earnings. Using a sample containing 319518 firm quarter observations during the period 1988-2012, Fan and liu (2017) argued that when managers are motivated to attain gross profit benchmarks, they tend to misclassify cost of goods sold but not selling, general, and administrative expenses as income decreasing special items. Besides, Poonawala and Nagar (2019) examined whether managers misclassify the cost of goods sold as operating expense in order to manipulate gross profits upwards. By utilizing a sample of 70,224 firm - year observations covering the period from 1988 to 2016, the results revealed that managers are more likely to shift cost of goods sold to operating expense to inflate their gross profits whereas they tend to shift cost of goods sold to research and development than to selling, general, and administrative expenses in order to just meet prior years gross profits. In the Egyptian context, Ezat (2021) examined CS practices through the manipulation of gross profit. He argued that gross profit figure is more comparable than core

earnings and that investors use this figure when making their decisions. The study was conducted on a sample of 494 Egyptian listed firm - year observation during the period 2016 to 2018 and the results reported that Egyptian firms tend to shift other operating income to total operating revenue to report a gross profit increase rather than shift cost of goods sold to selling, general, and administrative expenses.

Besides, Malikov, et al. (2018) expand the scope of expense classification shifting by investigating the misclassification of revenues items within the income statement based on a sample of 12804 firm year - observations covering the period from 1995 to 2014. They argued that managers are more likely to shift non-operating revenues as operating revenues to increase their operating revenue after mandatory IFRS adoption, showing a negative relationship between unexpected operating revenue and non-operating revenue.

On the other side, Nagar and Sen (2014), provided empirical evidence of CFCS, where firms shift operating cash outflows to investing and financing cash outflows, and investing and financing cash inflows to operating cash inflows, thereby increasing operating cash flows. Similarity, based on a sample of 3130 Korean firms that mandatorily adopt IFRS, Baik et al. (2016) denoted that managers shift interest payment from operating to financing cash flows to record an increase in operating cash flows. While tree et al. (2019) investigated the use of CS practices at the level of balance sheet statement. Based on a sample of 67045 firm year - observations during the period 1999 to 2019, they provided empirical evidence that firms are more likely to shift current liabilities to long term liabilities to avoid reporting working capital deficit.

Further, Bansal et al. (2021) addressed the three forms of CS practices which are revenue misclassification, expense misclassification, and cash flow misclassification. By utilizing a sample of 12870 Bombay Stock Exchange listed firm - year observation over the period from 2010 to 2018, the findings confirmed that managers are more inclined in misclassification of revenues rather than misclassification of expense. In addition, the results revealed the existence of cash flow misclassification among Indian firms by shifting investing and financing cash flow to operating cash flow.

1.6.2 Studies on the Impact of Earnings management practices on Firm Value

Arguably, the impact of EM practices (AEM, REM and CS) on firm value is unpredictable. Prior studies examined the managerial intent toward manipulation of earnings and so distinguished between opportunistic EM (to mislead investors) and informative EM (to view a good image to the stakeholders about a firm's performance), (Idris, 2012; Ali & Kamardin, 2018). In this context, most studies stated that EM is a negative behavior as misalignment of managers and shareholder's incentives could induce managers to manage earnings opportunistically thereby creating distortion in the reported earnings, which deteriorates the firm value and resource allocation. Alternatively, other studies concluded that EM practices may be efficient because it potentially enhances the quality, predictivity of information, makes earnings more informative to investors and subsequently enhance the value of the firm (Jiraporn, et al., 2008; Sirega & Utama, 2008; Ismail, 2017; Abbas & Ayad, 2019; Potharla et al., 2021). Relatedly, EM is "a collection of managerial decisions that result in not reporting the true short-term, value-maximizing earnings as known to management. EM can be beneficial: it signals long-term value; pernicious: it conceals short- or long-term value; neutral: it reveals the short-term true performance" (Ronen and Varada, 2008, p27).

Significantly, REM is detrimental to firms in the long term. That is REM activities can be beneficial for the firm value in the short term where managers increase cash flow from operations to satisfy their short-term motives. However, these short-term benefits are essentially at the cost of the long-term firm value (Abbas & Ayad, 2019).

Likewise, CS practices may affect the firm's valuation. Managers may be motivated to engage in CS due to its relatively low opportunity cost and its impact on stock price, moreover they believe that it lacks the negative economic consequences made by REM and thus it does not harm firm value (Chae & Nakano, 2015). Meanwhile, CS may have adverse implications for firm value according to the overvaluation of earnings. Thus, given the focus of the market participants on core items, CS practices could influence the decision-making of market participants, and their perceptions of persistence and growth associated with a firm (Nagar et al., 2021). Such results indicate that investors are potentially fooled by overvaluation of earnings due to the notion "*Higher earnings today implies higher earnings in the future*" (Zalata and Roberts, 2017, p.55). Accordingly, Lattanzio and Thomas (2019) demonstrated that the impact

of CS practices on firm value is conflicting. On one hand, CS may create additional value for shareholders by allowing firm to obtain lower cost financing or avoid costly debt covenants violations, in addition, managers may use CS to enhance the quality and predictivity of core items to help investors better predict future performance which in turn results in higher stock price informativeness (positive effect). On the other hand, managers may exploit CS for opportunistic behavior to increase their own compensation or delay turnover because of poor actual performance, eventually resulting in lowering firm value (negative effect).

Consistent with this discussion, prior studies provided mixed evidence about the impact of EM practices including AEM, REM, and CS on firm value. For instance, Challen and Siregar (2012) undertook a study aimed at examining the impact of EM practices (AEM and REM) on the value of public firms listed in the Indonesian stock exchange through the period 2006- 2009. AEM was measured by discretionary accruals (DA) following Kothari et al. (2005) model while REM was measured using abnormal cash flow value (CFO) estimated from sales manipulation model. Firm value was measured by market-to-book ratio. The results revealed that REM negatively affects firm value meanwhile AEM didn't have a significant impact on firm value.

Gill et al. (2013) tested the impact of EM practices on the firm performance measured by the rate of return on total assets (ROA) and the firm value captured by Tobin's Q, by utilizing a sample of 250 firms listed in the Bombay stock exchange during the period from 2009- 2012. EM was measured by four indicators including managed revenues, total accruals (balance sheet approach), total accruals (cash flow approach), and discretionary accruals (cash flow approach). The study concluded that EM practices affect the firm performance negatively. That is, the increase in EM activities, the greater its adverse impact on the firm's rate of return on assets in the following year. With regard to the impact of EM on market value, a significant and negative impact was found.

Suffian (2015) examined the impact of REM on the value of the firms in Malaysia by analyzing a sample 6215 firm years observations during the period from 2004 to 2011. REM was estimated using abnormal values of cash flow (CFO), discretionary expenditure (DISEXP), and cost of goods sold (COGS), meanwhile, firm value was estimated using Tobin's Q. The study revealed that COGS and DISEXP models have a negative relationship with firm value, whereas CFO model has a positive relationship with firm value. The reason behind that is

the existence of information asymmetry. That is, information asymmetry creates friction between management and shareholders investors.

Also, Yuniarti (2017) addressed the impact of EM practices on the value of state-owned firms listed in the Indonesia stock exchange during the period 2009–2013. In addition, the study examined the influence of corporate governance on the relationship between EM and firm value. The study showed an adverse impact of DA on the firm value measured by Tobin's Q. Moreover, the results revealed that corporate governance plays a moderating variable that influences the impact of EM on firm value. This is because good corporate governance helps mitigating opportunistic earnings manipulation practices.

The study of Mellado-Cid et al. (2018) aimed to investigate the relationship between REM and the value of the firms based on a sample comprised of non-financial firms covered in the standard & poor's Compustat data for the period 1990 to 2013. The study employed abnormal values of production costs (PROD) and discretionary expenditure (DISEXP) as a measure of REM. Tobin's Q has been used to proxy for firm value. The study found that low valuation firms are associated with abnormally high production costs and abnormally low discretionary expenditures.

Using a sample of 648 observations for non-financial firms listed in the Indonesia stock exchange during the period 2012 – 2015, Susanto (2017) examined the impact of AEM proxied with the modified Jones model and REM proxied with abnormal values of CFO, DISEXP, and PROD on firm value measured by market-to-book ratio. The results found a positive impact of AEM on firm value, in contrast, REM negatively affects firm value, implying that ease in detecting accruals manipulations helps investors to judge whether firm is worth or not, while difficulty in detecting REM practices makes investors not dare to speculate to invest in the firm.

Wibowo (2018) examined the relationship between AEM and REM on the value of firms in the manufacturing sector depending on a sample of 104 firms listed in the Indonesia stock exchange during the period from 2010 - 2014. The study employed the modified Jones model to estimate AEM practices and abnormal values of PROD, $DISEXP$ to estimate REM activities, while firm value was estimated using Rhodes–Kropf et al., 2005 measure. The results showed that PROD tends to increase firm value, however, DISEXP and AEM didn't influence market response. In addition, the study provided evidence that there were differences in the relationship between EM practices and firm value due to

the influence of other variables such as being audited by Big 4 auditors, the existence of institutional investors in the firm ownership and IFRS adoption.

While Sunardi (2018) tested the impact of EM on firm value before and after IFRS adoption, moderated by firm life cycle (sales growth, capital expenditure value and firm age) based on a sample of public manufacturing firms listed in the Indonesia stock exchange during the period 2010–2016. The study measured EM by DA using Kotharie et al., (2005) model. Firm value was represented by Tobin's Q proxy. The results revealed that before IFRS adoption EM didn't affect firm value. While during the application of IFRS the impact of EM on firm valuation was positive. Moreover, the study found that variables of sales growth, capital expenditure value and firm age strengthen this impact.

Similarly, Abbas and Ayub (2019) investigated both aspects of EM (AEM and REM) and tested the implications of these earnings manipulations for firm value in perspective of emerging economy of Pakistan for the period 2003-2017. The study classified 327 listed Pakistani non-financial firms into two categories based on their financial health, to differentiate the impact of EM practices on firm value. For AEM, Kaszink (1999) model was adopted and used whereas, REM was measured by the combination of abnormal values of CFO, PROD, and DISEXP. Both types of EM proxies were regressed against the firm's variables of non-discretionary net income, change in net income, and next year cash flow and therefore the results found a positive impact of these EM practices on firm value. However, the impact of AEM practices for Pakistan firms was more pronounced than REM activities.

Darmawan et al. (2019) empirically examines the impact of AEM and REM on firm value by testing a sample of all manufacturing firms listed in Indonesia stock exchange through the period 2013–2017. The study provided evidence that DA didn't affect firm value, meanwhile the impact of REM activities as measured by abnormal values of CFO, DISEXP, and PROD was negative.

Whereas Lattanzio and Thomas (2019) examined whether firm's use of CS is persistent over time, is influenced by peer firm's engagement in reporting strategy, and has consequences for firm value by utilizing a sample of 33106 firm year observations during the period 1990–2017. The study found a positive relationship between expense classification shifting (ECS) and firm valuation. More specially, the results provided evidence that firms engagement in ECS practices in the current year (t) relates positively to its Tobin's Q and stock returns in the following year (t +1). Furthermore, the study pointed out that the

positive relationship between CS and firm value is stronger for firms having higher levels of institutional ownership and when firms are leveraging up their capital structure. That is, the results revealed that the use of CS practices is consistent with the desire to increase debt capacity and generate additional firm value.

Kang and Kim (2019) investigated the relationship between EM practices including AEM and REM on the value of retail firms listed in the Korea Stock Exchange during the period from 2001 to 2016. AEM was measured by the Jones model and the modified Jones model. REM was estimated using the three abnormal values of CFO, PROD, and DISEXP. The results revealed a positive relationship between both AEM and REM practices and market to book as a proxy for firm valuation.

Moreover, Alsoboa (2019) examined the relationship between REM and the firm value of Jordanian industrial listed firm by testing a sample of 308 firm year observation for the period from 2011- 2017. Using the aggregated absolute values of abnormal CFO, abnormal PROD, and abnormal DISEXP to proxy for REM and adopting Tobin's Q as a measure of firm value, the results concluded that when firms engage more in REM activities, the firm value increased. More especially, the study found a positive relationship between abnormal cash flow, abnormal production cost and the firm value, whereas the relationship between abnormal discretionary expenditure and the firm value was insignificant.

Handayani and Ibrani (2020) undertook a study aimed at investigating the effect of audit committee characteristics on EM and its impact on the value of the firm captured by Tobin's Q. By using a sample of manufacturing firms listed in Indonesia stock exchange through the period 2016–2018, the findings showed that EM as measured by discretionary revenues (according to Stubben 2010 model) has a negative impact on the value of the firm. The study indicated that when firms decide to markup profits, there will be an increase in the profit figure and therefore it can temporarily enhance the value of the firm, but if such increase is executed continuously, it will deteriorate the firm value in the long term.

Al-Nsour (2020) undertook a study aimed at investigating the impact of AEM practices measured by modified jones model on the firm value throughout employing a sample of 170 observations for industrial firms listed in Amman stock exchange during the period 2012–2016. The results provided evidence that Jordanian firms engaged in AEM practices. In addition, the study showed no

significant impact of AEM on the market value and the market to book ratio of the firm.

Furthermore, Olatunji and Juwon (2020) examined the impact of AEM and REM on the value of listed manufacturing firms in Nigeria during the period 2008-2017. The study adopted the modified jones model to estimate DA, meanwhile REM was estimated using the three abnormal values of CFO, PROD, and DISEXP. Afterward, firm value was measured by return on equity (ROE). The results found a positive relationship between AEM and the value of the firm, whereas REM activities were inversely related to firm value.

Almari, et al. (2021) undertook a study to analyze the impact of EM among publicly traded firms listed in the Amman stock exchange on firm value as well as to investigate the impact of ownership structure on EM and the value of the firm during the period from 2012 to 2017. By utilizing DA as a proxy for EM and Tobin's Q formula to estimate firm value, the results showed a negative impact of EM on firm value especially in the long term.

In Indian context, Potharla, et al. (2021) tested the impact of REM on the future value of the firm and its persistence based on a sample consists of all listed non-financial firms through the period 2011-2018. The study employed the three abnormal values of CFO, PROD, and DISEXP to proxy for REM, whereas the firm value was measured with Tobin's Q. The results revealed that REM practices reduce the future value of the firm and its persistence. Moreover, the study found that REM activities through abnormal reduction in discretionary spending had more negative impact on firm value compared to the other models of REM.

In the Egyptian context, Osman (2022) undertook a study aimed at testing the effect of determinants of CFCS items on firm value based on a sample of 415 observations for Egyptian firms listed in the Egyptian Exchange over the period from 2016–2020. By employing an unexpected cash flow model to measure CS and Tobin's Q to measure firm value the study found that CS has an inverse impact on firm value.

While Deef (2022) examined the economic consequences of the relationship between EM by CS and managerial ability. i.e., whether highly competent managers are more likely to engage in CS practices than other managers and its influence on the value of the firms listed in the Suadi Capital Market based on a sample of 625 observations through the period from 2015 to 2019. By using Mc Vay 2006 model as a proxy for CS and Tobin's Q to measure the firm value, the

results provided empirical evidence that increased managerial ability can mitigate CS practices which positively reflected on firm value and might enhance financial reporting transparency as a result.

Further, Tulcanaza-Prieto and Lee (2022) analyzed the relationship between REM and firm value moderated by the role of corporate governance based on a sample of 3590 firm year observations of non-financial firms listed in the KOSPI through the period from 2011 to 2013. By using an aggregate REM proxy as calculated by the aggregation of abnormal values of CFO, PROD, and DISEXP and employing Tobin's Q to estimate the firm value, the study found a negative relationship between REM and firm value. Furthermore, the study concluded that corporate governance plays a moderating role in the negative relationship between REM and firm value. More importantly, the results pointed out that the presence of effective corporate governance in a firm increases its reliability and transparency through information disclosure and therefore, decreases the manager's selfish decisions. In addition, the implementation of strong corporate governance policies provides the implementation of high auditing standards and transparency in financial reports which increase shareholder's confidence, consequently, reduce opportunistic earnings manipulations.

Also, Tulcanaza-Prieto and Lee (2022) examined the impact of REM on the value of non-financial Korian firms based on a sample of 4681 firm quarter observations through the period 2010 – 2018. REM was measured using an aggregate REM proxy as measured by the aggregation of abnormal values of CFO, PROD, and DISEXP, while firm value was measured by Tobin's Q. The results revealed a negative impact of REM on firm value, showing that opportunistic behavior of managers to maximize their utility and compensation deteriorates the value of the firm and generates distortion in the reported earnings.

Further, Hussein (2023) investigated the relationship between REM and firm value measured by market value and determined the impact of managerial ability and management confidence as moderated variables on such relationship, based on sample firms listed in the Egyptian Exchange during the period 2016–2020. REM was measured by an aggregate REM proxy as measured by the aggregation of abnormal values of CFO, PROD, and DISEXP. The results revealed a negative relationship between REM and firm value.

Given the above discussion, one limitation of the existing prior studies is that they mainly concentrated on one or two tools of EM; especially AEM and REM, rather than considering all aspects of EM practices. Therefore, the current study provides a more comprehensive analysis that encompasses all three tools of EM together including AEM, REM, and CS then empirically investigates their impact on firm value. Majorly, it is evident that prior literature has a clear contradiction in results concerning the impact of EM practices on firm value. Some studies denoted that EM positively affects firm value while other studies found a negative or no significant impact. This inconsistency in the findings emphasizes the need for future analysis to investigate such impact in the Egyptian setting and developed countries in particular. A summary of the studies reviewed is shown in table (1.1):

Table (1.1) Summary of the Studies for the Impact of Earnings Management practices on Firm Value

Author	Variables Employed	Results
Challen and Siregar (2012)	▪ EM was measured by DA. ▪ REM was measured by abnormal CFO. ▪ Firm value was measured by market-to-book ratio.	▪ REM negatively affects firm value. ▪ AEM didn't have a significant impact on firm value.
Gill et al. (2013)	▪ EM was measured by four alternative EM measures including managed revenues, total accruals (balance sheet approach), total accruals (cash flow approach) and discretionary accruals. ▪ Firm performance was measured by the rate of return on total assets (ROA). ▪ Firm value was measured by Tobin's Q.	▪ A negative impact of AEM on firm performance and the value of the firm.
Suffian (2015)	▪ REM was measured by the abnormal values of CFO, PROD, and DISEXP . ▪ Firm value was measured by Tobin's Q.	▪ COGS and DISEXP models have a negative relationship with firm value. ▪ CFO model has a positive relationship with firm value
Yuniarti (2017)	▪ EM was measured by DA. ▪ Firm value was measured by Tobin's Q.	▪ Adverse impact of EM on firm value.

Author	Variables Employed	Results
Susanto (2017)	▪ AEM was measured by DA. ▪ REM was measured by the abnormal values of CFO, PROD, and DISEXP. ▪ Firm value was measured by market-to-book ratio.	▪ A positive impact of AEM on firm value. ▪ A negative impact of REM on firm value.
Mellado-Cid et al. (2017)	▪ REM was measured by an aggregate REM proxy as calculated by the aggregation of abnormal values of PROD, and DISEXP . ▪ Firm value was measured by Tobin's Q	▪ Firm value was inversely related to REM practices.
Wibowo (2018)	▪ AEM was measured by DA. ▪ REM was measured by the abnormal values of PROD, and DISEXP. ▪ Firm value measured by Rhodes – Kropf et al., 2005 model.	▪ Abnormal PROD tends to increase firm value. ▪ Abnormal discretionary expense and AEM didn't influence market response.
Sunardi (2018)	▪ EM was measured by DA. ▪ Firm value was measured by Tobin's Q.	▪ EM did not affect firm value before IFRS adoption. ▪ EM positively affects firm value after IFRS adoption.
Abbas and Ayub (2019)	▪ AEM was measured by DA. ▪ REM was measured by the combination of the abnormal values of CFO, PROD, and DISEXP . ▪ Firm value was measured by the firm's variables of non-discretionary net income, change in net income, and next year's cash flow.	▪ A positive impact of (AEM and REM) on firm value.
Darmawan et al. (2019)	▪ AEM was measured by DA. ▪ REM was measured by the abnormal values of CFO, PROD, and DISEXP. ▪ Firm value was measured by Tobin's Q.	▪ A negative impact of REM on firm value. ▪ AEM didn't affect firm value.
Lattanzio and Thomas (2019)	▪ CS was measured by unexpected core earnings and income decreasing special item. ▪ Firm value was measured by Tobin's Q and stock returns.	▪ A positive relationship between ECS and firm value.

Author	Variables Employed	Results
Kang and Kim (2019)	▪ AEM was measured by DA. ▪ REM was measured by the abnormal values of CFO, PROD, and DISEXP . ▪ Firm value was measured by market value and market-to-book ratio.	▪ A positive relationship between AEM and firm value. ▪ A positive relationship between REM and firm value.
Al-Nsour (2020)	▪ AEM was measured by DA. ▪ Firm value was measured by market value and market to book ratio.	▪ No significant impact of AEM on firm value.
Handayani and Ibrani (2020)	▪ EM was measured by discretionary revenues. ▪ Firm value was measured by Tobin's Q.	▪ A negative impact of EM on firm value.
Alsoboa (2020)	▪ REM was measured by the aggregated absolute values of abnormal CFO, abnormal PROD, and abnormal DISEXP ▪ Firm value was measured by Tobin's Q.	▪ A positive relationship between abnormal CFO, abnormal PROD and firm value. ▪ Insignificant relationship between abnormal DISEXP and firm value.
Olatunji and Juwon (2020)	▪ AEM was measured by DA. ▪ REM was measured by the abnormal values of CFO, PROD, and DISEXP. ▪ Firm value was measured by return on equity (ROE).	▪ A positive relationship between AEM and firm value. ▪ A negative relationship between REM and firm value.
Almari, et al. (2021)	▪ AEM was measured by DA. ▪ Firm value was measured by Tobin's Q.	▪ A negative impact of AEM on firm value.
Potharla, et al. (2021)	▪ REM was measured by the abnormal values of CFO, PROD, and DISEXP. ▪ Firm value was measured by Tobin's Q.	▪ REM practices reduce the future value of the firm and its persistence. ▪ REM activities through abnormal reduction in discretionary spending had more negative impact on firm value compared to the other models of REM.
Osman, et al. (2022)	▪ CS was measured by an unexpected cash flow. ▪ Firm value was measured by Tobin's Q.	▪ A negative impact of CFCS on firm value.

Author	Variables Employed	Results
Deef (2022)	▪ CS was measured by unexpected core earnings and income decreasing special items. ▪ Firm value was measured by Tobin's Q.	▪ Increased managerial ability can mitigate CS practices which positively reflected on firm value and enhance financial reporting transparency.
Tulcanaza-Prieto and Lee (2022)	▪ REM was measured by an aggregate REM proxy as calculated by the aggregation of abnormal values of CFO, PROD, and DISEXP. ▪ Firm value was measured by Tobin's Q.	▪ A negative relationship between REM and firm value.
Tulcanaza-Prieto and Lee (2022)	▪ REM was measured by aggregation of abnormal values of CFO, PROD, and DISEXP. ▪ Firm value was measured by Tobin's Q.	▪ REM activities negatively influence firm value.
Hussein (2023)	▪ REM was measured by an aggregate REM proxy as captured by the aggregation of abnormal values of CFO, PROD, and DISEXP. ▪ Firm value was measured by market value.	▪ A negative relationship between REM and firm value.

Source: made by the researcher.

Based on the aforementioned preceding literature, the present study aims to identify the impact of EM tools on firm value in the Egyptian context through the following hypothesis:

H1: AEM has an insignificant impact on the value of firms listed in the Egyptian Exchange.

H2: REM has an insignificant impact on the value of firms listed in the Egyptian Exchange.

H3: ICS and CFCS have an insignificant impact on the value of firms listed in the Egyptian Exchange.

***H4*:** AGG_EM has an insignificant impact on the value of firms listed in the Egyptian Exchange.

***H5*:** There are insignificant differences among sectors in the Egyptian Exchange regarding the study variables.

1.7 The Study Outline

The study is divided into five chapters as follows:

Chapter 1: provides the general framework of the study which includes an introduction, the main study problem, study objectives, study significance, study limitations, literature review, and finally the study outline.

Chapter 2: illustrates the conceptual framework of firm value and earnings management including firm value concept, its determinations, and approaches. Additionally, this chapter presents the definitions of EM, the different motivations behind such practices, and the two main approaches of EM and their determinants.

Chapter 3: presents the relationship among EM tools and the measurements of these different tools.

Chapter 4: includes the empirical study which illustrates the study methodology, the normal distribution test, descriptive statistics, validation tests of the models and the statistical analysis for testing hypotheses.

Chapter 5: includes the summary, conclusions, recommendations of the present study, and suggests some avenues for future research.

Chapter 2
Firm Value and Earnings Management: A Conceptual Framework

2.1 Introduction

The main objective of a firm is to maximize its profits; however, it is no longer considered as a comprehensive goal to meet the needs of all stakeholders. As a result, the concept of shareholders maximization has become a strategic goal for the firm since it reflects the success of management to maximize shareholder's wealth, so that appropriate decisions could be taken, especially those related to investment decisions and resource allocation that support the competitive advantage and maximizing value.

Arguably, the implications of EM are detrimental to firm valuation. Managers are more likely to use EM practices efficiently or opportunistically to attain their desired earnings targets. EM is a widely spread phenomenon, it has been extensively investigated in accounting literature. EM practices refer to the manager's intervention in financial reporting to manipulate reported earnings through accounting practices and/or deviating from normal operational activities to mask the firm's true economic performance. The driving forces of EM may include misleading stakeholders, achieving personal gains, reducing losses, influencing taxes, and meeting or beating the financial analysts' expectations. In this regard, two main approaches of EM were mostly addressed: the accounting earnings management approach and the REM approach.

Thus, this chapter presents the firm value concept, firm value determinants, discusses the firm valuation approaches, illustrates the EM concept, highlights the motivations and incentives behind EM practices, and discusses EM determinants. Finally, presents the EM approaches.

2.2 Firm Value Definition

Generally, the value concept represents the price that prospective buyers are willing to pay in exchange for an item (Ahmed, 2013). The Financial Regulatory Authority (2017) indicates that value is the financial returns estimated as a fair price for the entity being evaluated based on a set of future assumptions. The most important assumptions are the firm going concern and its ability to generate future benefits through the optimal use of its assets except for liquidation cases.

Firm value is one of the most crucial indicators used for many purposes in business like business valuation, portfolio analysis, and risk analysis (pratama,

2018). Extant literature has interested in defining firm value; however, it has not yet reached a specific definition, but it can be described depending on the perception of each stakeholder and expected interests that will be obtained.

Zalat (2015) defined firm value as a fair evaluation of what the firm's assets can generate in the future in terms of cash flows that are parallel to what buyers obtain to abandon such assets. Siboni and Pourali (2015) indicated that firm value refers to the firm's ability to generate profit in the future. According to Betty (2013); Ha and Minh (2020) firm value is an economic measure reflecting the market value of the whole business. As well, Lumapow (2019); Novatian (2021); Reschiwati et al., (2021) described firm value as the investor's perception of the firm that is often associated with stock prices. The higher the stock price, the higher the wealth of shareholders. While Shah and Khalidi (2020) argued that firm value is a measure of the financial worth of a firm at a specific time. Momon et al. (2021) defined firm value as the sum of the present value of future cash flows generated by the firm's current assets plus potential investment projects. The current share value not only reflects current earnings capacity but also reflects expectations of future operating and investment performance.

Inevitably, determining firm value has an importance to both management and stakeholders, especially investors and creditors. Firm value is considered as an indicator of the management's efficiency in managing the firm and making their investment decisions, which is reflected in its ability to obtain debts and choose the best sources of financing (Noagy, 2017). For investors, firm value reflects the investor's perception toward the firm's success. It provides a signal about the firm's current performance and its future prospects for investors to make better investment decisions in the capital market (Sukesti et al., 2019; Putri and Rahyuda, 2020). For creditors, the value of a firm gives them a sense of security in providing loans to such firms since it reflects the firm's ability to pay its debts and its ability to continue in the future (Antoro et al., 2020; Reschiwati et al., 2021).

2.3 Firm Value Determinants

Several studies have indicated that firm value is affected by several factors. Some factors are related to macroeconomic determinants that can't be controlled by the firm such as inflation, gross domestic product, monetary policy, and exchange rate. Whereas other factors are basically the firm characteristics which

include firm size, profitability, liquidity, leverage, dividend policy, investment opportunity, capital structure, and firm growth (Ha and Minh, 2020; Abba et al., 2020), as indicated in figure (2.1):

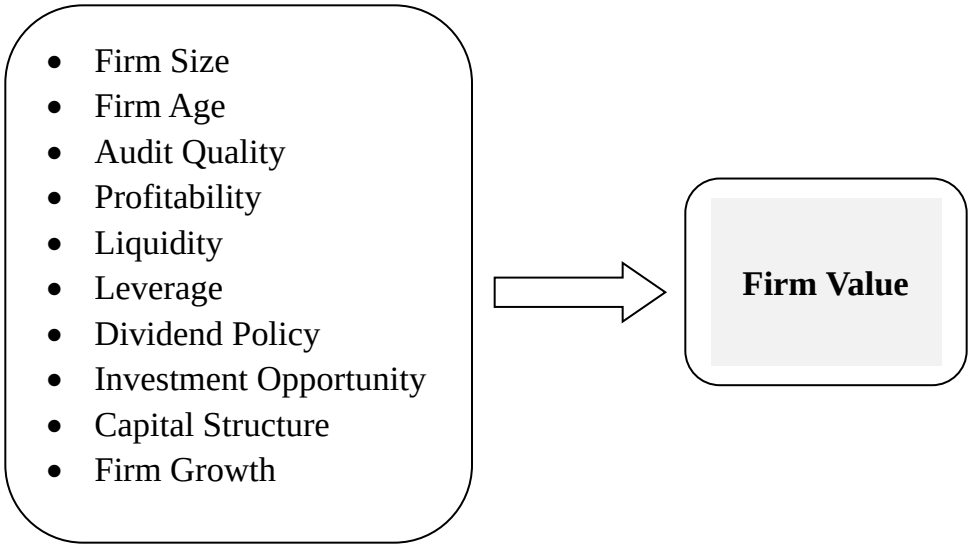


Figure 2.1: Factors Affecting Firm Value
Source: prepared by the researcher.

The following section will discuss the main firm value factors as follows:

2.3.1 Firm Size

Firm size is defined as a signal received by investors to make their investment decisions. Investors tend to be more interested in firms with large scale this is because large firms are in a more stable financial condition (Singh, 2021). More specially, they have more stable operations, are relatively more able to generate profits, and are more flexible in obtaining funds due to their accessibility to the capital market. In addition, large firms have reached their maturity stage and have positive cash flows and more strong funding sources (Marpaung et al., 2022). So, investors capture this as a positive signal concerning the firm's ability to provide a greater rate of return and therefore increase stock prices in the capital market which in turn increase the firm value (Bandanuji and Khoiruddin, 2020). This is supported by several studies such as (Mohamed, 2017; Atiningsih and Izzaty, 2021; Marpaung et al., 2022; Purwani and Santoso, 2023) who showed a positive impact of firm size on firm value. Meanwhile, Susanti and Restiana (2018);

Hirdinis (2019) stated that firm size had a negative impact on firm value. Antro et al. (2020) proved that firm size had an insignificant impact on firm value.

2.3.2 Firm Age

The firm age is how long a firm can survive, compete, and take business opportunities. It can be measured as measured by the natural logarithm of the time interval between the date of the firm's incorporation and the year under review (Susanti and Restiana, 2018). Research conducted by Leite and Carvalhal (2016); Susanti and Restiana (2018) demonstrated that older firms show higher value. Their results revealed that age has a positive impact on firm value. The longer the firms stand, the more information the public has gained about the firm which leads to investors' confidence in these firms. However, research conducted by Chay et al. (2015) showed a negative impact of firm age on firm value.

2.3.3 Audit quality

Audit quality plays an important role in determining firm value. Prior studies showed conflicting evidence relative to the impact of audit quality on firm value. Wijaya (2020) demonstrated that capital markets react positively to firms that have higher quality auditors. One possible explanation is the competencies and advantages possessed by BIG4 auditors. Moreover, in terms of human resources, assets, and networks, BIG4 auditors provide a guarantee of a higher quality audit process. Similarly, Afza and Nazir (2014) indicated that BIG4 auditing firms can enhance the firm's value and their reputation in the capital market. This is because investors perceive that firms who are audited by BIG4 firms provide proper, reliable, and authentic financial information which in turn improves investors' confidence and increase firm value accordingly. On the other side, Yolandita and Cahyonowati (2022) found a negative impact of audit quality on firm value, indicating that managers' attempt to signal a high firm value using one of the BIG4 Auditors fail to be communicated to investors.

2.3.4 Profitability

Profitability is considered as an indicator of the success of the firm's operations, it refers to the firm's ability to earn profits (Iswajuni et al., 2018). Profitability ratio measures the effectiveness of firm's management based on the returns generated from sales and investment income (Cordiaz et al., 2021). A high level of

profitability influences a firm's financial flexibility whereby firms can pay dividends and obtain a positive rating in the capital market. It shows the firm's prospects for the future, so that it helps investors to determine how efficiently a firm spends its funds on its operational activities to generate higher profits (Singh, 2021). Positive responses from these investors will increase stock prices and increase firm value (Antro et al., 2020). Research conducted by Tahu and Susilo (2017); Cordiaz et al. (2021); Marpaung et al. (2022) stated that firm value was positively affected by profitability. This result is inconsistent with the finding obtained by Dewingrat and Baskara (2020) who stated that profitability had a negative impact on firm value. While Hirdinis (2019); Rahayu et al. (2023) showed that profitability had an insignificant impact on firm value.

2.3.5 Liquidity

Liquidity refers to the firm's ability to meet short-term obligations that must be met immediately (Tahu and Susilo, 2017). It represents the amount of cash or current assets that can be quickly converted into cash for the firm's delayed operations and financial needs (Mohamed, 2017). Firms with a higher level of liquidity tend to be in a good position which attracts investors to invest in such firms, leading to an increase in share price and a further increase in firm value (Marpaung et al., 2022). This interpretation is in line with the results of Aggarwal and Padhan (2017); Nurmadi and Novietta (2022); Purwani and Santoso (2023) which showed a positive impact of liquidity on firm value. This finding is contrary to research conducted by Mohamed (2017); Crdiaz et al. (2021) which showed that liquidity had a negative impact on firm value. While Dewingrat and Baskara (2020) proved that liquidity had an insignificant impact on firm value.

2.3.6 Leverage

Leverage measures a firm's ability to meet all financial obligations. It shows how the firm's management uses the funds effectively and efficiently to achieve added value for the firm. Leverage in theory negatively related to firm value. The higher the leverage, the firm value is lower (Tahu and Susilo, 2017). This is consistent with the findings of Vo and Ellis (2017); Mohamed (2017); Gharaibeh and Qader (2017) who empirically proved that leverage had a negative impact on firm value. That is low leveraged firms are more likely to create value for shareholders. In contrast, leverage may have a positive impact on firm value

(Sukesti et al., 2019). That is more debt controls the amount of cash flows excessively used by managers, therefore reducing their ability to waste a firm's resources and avoid futile investment. In such a case, leverage is considered as a commitment since it forces managers to pay out cash for their creditors and lower agency cost, eventually firm value increases (Mugera, 2013). This is consistent with the findings of Radj and Artini (2020) which showed a positive influence of leverage on firm value.

2.3.7 Dividend policy

Dividend policy is a decision whether profits generated by the firm at the end of the year will be distributed to shareholders in the form of dividends or retained to increase capital to finance investment in the future (Trian and Tarmidi, 2019). It has been measured by extant research as the ratio of dividend per share to earnings per share. Dividend policy is considered an inseparable part of firm financing decision (Agung et al., 2021). Small or large dividends paid to shareholders depend on a firm's dividend policy and are related to consideration of various factors such as liquidity, cash position, debt covenants, and firm's growth prospects (Tahu and Susilo, 2017). Firm value can be seen in a firm's ability to pay dividends. An increase in dividend policy represents firm's better prospects, enhances investor confidence in such firms and ultimately increases firm value (Agung et al., 2021). This is in accordance with the findings of Siboni and Pourali (2015); Purwani and Santoso (2023) which showed that dividend policy had a positive impact on firm value. However, it contradicts the results of Gharaibeh and Qader (2017) who found an insignificant impact of dividend policy on firm value.

2.3.8 Investment Opportunity

Investment decisions are one of the crucial factors that reflect the firm's actions on the allocation of funds regarding asset investment to attain its long-term and short-term goals (Sherine et al., 2022). According to signaling theory, an investment decision is a positive signal that leads to an increase in firm value. That is management provides a signal in the form of information related to investment decisions that represent firm's growth prospects in the future which impacts profits and increases firm value ultimately (Suteja et al., 2023). This interpretation is in accordance with research conducted by (Siboni and Pourali, 2015; Afridi et al., 2022) who revealed that investment decisions had a positive impact on firm value.

On the contrary, Triani and Tarmidi (2019); Abba et al. (2020) proved that firm value wasn't affected by investment decisions.

2.3.9 Capital Structure

Capital structure is the allocation of firm equity and the use of debt to executive the firm's operational activities. The formation of capital structure has an important role in the firms' activities for determining its profits. The composition of the capital structure is shown as a signal for investors to assess the condition of the firm (Putri and Rahyuda, 2020). Determining the optimal capital structure is an important decision that affects firm value. That is capital structure management targets to create a permanent mix of sources of funds that is efficient and effective in increasing share prices and maximizing the firm value (Marpaung et al., 2022).

Studies show a contradictory impact of capital structure on firm value. A positive impact is determined by the success of the firm in utilizing debt in business expansion and obtaining a positive response from investors (Putri and Rahyuda, 2020). This is in accordance with the study of (Hiridinis, 2019; Rahayu et al., 2023; Purwani and Santoso, 2023). Meanwhile, negative impact means that any increase in debt can provide high risk and reduce firm value (Putri and Rahyuda, 2020). Such a result corresponds to the findings of (Paminto et al., 2016; Antoro et al., 2020; Cordiaz et al., 2021).

2.3.10 Firm Growth

Firm growth can be represented by increased assets or sales. Sales growth means that the firm has effectively and efficiently sold its products so that it can influence the rate of investment and market value. The higher sales growth provides a positive perception to investors, increases demand for shares, and drives up stock prices. This is because investors generally prefer firms that continue to show growth (Putri and Rahyuda, 2020). Moreover, asset growth is another factor that can influence firm value. It is measured by using changes in total assets owned by the firm. Increasing assets from one period to the next indicates that the firm's performance is improving and that provides a signal to investors in investing positively (Marpaung et al., 2022).

Accordingly, the existence of growth opportunities provides a positive signal about the firm's growth in the future (Singh, 2021). In this sense, firms that have

rapid growth opportunities are expected to have better value since these firms are favorable to investors who have higher prospects for recovering their investments (Mugera, 2013). This interpretation is supported by Putri and Rahyuda (2020); Marpaung et al. (2022) who stated that higher growth reflects an increase in firm value, and therefore their results revealed a positive impact of firm growth on firm value. However, this finding is contrary to Paminto et al. (2016) who found a negative impact of firm growth on firm value. While Antoro et al. (2020) showed an insignificant impact of firm growth on firm value.

2.4 Firm Value Approaches

There are many approaches to firm valuation. Accounting literature has determined three main approaches that are commonly used to estimate firm value as follows:

2.4.1 Asset Approach

This approach focuses on a firm's net asset value that is determined by subtracting total liabilities from total assets. There are three common ways of evaluating net assets: book value, adjusted net assets, and replacement value (The Egyptian standards for firms' evaluation, 2017).

The book value method uses the historical costs as recorded on the balance sheet. The main benefit of this method is its simplicity as it is easy to collect data, and it saves time and effort. However, one challenge that often accompanies this method is the impact of economic factors such as inflation and changes in purchasing power. Thus, according to adjusted net assets, the value of assets and liabilities are adjusted to the fair market value and therefore net value/ net worth is obtained (Hendawy, 2020). Under replacement value the buyer will pay no more for the targeted firm than it would cost to obtain a comparable set of substitute assets. That is, replacement value includes the cost of acquiring or replicating the assets and all costs related to replacement (Ameria, 2020).

2.4.2 Income Approach

This approach is based on converting the future cash flows into a present value by applying an appropriate discount rate that reflects the firm's conditions and circumstances and the rates of inflation and growth in the market in which it operates. The three main methods under this approach are: net cash flow (NCF),

discounted cash flows (DCF), and net present value (NPV) (The Egyptian standards for firms' evaluation, 2017).

2.4.3 Market Approach

This approach depends on market values; it reflects how investors perceive the firm. The assumption under this approach is that the firm being valued is comparable to other firms in a specific market. The most utilized methods of the market approach are market-to-book ratio and Tobin's Q (Ameria, 2020).

The market-to-book value ratio is commonly used in literature as a measure of firm value, it can be formulated as the ratio of the market value to book value of equity (Bandanuji and Khoiruddin, 2020; Atiningsih and Izzaty, 2021; Marpanung et al., 2022). Additionally, Tobin's Q is considered a widely used proxy for firm valuation, that was developed by James Tobin. It refers to the ratio of the market value of a firm to the replacement cost of its assets (lindenberg and Ross, 1981). That is, if Tobin's Q value is less than (1) it means that the cost to replace a firm's assets is higher than a firm's stock value. Conversely, a high Tobin's Q (greater than 1) interprets that a firm stock is overvalued and that investment in such firms is more attractive (Sajid and Safdar, 2023). Particularly, measuring replacement cost is more complex and difficult, therefore several prior studies (e.g., Gill et al., 2013; Hassanein, 2015; Lee and Yeo, 2016; Tulcanaza Prieto and Lee, 2022; Sajid and Safdar, 2023) depend on book value of total assets rather than replacement cost and so Tobin's Q value is calculated through the following ratio:
(Market value of equity + book value of total liabilities)/book value of total assets.

2.5 Earnings Management Definition

EM is a widely spread phenomenon, there is no specific definition of EM, extant studies have provided a lot of definitions for EM, Schipper (1989, p.92) is among the first who defined EM as the "*Purposeful intervention in the external financial reporting process with the intent of obtaining some private gains*".

Another definition commonly cited in the literature is by Healy and Wahlen (1999, p.368) "*Earnings management occurs when managers use judgement in financial reporting and in structuring transactions to alter financial reports to either mislead some stakeholders about the underlying economic performance of*

the company or to influence contractual outcomes that depend on reported accounting numbers".

An additional definition by Fields et al. (2001, p.260) noted that, "*Earnings management occurs when managers exercise their discretion over the accounting numbers with or without restrictions. Such discretion can be either firm value maximizing or opportunistic*".

Moreover, Wroblewski (2016) demonstrated that EM occurs when managers intervene in financial reporting to attain their desired earnings targets. Nevertheless, such practices might take place without necessarily violating accounting regulations, and otherwise take advantage of possibilities of choice in accounting policy.

Mohamed (2017) defined EM as the intentional intervention of managers in financial reporting to control or influence reported earnings by utilizing accounting practices and/ or real earnings activities to mislead some stakeholders about the underlying economic performance of the firm or to influence contractual outcomes that depend on reported accounting numbers.

Additionally, El Diri (2018) illustrated that EM practices stem from manager's desire to exhibit earnings in a way compatible with their private benefits (upward or downward). Particularly, managers exercise discretion over financial reporting by abusing some contracting deficiencies, stakeholders' bounded rationalities, and information asymmetry in the market, through some economic decisions, a change in the accounting treatment, or other sophisticated methods.

Furthermore, Abd El Kader (2020) indicated that EM practices refer to management deliberate intervention in financial reporting by selecting or changing accounting methods and treatments without violating the accounting standard and/or influencing firms' operational, investing, and financing decisions with the aim of misleading users of financial statements about the real economic performance to obtain their personal benefits or in response to contractual, political, and regulatory incentives.

Therefore, the current study concluded that EM practices occur when managers intentionally intervene in financial reporting to manipulate reported earnings

through accounting practices and/or real earnings activities to achieve specific targets. Such manipulation could be either firm value maximizing or opportunistic.

2.6 Earnings Management Motivations

EM motivations may arise from capital market expectations and valuations, contractual incentives, and political cost and regulatory incentives. These motivations will be discussed as follows:

2.6.1 Capital Market Expectation Motivation

Market incentives arise when managers perceive the connection between reported earnings and a firm's market value (Wroblewski, 2016). These incentives are recognized as the most important motivation for managers to manipulate earnings. Notably, managers have strong incentives to manipulate earnings due to the increased reliance of investors and financial analysts on accounting information for evaluating a firm's performance (Chen, 2020). That is, managers intend to exercise their manipulation in response to capital market pressure to achieve benchmarks and influence stock prices in the short run to satisfy investors' expectations and increase the firm's market value eventually (Ali, 2021). In other words, firms are more likely to manage earnings to satisfy market expectations to meet/beat analysts' forecast, achieve some essential thresholds, and preserve a desired credit rating. Taken together, they may smooth earnings to avoid negative market reactions (Jevasuwan, 2018).

2.6.2 Contractual Motivation

Managers may be incentivized to employ EM practices due to the contracts between firms and their stakeholder, whether they are internal (managers, workers) or outside the firm (especially lending contracts) as these parties rely primarily on the accounting information to monitor and evaluate the firm performance (Mohamed, 2017; Alfadhael and Jarraya, 2021).

Accounting data-based contracts create incentives for EM either to mislead the shareholders about the real performance or to influence the contractual outcomes like management compensation contracts and debt contracts (Ibrahim, 2012; Boghdady, 2019). That is, due to the positive relationship between bonus plans and reported earnings, managers tend to be more involved in EM practices to increase their personal wealth (El Sayed, 2019; Chen, 2020).

Also, Debt covenants may induce managers to manipulate earnings either to reduce the restrictiveness of accounting-based constraints imposed by creditors in debt agreements or to avoid the severe consequences of violating debt contracts (Orieby, 2017).

2.6.3 Regulatory Motivation

Since firms are subjected to numerous regulations related to accounting figures and ratios, managers may deliberately manipulate earnings to show how they comply with such regulations (Habbas and Alghamdi, 2015). Failure to comply with regulations may be the reason for government interference or may deprive the firm of obtaining government grants and subsidies (Mahmoud, 2020).

Accordingly, managers are more likely to manage their earnings to avoid the political and legislative costs especially taxes, government rules, and labor costs. They may increase or decrease their earnings (upward or downward) depending on their situation (Ali, 2021).

2.7 Earnings Management Determinants

In this section the current study presents the most important determinants of EM practices as follows:

2.7.1 Accounting Standards

Accounting standards help as a crucial infrastructure in capital markets, used as the yardstick for investors in making their investment decisions when assessing the business performance and fiscal status of firms (Ebaid, 2016).

The emergence of multinational firms and globalization of international capital markets has raised a need for a unified set of accounting standards to be applied around the world (Rad and Embong, 2014). These standards provide national and international decision makers with relatively homogenous information that is comparable and reliable (Aly and Nguyen, 2013). For this, International Financial Reporting Standards (IFRS) are introduced by ISAB which are increasingly being adopted across the world as a basis for a firm’s financial reporting (Hasan and Rahman, 2017).

Accounting regulators expect that IFRS can improve transparency and comparability of financial information across countries, reduce information

asymmetry, remove cross-border acquisition obstacles, result in more efficient allocation of resources and higher economic growth, reduce cost of capital, and enhance the quality level of financial reporting and thereby constrain managers' opportunistic discretion in determining the accounting figures which lead to a reduction in the level of EM practices (Rad and Embong, 2014; Doukakis, 2014; Mohamed, 2020). Despite that, prior studies contradicted such an explanation, providing evidence that the level of EM practices increases after the IFRS adoption. For instance, Zalata and Roberts (2017) pointed out that after the IFRS adoption in UK context, CS practices have become a common practice and more pervasive. More specifically, the results found that managers opportunistically engaged in CS practices due to less strict regulations on the disclosure of non-recurring items under IFRS. In the Egyptian context, El Garf (2017) revealed that after the development in the Egyptian accounting standards in 2015, the level of AEM decreased significantly, while the practice of REM increased. Also, Noh et al. (2017) found that classificatory shifting practices were more pervasive in the Korean mandatory IFRS adoption period. In the UK context, Malikov et al. (2017) indicated that firms employ revenue misclassification to a greater extent in the period following mandatory IFRS adoption. Conversely, Sellami and Fakhfakh (2013), confirmed that mandatory adoption of IFRS in the France context significantly reduced the use of AEM and REM practices, showing an improvement in earnings quality. Moreover, Hasan et al. (2022), revealed a lower magnitude of REM activities after the IFRS adoption.

2.7.2 Internal Corporate Governance

The financial scandals in several firms have triggered a crisis of confidence about the reliability of financial information and had a negative impact on stakeholders' behavior, this problem resulted from the conflict of interest between the owner and managers (Zgarni and Zehri, 2016).

The agency theory states that managers as agents will act for their own interest to maximize their private gains rather than for the benefits of their shareholders. This is because they have more inside information about the firm than the shareholders (El Madbouly, 2022). Thus, this managerial behavior emphasizes the crucial role of corporate governance as a monitoring system that aims primarily at

resolving agency problem and reducing agency cost by balancing between interests of all parties (Al-Haddad and Whittington, 2019).

According to the Egyptians Corporate Governance Code, corporate governance is "the set of rules, principles, and systems that govern the relationship between the board of directors, on one hand, and a company's owners and other stakeholders, on the other hand, to achieve the best protection and alignment of interests of these parties" (ECCG, 2016, p.12).

In reality, corporate governance reforms and guidelines reinforce investor's confidence by coordinating their interests with the preferences of managers, developing the trustworthiness, and enhancing the integrity of reliability, and accuracy of the reported earnings amounts disclosed in the financial statements (Attia, 2016). Accordingly, applying effective corporate governance mechanisms reduces agency costs, helps mitigating EM practices, and creates value for shareholders (Nazir and Afza, 2018).

More importantly, the role of corporate governance in mitigating EM practices is extensively researched. Almost studies reached a consensus that EM is more prevalent in countries with weak corporate governance. For instance, Zalata and Roberts (2016) observed that strong internal corporate governance curbs CS practices in the UK context. The results showed that CS is less prevalent in firms with boards comprising more independent directors and more directors with long tenure. Furthermore, when the audit committee is characterized by more meetings, more financial expert members, and more directors with long tenure, it can better mitigate CS. Also, Joo and Chamberlain (2017) concluded that strong governance such as having an independent board lead to a reduction in misclassification practices. Abdou et al. (2021), found that firms are less likely to engage in EM practices when they have larger boards (independent outside directors) and a low percentage of female directors. While Behery et al. (2019) provided evidence that in state owned manufacturing firms, corporate governance in the presence of board size, board independence, number of audit committee meetings, and financial expertise for audit committee members tends to act as an effective monitoring system for mitigating manipulation using production costs. While in privately owned manufacturing firms, only board independence and several audit committees are more likely to curb such practices. Azzam et al. (2021) indicated

that the implementation of corporate governance in Jordanian firms minimize the managerial discretion. The results found that corporate governance is negatively associated with AEM and REM practices due to the inclusion of more independent directors with expertise and the prohibition of a dual leadership structure. Usman et al. (2024) found a negative relationship between audit meetings, audit committee experts and CS, however, the audit committee size doesn't influence classificatory manipulation. While Bogdady (2019) revealed that corporate governance mechanisms act more effectively in monitoring and curbing accruals manipulation than REM practices.

2.7.3 Audit Quality

An agency problem arises from information asymmetry between the management and shareholder, this would lead to EM practices as the shareholders have insufficient resources or access to pertinent information to monitor and control the managers' activities (Al Zoubi, 2016).

The audit function assists in reducing this agency cost, auditors are responsible for providing reasonable assurance that the financial statements are free of material misstatements, identifying control risks in a client's accounting risks, restricting managerial discretion, and therefore reducing the information asymmetry and conflicts of interests (Mnif and Hamouda, 2020). Auditors of higher quality are not motivated to accept doubtful accounting methods and are more likely to detect errors and misstatements during the audit work. Hence the auditing process is considered as an effective monitoring mechanism that would restrict managers' incentives to engage in EM and so control managerial discretion (Al Zoubi, 2018; Yasser and Soliman, 2018; Khanh and Nhoung, 2018).

For the association of EM with audit quality, prior literature demonstrated whether high audit quality helps mitigate EM activities or otherwise increase it. Audit quality may be negatively related to EM practices; researchers revealed that large audit firms have greater motivation and competencies to ensure higher audit quality and avoid any reputational risks due to their large client base, they are not under pressure to accept fraudulent practices to retain or satisfy their clients, additionally, they have more financial and operational resources relating to a number of staff, auditor's expertise, budget and technology which allow them to implement more extensive procedures to detect earnings manipulation (Khanh and

Nguyen, 2018; Ismail, 2020; El Deeb and Ramadan, 2020). In contrast, audit quality may be positively related to the extent of EM due to the institutional setting in the country i.e., when the institutional settings don't motivate auditors to curb EM practices (El Deeb and Ramadan, 2020). Consistent with this, (Ahmad et al., 2016; Joo and Chamberlain, 2017; ElMadbouly, 2021; Nagar et al., 2021; Mahmoud, 2023) revealed a negative and significant relationship between audit quality and EM practices. However, such results contradict the findings of Chowdhury and Eliwa (2021) and therefore a positive and significant relationship was observed between audit quality and EM.

2.7.4 Firm Life Cycle

The conventional life cycle model reveals that firms progress monotonically from birth to decline and accordingly firm's strategies and activities change (Habib and Hasan, 2019). Firms face different opportunities and challenges during each stage and therefore they undertake different actions to compete strategically (Bansal, 2021). Moreover profitability, earnings persistence, cash flow volatility, and the cost of capital vary across stages so firms may have different incentives to engage in EM practices. In other words, their attitude to manage earnings varies across different stages of firm life cycle, at each stage managers may be motivated to manipulate earnings upwards or downwards to attain their targets and to respond to these requirements that emphasize their survival in the market (Hussain et al., 2020; Ezat, 2021).

In introduction stage firms entered the market for the first time and there is a high degree of risk and uncertainty about their continued survival. This stage is characterized by limited and informal structure, substantial information asymmetry, and high capital expenditures (Xie et al., 2022), in addition, there are relatively low levels of initial sales, profits, and negative operating cash flows due to startup costs and higher cost of debt, therefore earnings at this stage are expected to be more volatile and less persistent (Jaafar, 2011). As a result, managers may be motivated to use EM practices to present a better financial picture. They are heavily dependent on external financing and substantially wish to positively influence the perception of capital providers (Hussain et al., 2020; Bansal, 2024). On the other side, firms are less likely to have the pressure of reporting profits, they are motivated to spend more on innovation and marketing activities to form

the building blocks for the future prospects of the firm (Nagar and Radhakrishnan, 2017).

While in the growth stage, sales volume is expected to increase and therefore the gross margin is maximized to expand their market share over their competitor, and this leads to an increase in the operating cash flows. Accordingly, growth firms have a significant incentive to employ EM practices to meet earnings benchmarks to guarantee the continuity of their future performance (Ezat, 2021; Bansal, 2022). Moreover, growth firms still make large investment expenditures which in turn require large funding activities. This large need for funds provides room for managers to report positive and high profits to influence the perception of outside parties (Dewi et al., 2022).

At maturity stage, sales are stable, and profits are high. This stage is characterized by reduced risk and uncertainty about future earnings and (current and future) cash flows due to the decrease in cash flow volatility and the increased reliance of firms on their internal resources, in addition to reputational costs and a high level of governance (El Sayed, 2021). As a result, the managerial discretion at this stage is expected to be low as firms are more accountable and transparent as compared to other stages due to the high level of governance activities (Hussain et al., 2020). Moreover, in this stage, firms have a sustainable position in the market, have fewer potential projects investment, and rather focus more on better efficient operating, and therefore EM practices are not attractive to managers (Khuong et al., 2022).

In the shake-out and decline stage, firms deal with multiple issues like declining market share, a falling growth rate which impacts profitability, and an unstable CFO (El Sayed, 2021; Mulchandani et al., 2023). Generally, firms in the decline stage are less profitable, more leveraged, have lower sales growth and negative value of equity, carry maximum business risk, and accordingly have negative operating cash flows. Firms in the decline stage are likely to be in the process of liquidating assets, repaying, or renegotiating debt. (Nagar and Sen, 2016; Al Maleeh, 2019). Therefore, they are more probable to be financially distressed. In such a case, they have more incentives to exercise EM practices to enhance the image of their firm in the eyes of investors and creditors (El Deeb and Ramadan, 2020).

2.7.5 Firm Size

Firm size plays an active role in determining the behavior of the firm. There have been two contradictory viewpoints concerning the impact of the firm size on EM. The first viewpoint argued that there is a negative relationship between firm size and EM practices, several reasons exist to evidence this negative relationship. Large firms are more likely to have stronger internal control systems, better corporate governance mechanisms and more qualified internal auditors as compared to small firms which in turn reduce the ability of managers to engage in EM practices. In addition, large-sized firms are usually audited by higher audit quality firms which helps mitigate EM due to the efficient and effective audit performed (Bassiouny, 2016; Ali et al, 2015; Al Madbouly, 2021). Moreover, large firms may not be motivated to engage in EM due to their concern about their reputation and the financial reporting credibility they established over long periods among their shareholders (El Deeb and Ramadan, 2020). Furthermore, large firms are more politically sensitive than small firms and face higher political costs if they are engaged in EM practices (Desoky, 2016). In other words, this viewpoint demonstrates that large-sized firms can effectively control EM practices whereas small firms which have less experienced auditors and less strict internal control tend to practice higher levels of manipulation (Trang and Mai, 2021). Such interpretation is supported by the findings of (Ahmed and Azeem, 2014; El Hawary and Hassouna, 2021; Barus et al., 2019; El Sayed, 2019) which provided a negative and significant relationship between firm size and EM practices.

On the other side, an opposing viewpoint argues that the firm size is positively related to EM, as managers in large firms have more incentives to manipulate earnings because of their enormous transactions number as well as their complex business structures and therefore shareholders don't seem to understand their operations (El Shafey, 2020) also they are under greater pressure to meet or beat analysts' expectations. Moreover, they have more bargaining power with auditors i.e., they have better negotiating ability with their auditors to concede some EM practices in the financial reports (Khanh and Nguyen, 2018). In addition, since large firms have an effective internal control structure, managers may have more power to override this structure as a whole and be involved in EM practices to attain their targets (El Deeb and Ramadan, 2020).

Consistent with this, Ali et al. (2015) found a positive and significant relationship between firm size and AEM. In the same vein, Uwuigbe et al. (2015) indicated that large firms have more incentives to be involved in EM practices due to the complexity of their transactions and operations and accordingly it is difficult for external users to identify such practices. Thus, a positive and significant relationship was observed between firm size and discretionary accruals. Also, Trang and Mai (2021) provided evidence that firm size is positively related to discretionary accruals. That is the larger the firm size is, the greater motivation of managers to manipulate earnings. While Abiahu et al. (2020) found a significant positive relationship between firm size and unexpected core earnings. Furthermore, Bansal (2021) argued that the firm size influences the form of CS (i.e., revenue misclassification or expense misclassification). The results found that large firms have a larger magnitude of non-operating revenues along with operating revenue because they are more diversified and therefore, they may be motivated to engage in revenue misclassification to inflate their operating revenues and operating profits, in contrast, small firms may not exposure to frequent structural changes in their business operations and so they are more likely to practice expense misclassification. On the other side, another group of studies (e.g., Charfeddine et al., 2013; Bassiouny, 2016; Shahwan, 2021) found an insignificant relationship between firm size and EM practices.

2.7.6 Financial Leverage

Financial leverage is an indicator to assess the firm's ability to meet its financial obligations on the due date. It is also used to determine the financial resources needed to form a capital structure that aims to increase profits and maximize shareholder's wealth (Veronica, 2015; El Sayed, 2019).

Prior studies showed contradictory views relative to the impact of financial leverage on EM. On the one hand, highly leveraged firms may be reluctant to engage in EM practices. This is because they are subjected to high scrutiny from creditors and investors that constrain the manager's opportunistic earnings manipulation. Moreover, due to the contractually committed interests and debt repayments, these firms have low free cash flow available which in turn reduces manager's spending and limits EM activities (Houng and Phung, 2019).

On the other hand, highly leveraged firms may be motivated to engage in EM practices. They are more likely to manipulate earnings by shifting them from the future to the current period to increase the reported income to enhance the profitability level and therefore obtain low-cost financing (Desoky, 2016). More specifically, they tend to influence the perceptions of outside capital providers to improve the conditions at which they secure funding, improve the bargaining power with their creditors, avoid debt covenant violations and raise additional debt under better contracting terms (Anagnostopoulou and Tsekrekos, 2017). Consistent with this, Gao et al. (2017) found that the level of REM practices is higher for firms with a higher degree of leverage. Malikov et al. (2019) provide evidence of revenue misclassification to remain within EBITDA (earnings before interest, taxes, depreciation, and amortization) based on covenant limits. While Fan et al. (2019) provide evidence of expense misclassification to increase EBITDA and therefore avoid debt covenants violations. Moreover, Elsayed (2019) found a positive and significant relationship between leverage and REM practices. That is when firms rely more on debt, they are more likely to exercise REM activities to improve the bargaining power with their creditors and avoid debt covenant violations. In addition, Chung et al. (2021) demonstrated that private firms engage more in CS practices when they have outstanding debts.

However, these findings are inconsistent with the result of Zamari et al (2013) who found a negative and significant relationship between leverage and EM practices. Whereas the findings of (Vakilifard and Mortazavi, 2016; Awuye and Aubert, 2022) evidenced that when leverage increases, managers are prone to engage more in REM than AEM due to the strict audits and pressure of debt covenants. On the other side (Veronica, 2015; Santosa, 2023) found an insignificant relationship between leverage and EM.

2.7.8 Human Characteristics

According to social cognitive theory, personal characteristics affect an individual's behavior and help others to understand and predict their performance differences (Zalata and Abdelfattah, 2021). Gender-based psychological differences literature indicates that men and women exhibit distinct values and interests, and these characteristic differences may influence their behavior in the work life (Duong and Evans, 2016). That is, men and women are different in their

attitudes toward ethics, risk preferences and leadership styles (Abdullah and Ismail, 2016).

Management literature characterizes female directors as more cautious, risk averse and more conservative in the context of business decision-making, more likely to comply with rules and regulations, more likely to act more decisively and less overconfident (Duong and Evans, 2016; Chang et al, 2017; Ouyang et al., 2023). Additionally, from the moral development and social role theory view, female directors can strengthen the depth and breadth of discussion and deliberation that related to challenging issues (Zalata et al., 2021). Concluding that, female directors are less probable to engage in opportunistic practices to attain their self-motivated targets (Arun et al., 2015). Consistent with this, Lakhal et al. (2015) revealed that the increasing number of women on the board concurred with the reduction of AEM level. Suggesting that women play an effective monitoring role and are therefore considered as a crucial corporate governance mechanism. Duong and Evans (2016) indicated that female CFOs are conservative in both accruals and REM strategies and deliver higher reporting quality. In UK, Arun et al. (2015) found that firms with a higher number of female and independent female directors are more likely to exercise highly conservative accounting strategies. Zalata et al. (2021) found that the participation of female directors with relevant financial background could restrain managerial opportunism and improve the overall earnings quality. Alquhalf et al. (2017) indicated that firms with women directors on the board have fewer REM practices through accretive share buyback activities.

Besides gender diversity, managerial ability is one of the crucial personal characteristics, it refers to managerial knowledge, skills, and experience to make an optimal business decision (Simamora, 2021). Managerial ability indicates the managers' capability to better understand business activities and economic decisions that enable them to transform the firms' own resources into organizational revenues (Deef, 2022).

In accounting literature, there is a conflicting argument regarding the relationship between managerial ability and EM practices. On the one hand, managers with higher ability have greater knowledge and experience about their firms, competitors and industries that enable them to engage more in EM practices

to achieve their desired targets, also weak regulatory environment and a firm's unsatisfactory level of performance are other factors that encourage superior managers to be more inclined to manipulate earnings (Almaleeh et al., 2023). On the other hand, higher ability managers are more likely to maintain earnings quality and therefore they tend to engage less in opportunistic EM practices as they must maintain their reputation. i.e., they would devote greater effort and employ their capabilities to manage firm resources and enhance corporate operations instead of the inclusion of EM practices (Putra, 2021; Elhabashy et al., 2023).

2.8 Earnings Management Approaches

EM can be implemented by utilizing two main approaches. Firstly, the accounting earnings management approach which occurs when managers manipulate earnings using the flexibility allowed under accounting standards either in selecting accounting policies and methods or in making estimates to maximize the firm value or fulfill their opportunistic needs. In addition, this approach involves the misclassification of some items within the financial statements to attain desired reported earnings. Secondly, the REM approach which occurs when a manager's decision affects cash flow level as well as the reported earnings (Mohamed, 2017; Orieby, 2017; Abdelwahed, 2018). These approaches will be discussed in detail as follows:

2.8.1 Accounting Earnings Management Approach

Accounting earnings management occurs when managers intervene in the financial reporting to manipulate reported earnings without changing the underlying cash flows, it includes the following techniques:

2.8.1.1 The Flexibility of Accounting Standards

Managers may exploit the flexibility of accounting standards by selecting accounting methods according to their objectives or their motivations such as the choice between depreciation methods and the choice between inventory costing methods (Orieby, 2017).

However, this method is considered as the least used method for EM because it is more observable and easier to be detected by auditors as consistency and comparability may constrain such practices. Moreover, accounting standards require firms to disclose any changes in accounting policies, the reasons, and the

effects of such changes in the financial statements. So, it becomes easier for users of financial statements to distinguish between changes in earnings resulting from the firm's real performance and those resulting from changes in accounting policies (Mohamed, 2017; Abd El Kader, 2020).

2.8.1.2 Accrual based Earnings Management

The main objective of accruals is to present the true performance of the firm by recording revenues and expenses for the period in which they are incurred. However, managers could exploit these accruals for EM practices by exercising their discretion and judgment in financial reporting in a biased manner to influence the underlying economic performance (Mohamed, 2014).

Total accruals can be divided into two parts: non-discretionary and discretionary accruals. The non-discretionary accruals (normal or expected) arise from normal business transactions in the current period given the performance level, business strategy, industry conditions and macro-economic factors (Boghdady, 2019). Whereas the discretionary accruals are accruals that deviate from an entitys' expected normal levels of accruals. These accruals arise from transactions made or accounting treatments chosen to manage earnings (Idris, 2012).

More specifically, Managers may exploit such discretionary accruals to report income increasing when they estimate change or expect that future earnings will fall short of the required threshold, so that they employ income- increasing AEM to ensure that earnings benchmarks are met, or even slightly exceeded. On the other hand, they may manage earnings using income- decreasing AEM to make a reserve to report a profit in the following period, thereby attaining their future targets more easily (AlHadab, 2012, Al-Shattarat, 2017).

Essentially, AEM is an important EM tool that affects the reported earnings without any direct impact on cashflows, but it does reverse over time. Mainly, it occurs when managers deliberately intervene in the financial reporting at the end of the fiscal year to influence or control the reported earnings (Mohamed, 2017). Particularly, at the end of the fiscal year, managers are probable to determine whether earnings will meet or miss the desired thresholds, thus, they adjust reported earnings utilizing accruals according to their desired outcome (Al Hadab, 2012).

AEM practices can be achieved by exercising managerial discretion and judgment to estimate some events such as provisions and numerous future economic events that are reflected in the financial statements, including, expected lives and a salvage value of long-term assets, long-term contractual obligations, deferred tax and losses from bad debts and assets impairment (Healy and Wahlen, 1999; Mohamed, 2017).

Managers can also determine the desired reported earnings in a given period by altering the timing of revenues and expenses recognition (Ibrahim, 2012). In other words, managers may borrow earnings from future periods to improve current period earnings or the opposite pushing earnings from the current period to future period earnings, through the manipulation of revenues and expenses recognition (acceleration or deceleration) (Elkalla, 2017; Al Maleeh, 2019). For instance, when managers anticipate achieving losses in the subsequent period, they are likely to alter the recognition of revenues in a current profitable period to record an increased income in a loss period and vice versa (Boghdady, 2019).

2.8.1.3 Classification Shifting

Another tool of EM is CS, the primary incentive to engage in such practice comes from a manager's desire to meet or beat earnings benchmarks (Mc Vay, 2006). More importantly, managers may be motivated to engage in CS practices because of several reasons. Firstly, it is less costly than other EM tools as it is tough to be detected by auditors. Secondly, the firm is performing in weak corporate governance. Thirdly, the firm is in the decline stage, more specifically it is in a current or potential state of financial distress. Fourthly, when the ownership structure has some characteristics that incentivize managers to practice CS, especially family businesses or firms belonging to business groups (Al Maleeh, 2019).

CS was first defined by Mc Vay (2006) as deliberate misclassification of items within the income statement thereby inflating core earnings while bottom line income remains unchanged. Managers are more likely to engage in CS when they perceive that user of financial statements, especially investors value core earnings higher than non-core items. That is, being a top line item indicates a higher likelihood of persisting in the future (Alfonso et al., 2015; Bansal et al., 2021). Conversely, special items are unusual in nature or infrequent in occurrence, and

investors appear not to understand their nature and value income statement line items differently when making their investment decisions (Mc Vay, 2006; Zalata and Roberts, 2016; Darrough et al, 2019). Therefore, CS is less probable to be caught by auditors and regulators due to the transitory and infrequent nature of such non-recurring items (Usman et al., 2024). Additionally, managers are likely to practice CS using discontinued operations account "An item that can be viewed as transitory or non-recurring expense, it represents the income and cash flow of a component of a business that the firm has already discontinued or plans to discontinue" (Darrough et al, 2019). Consistent with this, Barua et al. (2010) revealed that managers achieve CS by misclassifying operating expense as income decreasing discontinued operations to record increased core earnings. These findings are in line with research by (Anthonius and Murwaningsari, 2018; Skousen and Wu, 2019; Inoue, 2021).

Though most of the previous studies have concentrated mainly on the manipulation of core earnings, recent studies have provided evidence of gross profit manipulation through the shifting of cost of goods sold to operating expenses. These studies argued that gross profit is perceived by investors as more sustainable than core earnings due to its closer proximity to sales and therefore it represents an important performance indicator signaling the efficiency of operations of a firm (Poonawala and Nagar, 2019; Ezat, 2021).

Apart from expense misclassification, other studies expand the scope of CS to revenue misclassification by shifting non-operating revenue to operating revenue. Such manipulation could be explained by the investor's tendency to pay more attention to operating revenue thus any increase in operating revenue is more valued by investors than a decrease in core expense (Malikov, et al. 2018).

On the other hand, CS practices may take the form of CFCS where managers misclassify items within the cash flow statement to record increased operating cash flows, holding the cash flow at the end with no change (Nagar and Raithatha, 2016; Baik et al. 2016; Nagar and Sen, 2016; Bansal, et al., 2021; Mahmoud, 2023; Hassan and Abdallah, 2024). The significance of cash flow information derives from its value relevance in determining the returns on investments, firm earnings generation power, and its dividend payment ability (Hassan and Abdallah, 2024). Particularly, managers may be motivated to manipulate operating cash flows to

meet or beat analyst forecasts of cash flows as markets react positively to stock price of firms with favorable cash flow surprise. Another reason is the investors tendency to value operating cash flow to a great extent. They perceive that cash flow from operations is likely to have an impact on firm value (Nagar and Raithatha, 2016).

In addition to ICS and CFCS, managers may engage in CS practices over the balance sheet statement without changing the aggregate figure of total assets, total liabilities, and total owner equity. For instance, they are likely to misclassify short term liabilities as long-term liabilities, especially when their firm suffers from financial distress or to avoid reporting working capital deficit and debt covenant violations (tree et al., 2019). Accordingly, CS practices can be summarized in the table (2.1) as follows:

Table (2.1) Summary of the Nature of Classification Shifting Practices

Definition	Deliberate misclassification of items within the financial statements (income statement, cash flow statement, and balance sheet statement) thereby inflating sub-aggregate figures while the bottom-line figure remains unchanged.
Forms	**1. <u>Income Classification Shifting</u>** • **Expense Classification Shifting** ***Examples:*** ➢ Shifting core expenses (COGS, SG&A) to income decreasing special items to inflate core earnings. ➢ Shifting core expenses (COGS, SG&A) to income decreasing discontinued operations to inflate core earnings. ➢ Shifting core expenses (COGS, SG&A) to non-operating expenses. ➢ Shifting COGS to operating expenses to inflate gross profit. • **Revenues Classification Shifting** ***Examples:*** ➢ Shifting non-operating revenues to operating revenues to inflate core earnings. ➢ Shifting other operating revenues to total revenues to inflate gross profit. **2. <u>Cash Flow Classification Shifting</u>** ***Example*** ➢ Shifting investing or financing cash flows to operating cash flows. **3. Balance Sheet Classification Shifting** ***Example*** ➢ Shifting current liabilities to long term liabilities.

Source: prepared by the researcher.

2.8.2 Real Earnings Management Approach

A second tool of EM occurs through real activity management. It was defined by Roychowdhury (2006, p.337) as "*departure from normal operational practices, deviated by manager's desire to mislead at least some stakeholders into believing certain financial reporting goals have been met in the normal course of operations*".

According to Gunny (2010) REM occurs when managers take actions that change the timing or structuring of an operation, investment, and or financing transaction to alter the reported earnings.

Hammad (2017) demonstrated two competing perspectives of REM activities. REM is an opportunistic managerial behavior as managers make various decisions related to real activities that deviate from optimal economic performance and mislead stakeholders when making their decisions. In contrast, the use of real activities to manage earnings is considered as an optimal response to adverse economic circumstances. That is, managers adapt to these casual circumstances by making decisions for a certain period to maintain the stable financial performance of the firm, implying that these decisions as part of their rational responsibilities.

El Sayed (2019) indicated that managers engaged in REM practices by making various decisions affecting operating or investing or financing activities with the objective of meeting or beating their earnings targets and to maintain the market value of the firm as well as increase the confidence of the different stakeholders.

More importantly, REM practices can be categorized according to the classification of activities in the cash flow statement into operating, investing, and financing activities. These activities will be discussed as follows:

2.8.2.1 Sales Manipulation

Roychowdhury (2006) indicated that managers can generate additional sales volume or accelerate sales from the next fiscal year to the current year by offering a price discount. However, the increased sales are likely to disappear once the firm retrieves the old prices. Such opportunistic use of price discount will increase information risk and raise customer's expectation of discount in the future period (Habib et al., 2022). Also, managers can boost sales by offering more lenient credit terms. For example, retailers consider offering easier credit terms as a better

way to increase sales volume and total earnings, thus, they often increase the period of credit repayment or lower or eliminate the interest paid on credit sales. Eventually, these REM activities will temporarily increase sales leading to higher total earnings but lower cash flow in the current period (Ezat, 2015; Alghemary et al., 2024).

2.8.2.2 Discretionary Expenditure Manipulation

Managers can boost the current period earnings by reducing discretionary expenditure including selling, general and administrative expense (SG&A), research and development expense (R&D) (Ismail, 2017). Management of discretionary expenditure involve managers deviation from the normal level of spending activity and deviation from a predetermined efficient spending plan to influence reported earnings and net cash flow and manipulate earnings upward or downward, which give more discretion to managers (Ibrahim, 2012). Nevertheless, such reduction is achieved at the risk of forgoing firms' future benefits and firm value in the long run (Li, 2019).

2.8.2.3 Overproduction

Managers can increase earnings by producing more than the necessary amount to meet expected demand, in this case, fixed overhead cost are distributed over an increased number of units of production, lowering fixed cost per unit and therefore declining the total cost per unit as long as the reduction in fixed cost per unit is not offset by inventory holding costs or any increase in marginal cost per unit. Such decreases lower reported cost of goods sold and increase operating margins (Ali, 2021; Al Madbouly, 2021; Potharla, et al., 2021).

2.8.2.4 The Timing of Asset Sale

The timing of asset sales can be used by managers as a way to manipulate earnings because gains are reported on the income statement at the time of the sale (the difference between the net book value and the current market value). Thus, managers may sell fixed assets and use the gains from such selling to avoid reporting losses or low earnings or avoid debt covenant violations (Gunny, 2010; Ali and Kamardin, 2018).

2.8.2.5 Stock Repurchases

Managers resort to manipulate earnings per share during a certain period through stock repurchase to reduce the number of common shares outstanding and therefore increase the earnings per share (Elsayed, 2019).

2.8.2.6 Securitization

Managers use gains from securitization for real manipulation purposes by estimating the retained interests and securitization gains along with the timing of securitization transactions or by securitizing receivables that are excluded from firms' books. As securitization gains or losses are recognized in the accounting books and reported in the financial statements in accordance with the management's targets (Kolsi and Matoussi, 2011; Hammad, 2017).

2.8.2.7 Financial Derivatives

Financial derivatives are an effective tool used to hedge away the uncertainty related to the volatility of firms' cash flows and earnings resulting from fluctuations in interest rates, foreign exchange rates, and commodity prices and other risks (Xu et al., 2007; Choi et al. 2015; Abd El Kader, 2020).

2.9 Chapter Summary:

This chapter addresses seven main points. Firstly, it presents the concept of firm value and argues that there is no unique definition of firm value, but it could be described depending on the perception of each stakeholder and expected interests that will be realized. Secondly, this chapter highlights the main determinants that could influence firm value including firm size, firm age, audit quality, profitability, liquidity, leverage, dividend policy, investment opportunity, capital structure, and firm growth. Thirdly, three main approaches of firm value are presented, which are asset approach, income approach, market approach. Fourthly, this chapter has endeavored to provide a general background about EM phenomenon in accounting literature, the concept of EM is demonstrated, and the study concluded that EM practices occur when managers intentionally intervene in financial reporting to manipulate reported earnings through accounting practices and/or real earnings activities to achieve specific targets. Such manipulation could be either firm value maximizing or opportunistic. Fifthly, the different motivations behind EM practices are presented. This point illustrates that EM motivations arise from capital market expectations and valuations, contractual incentives, and political cost and regulatory incentives. The sixth point sheds light on several determinants that could influence the extent of using an EM strategy including audit quality, internal corporate governance, auditing standards, firm life cycle, firm size, financial leverage, and human factors such as gender diversity and managerial ability. Finally, two main approaches of EM are presented. This point indicates that EM could be implemented by using the accounting EM approach and the REM approach. The Accounting EM approach occurs when managers intervene in the financial reporting to manipulate reported earnings without changing the underlying cash flows, it could be performed by exploiting the flexibility of accounting standards, exploiting accruals to manage earnings termed AEM, misclassifying some items within the financial statements (income statement, cash flow statement, and balance sheet statement) thereby inflating sub aggregate figures while bottom line figure remains unchanged. The second approach of EM is REM approach which occurs when manager’s decision affects the cash flow level as well as the reported earnings; it is categorized according to the classification of activities in the cash flow statement into operating, investing, and financing activities.

Chapter 3

Earnings Management Tools and their Models

3.1 Introduction

Prior literature has addressed three main tools of EM, namely AEM, REM, and CS. These strategies differ in their aim, costs, benefits, timing, and models. Managers have the capacity to use such tools jointly, simultaneously at the same time to attain the greatest desired earnings, but rather based on time, cost, and constraints managers can select among EM strategies, and therefore AEM, REM, and CS are perceived as substitutes.

Consequently, this chapter includes two sections. Section 3.2 presents the relationship among EM tools. Section 3.3 illustrates the measurement of EM practices.

3.2 The Relationship among Earnings Management Tools

The extensive literature on EM largely focuses on AEM and REM. They documented that if unconstrained by the reporting environment managers increasingly utilize accruals discretion because it is more accurate, has no direct impact on operating cash flows, and can be implemented at the fiscal year end to meet or beat earnings target. (Evans et al., 2015; Almand et al., 2023). However, managers may prefer REM instead of AEM for several reasons (Gunny, 2010; Alhadab, 2012; Hammad, 2017; Ali and Kamardin, 2018; El Sayed, 2019; Ali, 2021; Marisetty and Moturi, 2023): The main reason is that manipulation through accruals involves a high level of risk detection by control and audit procedures. Another one is the aggressive requirements on EM after the adoption of International Financial Reporting Standards (IFRS), more related studies on EM pointed out that the level of AEM declined significantly after the IFRS adoption due to their effectiveness to mitigate accruals manipulation. Further, the firm may have limited flexibility to manage accruals because it is constrained by the business operations and accruals manipulation in prior years. In addition, AEM has a limitation of timing, it occurs at the end of the year or quarterly so managers face uncertainty in which adjustments will be accepted by auditors and therefore they may not attain the desired outcome.

Conversely, REM activities are less susceptible to auditors' and regulators' scrutiny as it is related to the managerial decision, and it is difficult to differentiate between a willingness to manipulate earnings and real activities (Ali, 2021). According to time, REM occurs during the year whereby managers can adjust the

amount of AEM at the end of fiscal year i.e., (increase or decrease) the amount of accounting accruals depending on (high or low) REM outcome (Zang, 2012). Nevertheless, such activities may impose greater long-term costs on shareholders; significantly it has potential adverse economic consequences on cash flows and firm value in the long run. (Roychowdhury, 2006; Chi et al., 2011; Ali and Kamardin, 2018). Essentially, managers engage more in REM practices when their negative consequences on future cash flow and firm value are outweighed by the positive benefits derived from information conveyed to the capital markets (Habib, 2022). In this regard, such practices are attributed to managerial myopia (Marisetty and Moturi, 2023).

Recently, research has started to examine the use of CS practices due to the long time focus on AEM and REM. CS is a relatively less costly tool to manage earnings than other alternatives (AEM and REM), as it does not affect the bottom-line figure and therefore it is less susceptible to the scrutiny of auditors and regulators, moreover, it does not result in reversal of accruals like AEM nor foregone any future benefits like REM (Zalata and Roberts, 2017; Chung et al., 2021; Hwang et al., 2022). With respect to timing factor, CS is considered as a flexible strategy because it occurs outside of the accounting system and therefore it is more probable to be one of the last EM tools to attain desired earnings (Abernathy et al., 2014).

Arguably, CS appears to be a viable tool for managers who wish to manipulate the perceptions of outside stakeholders (pan et al. 2019). Nevertheless, numerous studies argued that CS may have several significant consequences. Ha and Thomas (2019) demonstrated that CS may have positive or negative consequences for earnings predictability. On the one hand managers may be incentivized to engage in CS practices to influence the perception of financial statements users about core performance to attain their self-motivated reporting targets, which worsen future earnings predictability. On the other hand, managers are prone to misclassify items to signal their future performance leading to an increase in earnings predictability. In addition, Mahmoud (2020) indicated that misclassification of revenues and expense to artificially inflate operating income inversely affect earnings quality and therefore influence investment and resource allocation decisions. Moreover, Anagnostopoulou et al. (2021) found strong and negative effects of CS on firm success and survival due to the improper signaling of actual repeatable core profitability. In a subsequent study, Anagnostopoulou et al. (2021) documented a

significant association between CS and inefficient firm level investment, the results revealed that misclassification may have significant information related consequences and influence the investors' expectations about the firm and so mislead them about the actual earnings and performance of the firm. A comparison among EM tools can be summarized in the table (3.1) as follows:

Table (3.1) A Comparison among Earnings Management Tools

	AEM	REM	CS
Effect	• Affect Net income. • No effect on cash flow.	• Affect Net income. • Affect cash flow.	• Affect core earnings /gross profit / operating cash flow. • Net income/ net cash flow unchanged
Timing	At the end of the fiscal year	During the fiscal year	Flexible strategy
Cost	High cost	Relatively low cost	Lowest cost method

Source: prepared by the researcher

Notably, managers can use the three tools of EM simultaneously at the same time, but may make substitution decisions on the basis of different cost, constraints, and timing of each strategy (Al Madbouly, 2021), this can be illustrated in figure (3.1):

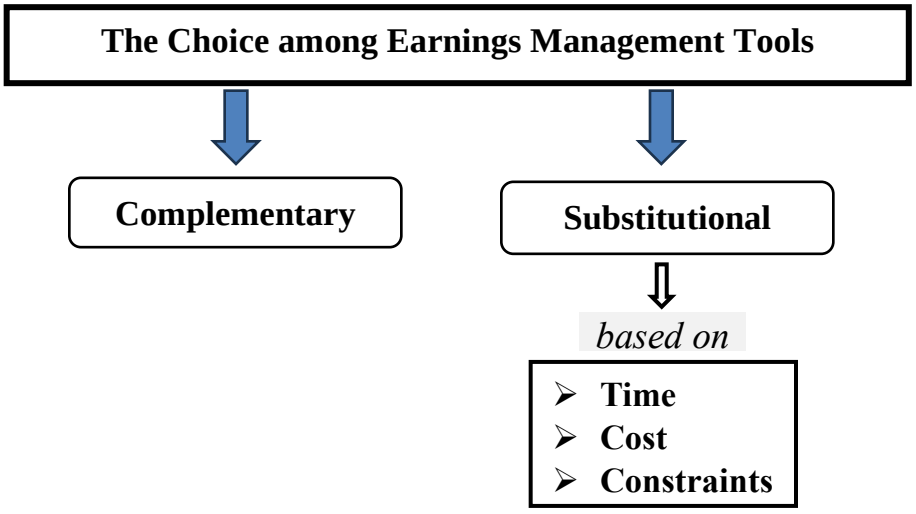


Figure 3.1: The Choice among Earnings Management Tools
Source: prepared by the researcher.

In this context, extant literature examined how managers use EM tools complementary to gain the greatest effect of EM reporting (either income increasing or income decreasing) or as a substitute under the relative cost benefit advantage. For example, Chen et al. (2012) indicated that REM is positively associated with the use of AEM, that managers manipulate real activities and discretionary accruals jointly and simultaneously. Similarly, Kou et al. (2014) found a long-term positive relationship between AEM and REM due to the weaker environment for investor protection and the lack of effective corporate governance. While Anagnostopoulou and Tsekrekos (2016) confirmed that high leveraged firms adopt both AEM and REM complementary. Ismail (2017) revealed that firms engage in AEM and REM practices evenly, no complete substitution relation between the two strategies employed to manipulate earnings. Also, the results of Al Absy et al. (2018) showed a positive and significant relationship between AEM and REM, implying that managers employ these EM tools at the same time. In the same vein, Al Haddad and Whittington (2019) found that managers use AEM and REM complementary (jointly) with an integrated (overall) EM strategy to attain the greatest desired earnings targets. Mohamed (2017) revealed that managers use both AEM and REM with a higher competition level. In the Egyptian context, Bogdady (2019) found no evidence support the trade-off between AEM and REM, this implies that managers use both EM strategies jointly and simultaneously to reach their desired objectives. In the study of Elkalla (2020), the results supported that AEM and REM tools complement each other rather than being used as substitutes. Hussain et al. (2020) revealed that managers of declining firms are more likely to use both AEM and REM strategies. While Ezat (2021) denoted that Egyptian firms adopt more than one EM tool for different stages of their life cycle. Al Madbouly (2021) demonstrated that managers involved in both AEM and REM practices evenly to attain their earnings targets. Similarly, Alghemary et al., (2024) highlighted the complementary relationship between AEM and REM. Moreover, the findings of Almaleeh (2023) revealed that managers with higher abilities are more likely to practise both AEM and REM activities jointly, they don't prefer one strategy over the other.

In contrast to the research findings of a complementary effect, several studies provide ample evidence that managers are more likely to make trade-off decisions among EM tools based on several factors. Graham et al. (2005) for example, found that managers tend to sacrifice long term economic value to attain their short-term

earnings targets, concluding that they would engage more in REM practices rather than accounting manipulation. Cohen et al. (2008) provided evidence that AEM has declined substantially after the passage of SOX due to increased auditors' and regulators' scrutiny, and therefore firms opted for REM as a substitute. Moreover, Cohen and Zarowin (2010) analyzed the firm's choice to use REM as a function of the costs of using AEM and the ability to do so, including the presence of a big auditor, longer auditor tenure, being in a high litigation industry and the level of net operating assets. They found that these factors are associated with an increased tendency to use REM. Zang (2012) analyzed the cost and timing differences between AEM and REM, she demonstrated that the trade-off decision is a function of the relative costs of the two strategies. That is, the level of each EM strategy decreases with its own cost and increases with the costs of the other. Therefore, the empirical results revealed that firms use REM when AEM is constrained due to a higher level of scrutiny of accounting practices and limited accounting flexibility because of accrual manipulation in prior years and shorter operating cycle. However, the level of AEM is higher when REM is constrained by firm's competitive status in the industry, financial health, scrutiny from institutional investors and the immediate tax consequences of manipulation. Relatedly, Abernathy et al. (2014) indicated that managers prefer to engage in one form of EM when the cost associated with another form is higher. Taken together, they found that firms are more inclined in CS when REM is restricted by poor financial performance, high institutional ownership, and low industry market share. Likewise, the level of CS increases when AEM is constrained by low accounting system flexibility and the provision of a cash flow forecast. Also, Mohamed (2017) revealed that AEM practices are constrained by industry specialist auditors, long tenured auditors, prior periods accruals, institutional ownership, and weakness of financial position. Whereas the level of REM decreased with the industry specialist auditors. While Makhaiel (2019) argued that political and economic context also could affect the managers trade-off EM approaches. Furthermore, Abd El-Kader (2020) demonstrated that managers make trade-off decisions between the accounting approach and REM approach based on their different costs. That is, when one approach is relatively more costly, firms opt for the other. This cost is determined by several factors that constrain the use of either of them including accounting standard, corporate governance and strong policies Moreover, the firm characteristics for instance, financial position,

accounting policies and competitive status in the market, and human factor such as, gender play a major role in determining the adopted EM tool.

More specifically, numerous prior literature has observed that due to stricter accounting standards, the cost of AEM become more costly, arising from increase in the level of monitoring and scrutiny of financial reports (Habib et al., 2022; Viana et al., 2022). Accordingly, managers may be motivated to switch to another EM strategy that is less detectable and less subject to scrutiny to offset the restrictions imposed by manipulations through AEM. Typically, they tend to be more inclined in REM activities as it is indistinguishable from other economic transactions. They would take economic transactions decisions and would abandon positive net present value projects to attain short-term earnings (Ipino and Parbonetti, 2016). Alternatively, they would resort to misclassification practices which is regarded as the lowest cost method to manipulate earnings. In this regard, Ho et al. (2015), evidenced that the higher quality accounting standards related to IFRS adoption leads to a reduction in the AEM level and therefore lead Chines firms to engage more in REM activities as a substitute to manipulate earnings. Similarly, Ferentionou and Anagnostopoulou (2016) investigated the existence of AEM and REM before and after the mandatory adoption of IFRS by Greek firms. The results observed that the presence of restricted alternatives in accounting policies and choices induces firms to substitute REM for AEM, implying that IFRS have particularly constrained available accounting choices for managers and reduced the possibility of AEM practices. In the same context, Ipino and Parbonetti (2016) analyzed the effect of mandatory IFRS adoption on the trade-off between AEM and REM. They argued that if IFRS are restrictive than domestic accounting standards, managers become more involved in REM practices at the expense of AEM. Similarly, Zalata and Roberts (2017) provided evidence that following IFRS adoption, CS practices became more preferable for managers because it is harder to be caught by auditors and regulators.

Additionally, prior studies have incorporated the impact of audit quality into the choice of an EM tool for example, Chi et al. (2011) found that audit specialization and the presence of big 4 auditors are both positively associated with more REM, implying that these factors could restrict the use of AEM and induce managers to engage more in REM practices. Moreover, Burnett et al. (2012) found that firms with high audit quality tend to use accretive stock repurchases and are less probable

to practise AEM to beat or meet analysts' forecast. Similarly, Choi et al. (2018) evidenced that managers opted to use REM than AEM due to the presence of big 4 auditors. While Eilifsen and Knivsfla (2020) showed that the level of CS practices increases as a substitute for AEM for firms with Big 4 audit and industry- specialized audit firm. Similarly, Mnif and Hamouda (2021) provided evidence that firms tend to switch from AEM to REM when they are audited by an industry expert. While Chung and Chae (2021) managers are more prone to switch to CS when their ability to manage earnings using other EM tools are restricted by higher audit quality. Taken together, internal corporate governance plays a crucial role in influencing the managers' choice among EM tools. Several studies (e.g., Ge and Kim, 2014; Gao et al., 2017; Aghab, 2020; El Diri et al., 2020) revealed that firms with strong corporate governance are related to less AEM practices but higher levels of REM activities.

Notably, the firm life cycle has a significant impact on the firm's financial performance, which in turn provides rooms for earnings manipulation (Durana et al., 2021). Prior literature stated that managers' attitude toward EM practices varies across the firm life cycle, in addition to that they would make substitution decisions among these different strategies. Relatedly, Nagar and Sen (2017) denoted that decline firms are prone to report more income decreasing special items than healthy firms which increase managers' incentives and opportunities to engage in ECS practices to meet or beat zero core earnings. Nagar and Radhakrishnan (2017) showed that mature firms are more likely to cut their discretionary expenditure to avoid reporting losses in contrast to those in the introduction and growth stages. While El Madbouly (2021) demonstrated that it could be difficult for managers to engage in REM practices at the decline stage if the firm is experiencing financial difficulties as REM activities need more economic resources and adequate funding. So, in such a case, the marginal cost of deviating from optimal business strategies is likely to be high and therefore REM is perceived by managers as relatively costly since their primary goal is to survive and improve operations (Abernathy et al., 2014). These findings contradict the results of Xie et al., (2022) who revealed that decline firms have significantly higher levels of REM activities as compared to mature firms. While Dewi et al., (2022) demonstrated that EM strategies fluctuates at each stage of the firm life cycle, concluding that growth stage concurred with a high level of AEM whereas at maturity stage managers are more likely to use AEM and REM practices, moreover, at stagnant stage managers are more inclined to

employ the three EM strategies including AEM, REM, and CS to maintain their profits since sales start to decrease steadily. Bansal (2021) argued that firm life cycle could influence manager's trade-off decisions concluding that mature firms are prone to prefer revenue shifting over expense shifting whereas introductory firms prefer expense shifting over revenue shifting. That is, in the introductory stage, firms are required to make numerous one-time expenses to set up the business, thus, the larger the magnitude of income decreasing special item increase the managers incentives to engage in expense misclassification. On the other side, in the maturity stage, firms are highly motivated to inflate their operating revenues to meet analyst's sales and profitability benchmarks, so they are highly incentivized to engage in revenue misclassification.

Referring to managerial characteristics, Huang and Sun (2017) indicated that more capable managers are less likely to be engaged in REM activities upon AEM and CS practices because they realize the value destroying nature of REM to future performance and firm value in the long term. This contradict with the results of Hasan et al (2022), who stated that higher ability managers are prone to engage more in AEM practices rather than REM activities as they better make trade-off decisions between AEM and REM concerning cost and benefit of these alternative strategies available to them. Regarding gender diversity extant literature has investigated the influence of female directors on EM practices. Almost studies argued that female directors are more cautious and risk aversion than men in making their decisions and therefore they are less likely to manipulate earnings or otherwise tend to prefer the less costly EM strategy to reach their targets. For instance, Liu et al. (2016) examined the impact of CFO gender on AEM and REM respectively in Chines listed firms. They argued that female directors are more conservative in making financial decisions regardless the institutional and business environment within which they operate. As a result, they showed that female CFO firm-years have significantly lower accruals manipulation and lower abnormal production costs but higher abnormal discretionary expense. While Zalata and Abdelfattah (2021) found that the presence of non-executive female directors on firm's board was positively associated with the misclassification of expenses. That is, non-executive female directors are less likely to challenge such practices. While Ouyang et al. (2023) concluded that female CEOs substitute REM with revenue misclassification as the latter is less costly and more viable to manipulate earnings under stricter monitoring.

To summarize, figure (3.2) illustrates the factors that affect the choice of an EM tool as follows:

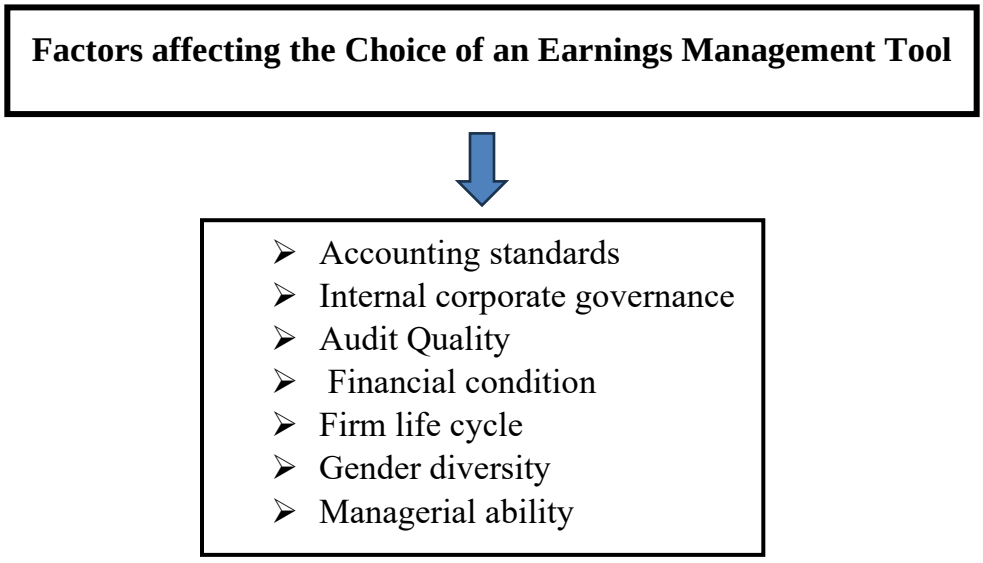


Figure 3.2: Factors Affecting the Choice of an Earnings Management Tool
Source: prepared by the researcher.

3.3 Measurement of Earnings Management Tools

Several models have been developed by researchers to detect and measure EM practices including AEM, CS, and REM. This section will present in detail the most common developed models as follows:

3.3.1 Accrual based Earnings Management Models

According to past EM studies, there are three relevant approaches to measure AEM: specific accruals approach, frequency distribution of earnings and aggregated accruals approach. The first focus on specific accruals of a firm (bad debt provisions), in contrast frequency distribution of earnings attempts to detect EM practices through earnings distribution, on the other hand, aggregate accruals approach analyzes the total accruals of the firm, this approach has been extensively employed in the extant literature (Birk, 2018; Khaled, 2018; Aghab, 2020).

Consistent with the aggregate accruals approach, total accruals are partitioned into non-discretionary and discretionary accruals (NDA + DA). Nondiscretionary accruals are defined as items resulting from the application of fundamental

accounting rules that can't be changed by managers whereas discretionary accruals derive from accounting choice, managerial decisions, and various estimates (Anton and Carp, 2020). Particularly, to measure AEM, total accruals are estimated by two alternative approaches balance sheet approach and cash flow approach.

According to balance sheet approach total accruals are estimated as follows:

$$TA_t = \Delta CA_t - \Delta Cash_t - \Delta CL_t + \Delta DCL_t + DEP_t$$

Where:

TA_t = Total accruals in year (t).

ΔCA_t = Change in current assets in year (t).

ΔCL_t = Change in current liabilities in year (t).

$\Delta Cash_t$ = Change in cash and cash equivalents in year (t).

ΔDCL_t = Change in debt included in current liabilities in year (t).

DEP_t = Depreciation and amortization expense.

While, according to cash flow approach total accruals are estimated as follows:

$$TA_t = EPEX_t - CFO_t$$

Where:

TA_t = Total accruals in year (t).

$EPEX_t$ = Earnings before extraordinary items in year (t).

CFO_t = Operating cash flow in year (t).

Further, Nondiscretionary accruals are estimated through alternative models developed by many researchers such as: Healy (1985); DeAnglo (1986); Jones (1991); modified Jones (1995); Kothari (2005); Dechow and Dichav (2002); and others.

Eventually Discretionary accruals are estimated by the difference between total accruals and nondiscretionary accruals (TA - NDA). These accruals are used as a proxy for detecting AEM.

To illustrate, studies have been evolved over decades to develop models of AEM, these models can be presented in detail as follows:

3.3.1.1 The Healy (1985) Model

Healy (1985) introduced one of the earliest studies in the accounting literature for measuring EM practices, the study investigated the compensation theory which is

concerned with the effect of earnings-based bonus contracts on manager's incentives toward managing earnings upwards or downwards (Idris, 2012). The Healy (1985) model is considered as the simplest model to measure EM whereby total accruals include both discretionary accruals and non-discretionary accruals and are estimated by the difference between firm's reported earnings and operating cash flow (Khaled, 2018). Basically, the model assumed that systematic EM occurs in every period and estimated the extent of manipulation as the deviation from the normal level of accruals (Kourdoumpalou, 2017).

According to the Healy (1985) model the normal level of accruals (non-discretionary accruals) is estimated as the mean total accruals from the estimated period (scaled by lagged total assets). In other words, Healy (1985) used total accruals to proxy for discretionary accruals based on the assumption that change in non-discretionary accruals are constant over time. Hence the model specification is the following:

$$NDA_t = \sum_n TA_t / AT_{t-1}$$

Where:

NDA_t = Non-discretionary accruals in year (t).

$\sum_n TA_t$ = Mean total accruals from the estimation period.

AT_{t-1} = Total assets in year (t-1).

However, the Healy (1985) model, was criticized as the assumption that change in non-discretionary accruals are constant over time is restrictive, Kaplan (1985) pointed out that expected level of non-discretionary accruals may fluctuate in response to the economic circumstances of a firm, rather than the opportunistic behavior of managers , furthermore, total accruals level are likely to be negative for some firms due to the impact of depreciation expense (Aghab, 2020; Khalil, 2010).

3.3.1.2 The DeAnglo (1986) Model

Unlike Healy (1985) who used mean total accruals of the year in the estimation period (scaled by lagged total assets) to proxy for discretionary accruals, DeAnglo (1986) used the last period accruals (scaled by lagged total assets) to estimate non-discretionary accruals, the model specification is the following:

$$NDA_t = \frac{TA_t}{A_{t-1}}$$

Where:

NDA_t = Non-discretionary accruals in year (t) scaled by lagged total assets.

TA_{t-1} = Total accruals in year (t-1).

A_{t-1} = Total assets in year(t-1).

The DeAnglo (1986) model can be viewed as a specific case of the Healy (1985) model, the two are quite different, the DeAnglo model adopted the assumption that NDA follow a random walk process whereas the Healy model assume that NDA follow a mean reverting process (Batrov et al. 2000). A common feature in the Healy and DeAnglo models was that if NDA were constant over time and DA has a mean of zero in the estimation period then both models would measure NDA without error (Dechow et al., 1995). In line with Healy, DeAnglo based his model on the assumption that change in non-discretionary accruals are constant over time, nevertheless, several studies criticized this assumption and stated that non-discretionary accruals affected by the changes in the economic circumstances of the firm (Khaled, 2018).

3.3.1.3 The Jones (1991) Model

Jones (1991) presented one of the pioneer studies in measuring AEM, in contrast with Healy (1985) and DeAnglo (1986), who adopted the assumption that average change in non-discretionary accruals is constant over time, Jones (1991) attempted to control for the effects of change in firm's economic circumstances (Batrov et al., 2001). Accordingly, jones inserted change in revenues and property, plant and equipment within the regression model. Revenues were included to control changes in working capital accruals such as account receivable, inventory, and account payable, whereas property, plant and equipment were added to control for non-discretionary depreciation expense (Idris, 2012). Hence the model specification is the following:

$$\frac{NDA_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\left(\frac{\Delta REV_t}{A_{t-1}}\right) + \beta 3\left(\frac{PPE_t}{A_{t-1}}\right) + \varepsilon_t$$

Where:

NDA_t = Non-discretionary accruals in year (t).

A_{t-1} = Total assets in year (t-1).

ΔREV_t =Revenues in year (t) less revenues in year (t-1).

PPE_t=Gross property, plant, and equipment in year (t).

β_1, β_2, β_3: Estimated parameters, obtained from the following regression model:

$$\frac{TA_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\left(\frac{\Delta REV_t}{A_{t-1}}\right) + \beta 3\left(\frac{PPE_t}{A_{t-1}}\right) + \varepsilon_t$$

Where:

TA_t =Total accruals in year (t).

However, prior studies argued that jones (1991) model has some limitation. The main limitation of the model is that all revenues are non-discretionary accruals, while firms can manipulate earnings through discretionary revenues, (Dechow et al., 1995). Another limitation of Jones model is the need for a long time series of data to allow effective estimation of the regression parameter, which excludes firms without enough observations from the sample (wrobewski, 2016).

3.3.1.4 The Modified Jones Model (Dechow et al. 1995)

Due to the implicit assumption underlying the jones model that all revenues are non-discretionary Dechow et al. (1995) developed a new version of the jones model to overcome its deficiency by adjusting the change in revenues with the change in receivables (Aghab, 2020). Hence the model specification is the following:

$$\frac{NDA_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\left[\frac{(\Delta REV_t - \Delta REC_t)}{A_{t-1}}\right] + \beta 2\left(\frac{PPE_t}{A_{t-1}}\right) + \varepsilon_t$$

Where:

NDA_t = Non-discretionary accruals in year (t).

A_{t-1} = Total assets in year (t-1).

ΔREV_t =Revenues in year (t) less revenues in year (t-1).

ΔREC_t = Net receivables in year (t) less net receivables in year (t-1)

PPE_t=Gross property, plant, and equipment in year (t).

Obviously, Dechow et al. (1995), assumed that all changes in credit sales in the event period are due to earnings manipulation as these earnings are likely to be discretionary than cash sales, as a result the only adjustment relative to the original jones model is the inclusion of change in receivable (ΔREV- ΔREC).

3.3.1.5 The Kotharie et al. (2005) Model

Further, Kotharie et al. (2005) argued that accruals are correlated with firm's contemporaneous and past performance. As a result, when discretionary accruals are measured by jones and modified jones model, they may include error as these models disregarded the performance of the firm (Aghab, 2020). Consequently Kothari et al. (2005) extended the modified jones model through two modifications: first inserting either current or last year's return on assets (ROA) to the regression model. Second: adjusting firm's discretionary accruals to matched firm. They identify a firm from the same industry and with closest ROA to that of the sample firm and deduct the control firm's discretionary accruals from those of the sample firm to generate performance matched residuals (Jackson, 2018; Mohamed, 2017) the model specification is the following:

$$\frac{NDA_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\left[\frac{(\Delta REV_t - \Delta REC_t)}{A_{t-1}}\right] + \beta 3\left(\frac{PPE_t}{A_{t-1}}\right) + \beta 4\left(\frac{ROA_{t\,(ort-1)}}{A_{t-1}}\right) + \varepsilon_t$$

Where:

$\mathrm{ROA_{t\,(ort-1)}}$ = Return on assets either in the current period or in the prior period, all other variables were defined in the previous equation.

A number of studies demonstrated that Kotharie et al., (2005) model is well specified and provides a powerful test of EM, nevertheless it suffers from some limitations: the performance matched approach require a large sample for the comparison study i.e., enough observations (sample firm and control firm) to group firms and matching each firm within its group firm so it is difficult to be applied when the population size is small and the number of firms within the same industry is not enough (Abd El Kader, 2020). In addition to this, performance matching based on ROA may induces measurement error in estimated discretionary accruals, as the high (or low) value of ROA may be the result of managerial discretion not the reflection of high (or low) firm's ROA, therefore this approach requires the knowledge of original (honest) ROA instead of the reported ROA (Abd El Kader, 2020; Keung and Shin, 2014).

3.3.1.6 The Dechow and Dichev (2002) Model

Later, Dechow and Dichev (2002), presented a model for measuring the quality of accruals and earnings based on the extent to which working capital accruals map into cash flow (Al-Sayed, 2021), this model regress capital accrual on cash flow from operations in the previous, current, and next period (Kamarudin and Ismail, 2014) the model specification is the following:

$$\Delta WC_t = \alpha + \beta_1 CFO_{t-1} + \beta_2 CFO_t + \beta_3 CFO_{t+1} + \varepsilon_t$$

Where:

ΔWC_t = The change in working capital in year (t) scaled by lagged total assets.

CFO_{t-1} = Operating cash flow in year (t-1) scaled by lagged total assets.

CFO_t = Operating cash flow in year (t) scaled by lagged total assets.

CFO_{t+1} = Operating cash flow in year (t+1) scaled by lagged total assets.

The change in working capital from year (t-1) to year (t) is computed as:

$$\Delta WC_t = \Delta AR + \Delta Inventory - \Delta AP - \Delta TP + \Delta Other\ assets\ (net).$$

Where:

ΔWC = Change in working capital.

ΔAR = Change in accounts receivable.

Δ Inventory = Change in inventory.

ΔAP = Change in accounts payable.

ΔTP = Change in taxes payable.

According to Dechow and Dichev (2002) model, the residuals from the regression equation are the measure of accruals quality and therefore the level of managerial discretion, where a lower standard deviation of these residuals indicate higher quality of accruals and earnings (saadah, 2021). However, the Dechow and Dichev (2002) model was criticized by McNichols (2002) and Francis et al. (2005) as this model focus on capital accruals (short term accruals) rather than long term accruals so that it doesn't reflect an objective measure of accruals quality (Al Sayed, 2019).

To summarize, in accordance with aggregate accrual approach, there are many structures and models to measure AEM, these models began with simplest models to the relatively sophisticated regression (Attia, 2016; Orieby, 2017) among these models the modified jones (1995) model, which is suggested by several research as

a more powerful model for detecting AEM, therefore the present study will rely on this model.

3.3.2 Classification Shifting Models

Studies have evolved to measure CS practices including ICS and CFCS, these can be illustrated as follows:

3.3.2.1 The Mc Vay (2006) Model

Mc Vay (2006) introduced the first model to measure ICS. The study argued that managers might opportunistically shift core expenses to special items and therefore inflate core earnings instead of bottom-line net income.

According to this model, firm's core earnings are decomposed into expected and unexpected core earnings components by modeling expected core earnings in a similar vein as the accrual model jones (1991). Mc Vay (2006) predicted a positive relationship between unexpected core earnings and income decreasing special items if managers are classification shifter, subsequently presented empirical evidence of CS based on the following two-steps equations:

The first step was to measure expected core earnings, that is, expected core earnings for a given firm are based on prior period core earnings, asset turnover, change in sales, contemporaneous and previous periods accruals, particularly Mc Vay (2006), developed the following model to estimate expected core earnings:

$$CE_t = \alpha + \beta_1 CE_{t-1} + \beta_2 ATO_t + \beta_3 Accruals_{t-1} + \beta_4 Accruals_t + \beta_5 \Delta sales_t + \beta_6 NEG_\Delta sales_t + +\varepsilon_t \quad (1)$$

Where:

CE_t = Core earnings (before special items and depreciation) computed as (sales cost of goods sold - selling, general, and administrative expenses) / $sales_t$.

$\Delta sales_t$ = Percent change in sales computed as ($sales_t$ - $sales_{t-1}$) /$sales_{t-1}$.

$NEG_\Delta sales_t$ =1 if $\Delta sales_t$ is less than 0, and 0 otherwise.

$Accruals_t$=Operating accruals computed as (Net income before extraordinary items - cash from operations) / $sales_t$.

ATO_t = Asset turnover ratio, computed as ($sales_t$ / (Average NOA_t), where: NOA, or Net operating assets, is equal to the difference between operating

Assets - operating liabilities. operating assets are calculated as total Assets less cash and short-term investments. Operating liabilities are computed as total assets less total debt and, less book value of common and preferred equity and, less minority interests.

The expected core earnings model includes lagged core earnings CE_{t-1} because core earnings tends to be persistent, Assets turnover ratio ATO_t is also included as it is inversely related to profit margin, the model also includes both contemporaneous and previous accruals as control variables because they are associated with firm performance, Sales growth is added in order to control for the impact of sales growth on fixed costs(as sale grow, fixed cost per unit decline), whereas negative sales is added to the model to allow for different slopes for sales increase and decrease.

Next unexpected core earnings $UECE_t$ were calculated as the difference between reported core earnings and expected core earnings.

In the second step Mc Vay, (2006) predicted that managers shift core expense to special item (income decreasing special items) and anticipated that if managers are classification shiftor, then unexpected core earnings in year t is expected to be increasing with special item in year t, therefore estimated the following regression model:

$$UECE_t = \alpha + \beta_1 \%SI_t + E_t$$

Where:

$UECE_t$= Unexpected change in core earnings in year (t).

SI_t= Income decreasing special items scaled by sales in year (t).

In this equation, the coefficient β_1 is expected to be positive if CS does occur.

One of the most crucial limitations of Mc Vay (2006), is that the expectation model may have a bias because it uses a contemporary accrual. The inclusion of current period accruals (a control performance) could induce a mechanical relation between unexpected core earnings and special items. More specifically, accruals may also contain special items accruals so high special items accruals, might drive down expected core earnings and result in higher unexpected core earnings which is positively related to special items causing evidence on CS (Pan, 2013; Joo and Chamberlain, 2017; Li and Guo, 2018; Al-Haddad and Whittington, 2019). For that

reason, a subsequent study conducted by Fan et al. (2010), removed contemporaneous accruals from their model of expected core earnings.

3.3.2.2 The Poonawala and Nagar (2019) Model

Poonawala and Nagar (2019) concentrated mainly on gross profit figure rather than core earnings to measure ICS. The study stated that managers engaged in ICS practices through the shifting of (COGS) to operating expense in order to inflate gross profit. Therefore, the study estimated expected gross profit as follows:

$$\frac{GP_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\left(\frac{GP_{t-1}}{A_{t-1}}\right) + \beta 3\left(\frac{TACC_{t-1}}{A_{t-1}}\right) + \beta 4\left(\frac{TACC_t}{A_{t-1}}\right) + \beta 5\left(\frac{SALE_t}{A_{t-1}}\right) + \beta 6\left(\frac{CH_SALE_t}{A_{t-1}}\right) + \beta 7\left(\frac{NEG\ CH_SALE_t}{A_{t-1}}\right) + \varepsilon_t$$

Where:

GP_t= Reported gross profit in year (t).

A_{t-1} = Total assets in year (t-1).

GP_{t-1}= Gross profit in year (t-1).

$TACC_t$= Accruals in year (t).

$TACC_{t-1}$= Accruals in year (t-1).

$SALE_t$= Sales in year (t).

CH_SALE_t= Percentage change in sales (sales growth).

$NEG\ CH_SALE_t$ = (1) if $\Delta sales_t$ is less than 0, and (0) otherwise.

The residuals from the equation were termed unexpected gross profit (UEGP).

Next operating expense were represented as the sum of (SG&A) & (R&D) and therefore the expectation models were estimated as follows:

$$\frac{SG\ \&A_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2(MV_t) + \beta 3(Q_t) + \beta 4\left(\frac{INT_t}{A_{it-1}}\right) + \beta 5\left(\frac{CH_SALE_t}{A_{t-1}}\right) + \beta 6\left(\frac{NEG\ CH_SALE_t}{A_{it-1}}\right) + \varepsilon_{it}$$

Where:

SG &A $_t$= Selling, general and administrative expenses in year (t).

AT_{t-1}= Total assets in year (t-1).

MV_t= The log of market value in year (t).

Q_t= Tobins Q in year (t).

INT_t= Available internal fund in year (t).

The residuals from the equation were termed unexpected selling, general and administrative expense (UESGA).

$$\frac{RD_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\ (MV_t) + \beta 2\ (Q_t) + \beta 3\left(\frac{INT_t}{A_{t-1}}\right) + \beta 5\left(\frac{RD_{t-1}}{A_{t-1}}\right) + \varepsilon_t$$

Where:

RD $_t$ = Research and development in year (t).

RD $_{t-1}$ = Research and development in year (t-1).

The residuals from the equation were termed unexpected research and development *(UERD).*

Further, the unexpected gross profit (UEGP) was regressed on the sum of (UESGA +UERD) as follows:

$$UEGP_t = \alpha + \beta 1\ UEOE_t + \varepsilon_t$$

If the coefficient $\beta 1$ is positive, then managers engage in CS practices.

Later, following Fan and Liu (2017) model, Ezat (2021) employed an alternative model to estimate expected selling, general and administrative expense (SG&A) as follows:

$$\frac{SG\ \&A_{\ t}}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\left(\frac{SG\&A_{t-1}}{A_{t-1}}\right) + \beta 3\left(\frac{TACC_{t-1}}{A_{t-1}}\right) + \beta 4\left(\frac{TACC_t}{A_{t-1}}\right) + \beta 5\left(\frac{S_t}{A_{it-1}}\right) + \beta 6\left(\frac{RET_t}{A_{t-1}}\right) + \beta 7\left(\frac{RET_{t-1}}{A_{t-1}}\right)\beta 8\left(\frac{CH_S_t}{A_{t-1}}\right) + \beta 9\left(\frac{NEG\ CH_S_t}{A_{t-1}}\right) + \varepsilon_t$$

Where:

SG &A $_t$= Selling, general and administrative expenses in year (t).

A_{t-1}= Total assets in year (t-1).

$SG\&A_{t-1}$= Selling, general and administrative expenses in year (t-1).

$TACC_{t-1}$= Total accruals in year (t-1).

$TACC_t$= Total accruals in year (t).

RET_t= Stock returns in year (t).

RET_{t-1}= Stock returns in year (t-1).

S_t= Total sales in year (t).

CH $_S_t$= Percentage change in sales (sales growth).

NEG CH $_S_t$= (1) if $\Delta sales_t$ is less than 0, and (0) otherwise.

The residuals from the equation were termed unexpected selling, general and administrative expense (UESGA).

In conclusion, the current study will depend on this model to measure ICS practices in the Egyptian context.

3.3.2.3 The Malikov, et al. (2019) Model

Malikov, et al. (2019), developed a methodology to measure ICS, the study stated that firms can also overstate their core earnings by shifting non-operating revenues to operating revenues and so results in CS practices. For that reason, Malikov et al. (2019), developed an expectation model for decomposing operating revenues into expected and unexpected components similar to Mc Vay (2006) core earnings model. Then anticipated that unexpected operating revenues as measured by reported operating revenues in year (t) less expected operating revenues in year (t) increase as non-operating revenues in year (t) decrease if CS does occur.

Hence Malikov, et al. (2019), developed the following model to estimate the expected level of operating revenues as follows:

$$\frac{OR_t}{AT_{t-1}} = \alpha + \beta_1 \frac{1}{AT_{t-1}} + \beta_2 \frac{OR_{t-1}}{AT_{t-2}} + \beta_3 MTB_{t-1} + \beta_4 \frac{AR_{t-1}}{AT_{t-2}} + \beta_5 \frac{AR_t}{AT_{t-1}} + \varepsilon_t$$

Where:

OR_t= Operating revenues in year (t).

AT_{t-1}= Total assets in year (t-1).

OR_{t-1}= Operating revenues in year (t-1).

MTB_{t-1}= Market to Book ratio in year (t-1).

AR_{t-1}= Account receivable in year (t-1).

Next unexpected operating revenues ($UEOR_t$) were calculated as the difference between reported and expected operating revenues.

Further, unexpected operating revenues were regressed with the non-operating revenues to test the existence of classification shifting of revenues as follows:

$$UEOR_t = \alpha_0 + \alpha_1 NOR_t + E_t$$

Where:

$UEOR_t$ = unexpected operating revenues in year (t).

NOR_t= Non-operating revenues in year (t).

This equation expects the coefficient of NOR to be negative if the firms misclassify the NOR to operating revenues.

3.3.2.4 The Nagar and San (2014) Model

Referring to CFCS, Nagar and San (2014), decomposed cash flow from operations into expected and unexpected cash flow components following the model proposed by Lee (2012) as follows:

$$\frac{CFO_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\left[\frac{S_t}{A_{t-1}}\right] + \beta 3\left(\frac{\Delta S_t}{A_{t-1}}\right) + \varepsilon_t$$

Where:

CFO_t = Cash flow from operations in year t.

A_{t-1} = Total assets in year t-1.

S_{it} = Total sales in year t.

ΔS_{it} = Change in sales in year t.

The residuals from the equation were termed unexpected cash flow from operations (UNCFO).

Further, the study regress unexpected cash flows with investing and financing activities to test whether firms engage in CFCS practices through the following equation:

$$UCFO_t = \alpha + \beta 1\left(\frac{INV_t}{A_{t-1}}\right) + \beta 1\left(\frac{FIN_t}{A_{t-1}}\right) + \varepsilon_t$$

Where:

UCFO = Unexpected operating cash flows in year (t).

A_{it-1}= Total assets in year (t-1).

INV_{it} = Investing cash flows in year (t).

FIN_{it} = Financing cash flows in year (t).

If firms misclassify operating cash flows as investing or financing cash flows, then the coefficient B_1 and B_2 should be negative.

Accordingly, to investigate CFCS practices in Egyptian firms, the current study will depend on such model.

3.3.3 Real Earnings Management Models

Based on the study of Dechow et al (1995), Roychowdhury (2006), developed an empirical model that represents one of the most commonly used measures of REM. Subsequent studies (Cohen, 2008), (Gunny, 2010), (Zang, 2012), (Cohen and Zarowin, 2010), provided further evidence of the validity of these proxies.

This model estimates the normal levels of real business activities as reflected in cash flow from operations, production cost, and discretionary expenditure respectively, the abnormal levels of each operational activity can be measured by the difference between the actual and expected components of each respective proxy. Therefore, the abnormal levels of these proxies indicate that firms engage in real activities manipulation by accelerating sales or generating unsustainable sales with price discount and more lenient credit terms, cutting discretionary expense and over production of inventory (Idris, 2012; Al-Shattarat, 2017; El Diri, 2018; El Sayed, 2019). Furthermore, built on Roychowdhury (2006) model, Gunny (2010), modified these measures slightly to specify other measures for the level of R&D, the level of SG & A and production cost, moreover the study expanded the literature with a fourth model of gain on asset sales model (Cohen et al., 2020). The four proxies will be presented in detail as follows:

3.3.3.1 The Sales Manipulation Model

To detect sales-based manipulation, Rochowdhury (2006), expressed the normal cash flow from operations as a linear function of sales and change in sales and estimated the model through the following cross- sectional regression for each firm and year:

$$\frac{CFO_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\left[\frac{S_t}{A_{t-1}}\right] + \beta 3\left(\frac{\Delta S_t}{A_{t-1}}\right) + \varepsilon_t$$

Where:

CFO_{it} = Cash flow from operations in year (t).

A_{it-1} = Total assets in year (t-1).

S_{it} = Total sales in year (t).

ΔS_{it}= Change in sales in year (t).

3.3.3.2 The Over Production Model

For over production model, Roychowdhury suggested that deviations in production costs from otherwise optimal operating decisions represent managers attempt to manipulate earnings (Srivastava, 2019), as a result he estimated the normal production costs through the following cross-sectional regression for each year and firm and therefore the estimated residuals from the equation are the abnormal level of production:

$$\frac{PROD_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\left[\frac{S_t}{A_{t-1}}\right] + \beta 3\left(\frac{\Delta S_t}{A_{t-1}}\right) + \beta 2\left(\frac{\Delta S_{t-1}}{A_{t-1}}\right) + \varepsilon_t$$

where:

$PROD_t$ = Production cost, calculated as the sum of COGS in year t plus the change in inventory in year (t).

A_{t-1} = Total assets in year (t-1).

S_t = Total sales in year (t).

ΔS_t = Change in sales in year (t).

ΔS_{t-1} = Change in sales in year (t-1).

On the other side, Gunny (2010), adjusted the overproduction model by inserting market value and Tobin's Q to the regression model. Hence the model specification is the following:

$$\frac{PROD_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\, MV_t + \beta 2\, Q_t + \beta 3\left(\frac{S_t}{A_{t-1}}\right) + \beta 4\left(\frac{\Delta S_{t-1}}{A_{t-1}}\right)\beta 5\left(\frac{\Delta S_{t-1}}{A_{t-1}}\right) + \varepsilon_t$$

Where:

MV = The natural logarithm of market value.

Q_t = Tobin's Q in year (t).

3.3.3.3 The Discretionary Expense Model

Rochowdhury (2006), estimated the normal discretionary expense by the following firm year regression:

$$\frac{DISEXP_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\left[\frac{S_t}{A_{t-1}}\right] + \varepsilon_t$$

Where:

$DISEXP_t$= Discretionary expenditures in year t and for firm i.

A_{t-1} = Total assets in year (t-1).

S_t = Total sales in year (t).

However, Gunny (2010), modified the previous model by separating between R&D expense and SG&A expense. Another modification on the model was the inclusion of the market value of equity, Tobin's Q and internal funds variables to the regression model as follows:

$$\frac{RD_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\, MV_t + \beta 2\, Q_t + \beta 3\left(\frac{INT_t}{A_{t-1}}\right) + \beta 5\left(\frac{RD_{t-1}}{A_{t-1}}\right) + \varepsilon_t$$

Where:

RD_t= Research and development expense in year (t).

RD_{t-1} = Research and development expense in year (t-1).

A_{t-1}= Total assets in year (t-1).

MV =The natural logarithm of market value.

Q_t =Tobin's Q in year (t).

INT_t = Internal funds in year (t).

$$\frac{SGA_t}{A_{t-1}} = \alpha + \beta 1\left(\frac{1}{A_{t-1}}\right) + \beta 2\, MV_t + \beta 3\, Q_t + \beta 4\left(\frac{INT_t}{A_{t-1}}\right) + \beta 5\left(\frac{\Delta S_t}{A_{t-1}}\right) + \beta 6\left(\frac{\Delta S_t}{A_{t-1}}\right) * DD + \varepsilon_t$$

Where:

SGA_t = Selling, general, and administrative expense in year (t). expense.

A_{t-1} = Total assets in year (t-1).

MV_t = The natural logarithm of market value in year (t).

Q_t = Tobin's Q in year (t).

INT_t = Internal funds in year (t).

S_t = Total sales in year (t).

DD = Indicator variable equal to (1) when total sales decrease between (t-1) and (t), zero otherwise.

3.3.3.4 The Selling of Long-Term Assets Model

Gunny (2010), developed a fourth model to estimate the normal level of gain on assets sales, abnormal values of the model were obtained from cross sectionally

regression for each year and firm, subsequently, Gunny (2010), interpreted high residuals from the model as indicative of asset sales manipulation as follows:

$$\frac{GainA_t}{A_{t-1}} = \alpha \ + \beta 1\ MV_t + \beta 2\ Q_t + \beta 3\left(\frac{INT_t}{A_{t-1}}\right) + \beta 5\left(\frac{ASales_t}{A_{t-1}}\right) + \beta 5\left(\frac{ISales_t}{A_{t-1}}\right) + \varepsilon_t$$

Where:

$GainA_t$ = Income from asset sales in year (t).
A_{t-1}= Total assets in year (t-1).
MV_t = The natural logarithm of market value in year (t).
Q_t= Tobin's Q in year (t).
INT_t= Internal funds in year (t).
$ASales_t$ = Long – lived assets sales in year (t).
Isales= Long – lived investment sales in year (t).

Despite the increasing and widely adoption of Rochowdhury (2006) models in literature, it suffers from a key limitation associated with the proxy of sales manipulation activities, that is boosting sales by offering price discounts and more lenient credit terms increases current period's earnings but also results in lower cash flow in the current period (Idris, 2012). Moreover, inconsistency between sales behavior and operating cash flow is not necessarily related to EM practices but may be related to other factors such as: uncertainty risks about the nature of the product and market characteristics, in addition to the macroeconomy risks such as exchange rate, inflation rate and fiscal and monetary policies. However, the other models of overproduction and discretionary expense model are less biased compared to the sales manipulation model. (Saadah, 2021).

3.4 A Proposed Multidimensional Model for Measuring Earnings Management and Its Impact on Firm Value

The current study seeks to propose a multidimensional model for measuring EM practices and its impact on firm value. Two alternative measures will be used to proxy for firm value; MTB and ROA. Using the methodology employed by Dechow et al. (1995), AEM will be proxied by the Modified Jones Model. To measure REM the current study will rely on an aggregate REM proxy as computed by the aggregation of abnormal values of operating cash flow, production cost, and discretionary expenditure, following Rochowdhury (2006); Cohen (2008). ICS will be measured by unexpected gross profit similar to Poonawala and Nagar (2019);

Ezat (2021) whereas CFCS will be measured by unexpected cash flow from operations consistent with Lee (2012); Nagar and San (2016); Nagar and Raithatha (2016). AGG_EM will be captured by averaging the ranks of AEM, REM, ICS, and CFCS. Furthermore, the study will incorporate five control variables that could affect the firm value including BIG4, firm age, firm size, leverage, and asset growth. These can be illustrated by the following figure as follows:

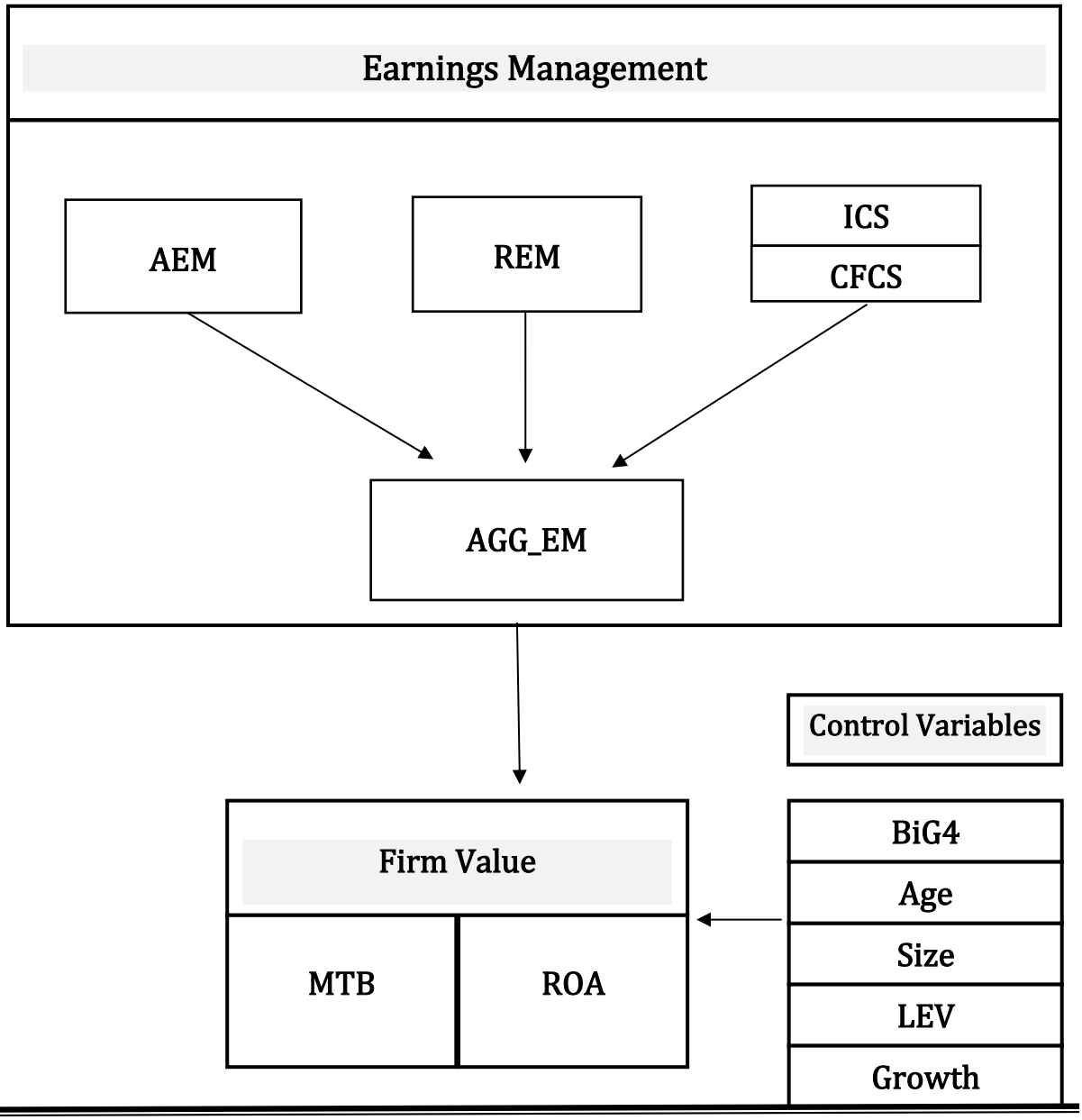


Figure 3.3: A Proposed Multidimensional Model for Measuring Earnings Management and Its Impact on Firm Value

Source: prepared by the researcher.

3.5 Chapter Summary:

This chapter highlights two main points. Firstly, the relationship among EM tools. This point indicates that EM practices includes three main tools namely, AEM, CS, and REM. Managers can implement the three EM strategies complementary, simultaneously at the same time to attain the greatest desired earnings, implying that they are not restricted to a specific strategy, however they can make trade off decisions based on cost, time, and constraints of each strategy ie., they may prefer one strategy over the other which is more difficult to be detected, relatively less costly, and realized the desired targets of the firm. Also, this part investigates the alternative EM tools concluding that AEM is a more accurate strategy which occurs at the end of the year and has no direct impact on operating cash flows, however it involves a high level of risk detection by control and audit procedures, moreover it is constrained by the business operations and accruals manipulation in prior years. For REM, it occurs during the fiscal year, it is less likely to be detected by auditors and regulations since it is related to managerial decisions, and it is difficult to differentiate between real activities and willingness to manipulate earnings. Nevertheless, this strategy imposes greater long-term costs on shareholders due to its adverse economic consequences on cash flows and the value of the firm. Referring to CS, it is considered as the lowest cost method to manipulate earnings than AEM and REM because it has no effect on bottom line figure and therefore, it is less likely to be detected. Additionally, it doesn't result in reversal of accruals nor foregone any future benefits.

Secondly, the measurements of EM are illustrated. For AEM many studies have evolved over decades to develop models in order to measure and detect such practices. The most common published models are Healy (1985), DeAnglo (1986), Jones (1991), Modified Jones (1995), Dechow and Dichev (2002), and Kotharie et al. (2005). Referring to CS practices, Mc Vay (2006) is the pioneer model to measure expense misclassification, further studies provided additional evidence regarding gross profit manipulation (Poonawala and Nagar, 2019), revenue misclassification (Malikov, et al. 2019), cash flow misclassification (Nagar and San, 2014). Finally, four proxies are presented to measure REM practices including sales manipulation model, over production model, discretionary expense model, and selling of long-term assets model.

Chapter 4
The Empirical Study

4.1 Introduction

The main goal of this chapter is to propose a multidimensional model for measuring EM practices and investigate its impact on firm value. This chapter is divided into two sections. The first section provides the study methodology including sample selection, data collection method, variables description, model specification, and statistical techniques. The second section illustrates the empirical results and testing of hypotheses. To achieve this, the following sub-points were conducted as follows:

4.2 The Study Methodology

The study methodology comprises of sample selection, data collection method, variables description and measurement, model specification, and statistical techniques used in the study. It can be illustrated as follows:

4.2.1 Sample Selection

The population includes all Egyptian firms listed in the Egyptian Exchange during the period from 2017 to 2021. Banks and financial services firms have been excluded from the sample because of the unique nature of their activities; moreover, they are subjected to different rules and regulations. Four main criteria are employed for inclusion in the sample: Firstly, all firms pulled from the population must have been listed in the Egyptian Exchange during the period from 2017 to 2021. Secondly, financial statements must have been issued in the Egyptian pound. Thirdly, financial statements must have been prepared on 31/12 to achieve the consistency of the financial year. Fourthly, the study included only sectors whose number of firms are greater than or equal to 6 in the sample. According to these criteria, the final study sample consists of 48 firms from 7 sectors with 240 firm-year observations. Table (4.1) summarizes the study sample as follows:

Table (4.1): The Study Sample

No.	Sector	Total Firms Number	Excluded Firms Number	Final Sample
1	Contracting and Construction Engineering	11	5	6
2	Health care, pharmaceuticals	19	12	7
3	Basic resources	13	6	7
4	Travel & Leisure	9	2	7
5	Building materials	11	5	6
6	Industrial goods, services, and Automobiles	6	--	6
7	Real estate	33	24	9
	total	102	54	48

4.2.2 Data Collection Method

The study used secondary data. Data for all study variables are obtained from annual financial statements of Egyptian listed firms and reports from the Egyptian Exchange to obtain the stock prices. The required data are often available on the firm's website, The Mubasher Info Egypt website, The Egyptian Exchange website, and The Investing website.

4.2.3 Variables Description and Measurement

The study relied on a set of variables. The independent variables are AEM, REM, ICS, CFCS, and AGG_EM. The dependent variable is represented by the firm value which is measured by market-to-book ratio and return on assets. Regarding control variables, the study incorporated five control variables that could affect the firm value such as BIG4, Firm age, Firm size, Leverage, and Asset growth.

4.2.3.1 Dependent Variable

The study employed two alternative measures of firm value as dependent variables: The first measure is the Market-to-Book ratio (MTB) which is measured as the ratio of the market value to book value of equity (Bandanuji and Khoiruddin, 2020; Atiningsih and Izzaty, 2021; Marpanung et al., 2022; Sutejaet al., 2023) .The second measure is return on assets (ROA) which can be computed by dividing net income after tax over total assets (Agung et al., 2021; Atiningsih and Izzaty, 2021).

4.2.3.2 Independent Variables

- **Accrual based Earnings Management (AEM)**

The present study adopted the modified Jones model (1995) to estimate discretionary accruals (DA) as a proxy for AEM; this can be illustrated by the following steps:

1. Total accruals are calculated as the difference between net income before extraordinary items and cash flow from operations.

$$TA_{it} = NI_{it} - CFO_{it}$$

Where:

TA_{it} = Total accruals for firm (i) in year (t).

NI_{it} = Income before extraordinary items for firm (i) in year (t).

CFO_{it} = Operating cash flow for firm (i) in year (t).

2. The following regression model is used to obtain the specific parameters $\alpha, \beta_1, \beta_2, \beta_3$ at the level of each sector as follow:

$$\frac{TA_{it}}{A_{it-1}} = \alpha + \beta 1\left(\frac{1}{A_{it-1}}\right) + \beta 2\left[\frac{(\Delta REV_{it} - \Delta REC_{it})}{A_{it-1}}\right] + \beta 2\left(\frac{PPE_{it}}{A_{it-1}}\right) + \varepsilon_{it}$$

Where:

TA_{it} = Total accruals for firm (i) in year (t).

A_{it-1} = Total assets for firm (i) in year (t-1).

ΔREV_{it} = Revenues in year (t) less revenues in year (t-1) for firm (i).

ΔREC_{it} = Net receivables in year (t) less net receivables in year (t-1) for firm (i).

PPE_{it}= Gross property, plant, and equipment for firm (i) in year (t).

3. The coefficients estimated from the previous equation are used to estimate nondiscretionary accruals as flows:

$$\frac{NDA_{it}}{A_{it-1}} = \alpha + \beta 1\left(\frac{1}{A_{it-1}}\right) + \beta 2\left[\frac{(\Delta REV_{it} - \Delta REC_{it})}{A_{it-1}}\right] + \beta 2\left(\frac{PPE_{it}}{A_{it-1}}\right) + \varepsilon_{it}$$

Where:

NDA_{it}= Nondiscretionary accruals for firm (i) in year (t).

4. Discretionary accruals (*DA*) are then computed as the difference between total accruals scaled by lagged total assets (*TA*) and non-discretionary accruals scaled by lagged total assets (*NDA*) as follows:

$$DA = TA - NDA$$

Determining the absolute value of discretionary accruals and the average of these values at the level of each sector. If the absolute value of discretionary accruals is higher than the average of the sector, this indicates the firm's greater involvement in AEM practices, and therefore takes the value of (1), and zero otherwise.

- **Real Earnings Management (REM)**

Based on Roychowdhury (2006), the present study used three proxies: abnormal level of cash flow, abnormal level of production cost, and abnormal level of discretionary expenditure respectively to measure the level of REM activities. The study estimates the normal levels of such activities through regression equations then the abnormal levels can be measured by the difference between the actual and expected components of each respective proxy. The three REM proxies can be measured through the following models:

- **Sales Manipulation Model**

Using the Rochowdhury (2006) model, estimating the normal level of cash flows is possible as follows:

$$\frac{CFO_{it}}{A_{it-1}} = \alpha + \beta 1\left(\frac{1}{A_{it-1}}\right) + \beta 2\left[\frac{S_{it}}{A_{it-1}}\right] + \beta 3\left(\frac{\Delta S_{it}}{A_{it-1}}\right) + \varepsilon_{it}$$

Where:

CFO_{it} = Cash flow from operations for firm (i) in year (t).

A_{it-1} = Total assets for firm (i) in year (t-1).

S_{it} = Total sales for firm (i) in year (t).

ΔS_{it}= Change in sales for firm (i) in year (t).

The difference between actual cash flow from operations and normal levels of cash flows from operations is used as a proxy for abnormal cash flow from operations (*Ab.CFO*).

- **Over Production Model**

Following Rochowdhury (2006), the normal levels of production costs are estimated as follows:

$$\frac{PROD_{it}}{A_{it-1}} = \alpha + \beta 1\left(\frac{1}{A_{it-1}}\right) + \beta 2\left[\frac{S_{it}}{A_{it-1}}\right] + \beta 3\left(\frac{\Delta S_{it}}{A_{it-1}}\right) + \beta 2\left(\frac{\Delta S_{it-1}}{A_{it-1}}\right) + \varepsilon_{it}$$

Where:

$PROD_{it}$ = Production cost, calculated as the sum of COGS in year t plus the change in inventory for firm (i) in year (t).

A_{it-1}= Total assets for firm (i) in year (t-1).

S_{it} = Total sales for firm (i) in year (t).

ΔS_{it} = Change in sales for firm (i) in year (t).

ΔS_{it-1}= Change in sales for firm (i) in year (t-1).

Abnormal production cost is measured as the difference between actual production costs and normal levels of production costs *(Ab.PROD*)

- **Discretionary Expense Model**

Based on Rochowdhury (2006), the normal levels of discretionary expense are expressed by the following regression model:

$$\frac{DISEXP_{it}}{A_{it-1}} = \alpha + \beta 1\left(\frac{1}{A_{it-1}}\right) + \beta 2\left[\frac{S_{it}}{A_{it-1}}\right] + \varepsilon_{it}$$

Where:

$DISEXP_{it}$= Discretionary expenditures for firm (i) in year (t).

A_{it-1} = Total assets for firm (i) in year (t-1).

S_{it} = Total sales for firm (i) in year (t).

The difference between actual discretionary expenses and normal levels of discretionary expenses is used as a proxy for abnormal discretionary expenses *Ab.DISEXP*.

Finally, consistent with prior studies (Cohen, 2008; Ferentinou and Anagnostopoulon, 2014) these three individual measures (abnormal *CFO*, abnormal *PROD* and abnormal *DISEXP*) can be combined into an aggregate measure of REM to capture the total effects of REM as follows:

$$Agg.REM = Ab.CFO + Ab.DISEXP + Ab.PROD$$

Where:

Agg.REM = Aggregate measure of real earnings management

Ab.CFO = Abnormal cash flow from operations.

Ab.DISEXP = Abnormal discretionary expenditure.

Ab.PROD = Abnormal production cost.

This measure is computed by multiplying *Ab_CFO* and *Ab_DISEX* by negative one (-1) so that the larger their value, the higher their upward REM, then all results values are added together plus the value of *Ab_PROD* to capture one single comprehensive measure of REM, this aggregate measure is expected to proxy for REM more accurately than adopting one or two of the three above measures.

- **Classification Shifting**

The present study employed two measures of CS (ICS and CFCS); this can be illustrated as follows:

- **Income Classification Shifting**

Following (Fan and Liu, 2017; Poonawala and Nagar, 2019; Ezat, 2021), the present study concentrates mainly on unexpected gross profit manipulation proxy for ICS. Expected gross profit is estimated through the following regression model:

$$\frac{GP_{it}}{A_{it-1}} = \alpha + \beta 1\left(\frac{1}{A_{it-1}}\right) + \beta 2\left(\frac{GP_{it-1}}{A_{it-1}}\right) + \beta 3\left(\frac{TACC_{it-1}}{A_{it-1}}\right) + \beta 4\left(\frac{TACC_{it}}{A_{it-1}}\right) + \beta 5\left(\frac{S_{it}}{A_{it-1}}\right) + \beta 6\left(\frac{CH_S_{it}}{A_{it-1}}\right) + \beta 7\left(\frac{NEG\ CH_S_{it}}{A_{it-1}}\right) + \varepsilon_{it}$$

Where:

GP_{it}= Reported gross profit for firm (i) in year (t).

A_{it-1} = Total assets for firm (i) in year (t-1).

GP_{it-1}= Reported gross profit for firm (i) in year (t-1).

$TACC_{it}$= Total accruals for firm (i) in year (t).

$TACC_{it-1}$= Total accruals for firm (i) in year (t-1).

S_{it}= Total sales for firm (i) in year (t).

CH_S_{it}= Percentage change in sales (sales growth).

$NEG\ CH_S_{it}$= (1) if Δsales_t is less than 0, and (0) otherwise.

Next unexpected gross profit (UGP) is measured as the difference between reported gross profit and expected gross profit.

Afterward the current study basically investigates whether Egyptian firms engage in CS practices through misclassifying cost of goods sold (COGS) as operating expense OEXP and misclassifying other operating income (OI) as total revenues in order to inflate gross profit.

Accordingly, the study employs the below expected selling, general and administrative expense model to estimate unexpected selling, general and administrative expense (USGA) as computed by the difference between reported and expected values. The model specification is the following:

$$\frac{SG\ \&A_{\ it}}{A_{it-1}} = \alpha + \beta 1\left(\frac{1}{A_{it-1}}\right) + \beta 2\left(\frac{SG\&A_{it-1}}{A_{it-1}}\right) + \beta 3\left(\frac{TACC_{it-1}}{A_{it-1}}\right) + \beta 4\left(\frac{TACC_{it}}{A_{it-1}}\right) + \beta 5\left(\frac{S_{it}}{A_{it-1}}\right) + \beta 6\left(\frac{RET_{it}}{A_{it-1}}\right) + \beta 7\left(\frac{RET_{it-1}}{A_{it-1}}\right)\beta 8\left(\frac{CH_S_{it}}{A_{it-1}}\right) + \beta 9\left(\frac{NEG\ CH_S_{it}}{A_{it-1}}\right) + \varepsilon_{it}$$

Where:

$SG\ \&A_{\ it}$= Selling, general and administrative expenses for firm (i) in year (t).

A_{it-1}= Total assets for firm (i) in year (t-1).

$SG\&A_{it-1}$= Selling, general and administrative expenses for firm (i) in year (t-1).

$TACC_{it-1}$= Total accruals for firm (i) in year (t-1)

$TACC_{it}$= Total accruals for firm (i) in year (t).

RET_{it}= Stock returns for firm (i) in year (t).

RET_{it-1}= Stock returns for firm (i) in year (t-1).

S_{it}= Total sales for firm (i) in year (t).

CH_S_{it}= Percentage change in sales (sales growth).

$NEG\ CH_S_{it}$= (1) if $\Delta sales_t$ is less than 0, and (0) otherwise.

Next unexpected gross profit is regressed with operating expenses (*OEXP*) and other operating income (*OI)* as follows:

$$UGP_{it} = \alpha + B_1\ OEXP_{it} + \beta 2\left(\frac{OI_{it}}{A_{it-1}}\right) + \varepsilon_{it}$$

Where:

UGP_{it}= Unexpected gross profit for firm (i) in year (t).

$OEXP_{it}$ = Operating expense for firm (i) in year (t) computed as [unexpected selling, general and administrative expense (*SG&A)* + other operating expenses *(OOEXP)]* scaled by previous total assets.

OI_{it}= other operating income for firm (i) in year (t).

A_{it-1} = Total assets for firm (i) in year (t-1).

If the Egyptian firms misclassify *COGS* as *OEXP*, then the coefficient B_1 should be positive while, if these firms misclassify *OI* as total revenues, then the coefficient B_2 should be negative.

- **Cash Flow Classification Shifting**

Based on prior studies (Lee, 2012; Nagar and Raithatha, 2016; Nagar and San, 2016; Bansal et al., 2020; Mulchandani et al., 2023) the present study measure CFCS through unexpected operating cash flows model as follows:

$$\frac{CFO_{it}}{A_{it-1}} = \alpha + \beta 1\left(\frac{1}{A_{it-1}}\right) + \beta 2\left[\frac{S_{it}}{A_{it-1}}\right] + \beta 3\left(\frac{\Delta S_{it}}{A_{it-1}}\right) + \varepsilon_{it}$$

Where:

CFO_{it} = Cash flow from operations for firm (i) in year (t).

A_{it-1} = Total assets for firm (i) in year (t-1).

S_{it} = Total sales for firm (i) in year (t).

ΔS_{it} = Change in sales for firm (i) in year (t).

The difference between actual cash flow from operations and expected cash flows from operations is used as a proxy for unexpected cash flow from operations (*UNCFO).*

Next the study tests whether Egyptian firms misclassify operating cash flow as investing or financing cash flows through the following regression model:

$$UCFO_{it} = \alpha + \beta 1\left(\frac{INV_{it}}{A_{it-1}}\right) + \beta 1\left(\frac{FIN_{it}}{A_{it-1}}\right) + \varepsilon_{it}$$

Where:

$UCFO$ = Unexpected operating cash flows for firm (i) in year (t).

INV_{it} = Investing cash flows for firm (i) in year (t).

FIN_{it} = Financing cash flows for firm (i) in year (t).

If the Egyptian firms misclassify operating cash flows as investing or financing cash flows, then the coefficient B1 and B2 should be negative.

- **Aggregate Earnings Management**

AGG_EM is calculated by averaging the ranks of AEM, REM, ICS, and CFCS.A higher rank of AGG_EM implies a higher degree of earnings manipulation by the firm and vice versa (Leuz et al., 2003; Habib, 2004; Miletic and Vuckovic, 2020).

4.2.3.3 Control Variables

The study used five control variables that are indicated in the literature as significantly influencing firm value. These variables could be measured as follows:

BIG4: is a dummy variable equals one if the audit firm is a big 4 firm and zero otherwise.

Age: is the firm's age measured by the natural logarithm of firm age.

Size: is the firm's size measured by the natural logarithm of total assets.

LEV: a firm's financial leverage calculated as total debt divided by total assets.

Growth: is the assets growth computed as total assets of the current year less total assets of the previous year divided by total assets of the previous year.

Table (4.2): Description and Measurement of Variables

Abbreviation	Variable	Measurement	References
Dependent Variables			
MTB	market-to-book ratio	The ratio of the market value to book value of equity.	(Bandanuji and Khoiruddin, 2020) (Atiningsih and Izzaty, 2021) (Marpanung et al. 2022) (Nurdiansari et al. 2022) (Suteja et al., 2023)
ROA	Return on Assets	Net income after tax /total assets	(Agung et al., 2021) (Atiningsih and Izzaty, 2021)

Independent Variables			
AEM	Accruals based Earnings Management	Is measured by discretionary accruals model.	(Dechow et al. 1995)
REM	Real Earnings Management	Is measured by an aggregate value of abnormal cash flow, abnormal discretionary expenditure, and abnormal production cost.	(Rochowdhury 2006) (Cohen, 2008) (Ferentinou and Anagnostopoulon, 2014)
ICS	Income Classification Shifting	Is measured by unexpected gross profit model	(Fan and Liu, 2017) (Poonawala and Nagar, 2019) (Ha and Thomas, 2019) (Anagnostopoulou et al. 2021) (Ezat, 2021)
CFCS	Cash flow Classification Shifting	Is measured by unexpected cash flow from operations model.	(Lee, 2012) (Nagar and San, 2016) (Nagar and Raithatha, 2016) (Bansal et al. 2020) (Mulchandani et al., 2023)
AGG_EM	Aggregate Earnings Management	Is calculated by averaging the ranks of AEM, REM, ICS, and CFCS. A higher rank of AGG_EM implies a higher degree of earnings manipulation and vice versa.	(Leuz et al., 2003) (Habib, 2004) (Miletic and Vuckovic, 2020)
Control Variables			
BIG4	Big 4 auditing firms	A dummy variable equals one if the audit firm is a big 4 firm and zero otherwise.	(Imeni et al., 2021) (Darmawan et al. 2019) (Suffian et al. 2014)
Age	Firm Age	The natural logarithm of firm age.	(Imeni et al. 2021) (Deef, 2022)
Size	Firm Size	The natural logarithm of total assets.	(Atiningsih and Izzaty, 2021) (Agung and Huzaimal, 2021) (Marpanung et al. 2022) (Suteja et al., 2023)
LEV	Leverage	Total debt/total assets	(Gill et al., 2013) (Siboni and Pourali, 2015) (Imeni et al., 2021)
Growth	Asset Growth	(Total assets of the current year- total assets of the previous year) / total assets of the previous year.	(Fajari and Isnalita, 2018) (Nurdiansari et al. 2022) (Marpanung et al., 2022)

Source: prepared by the researcher

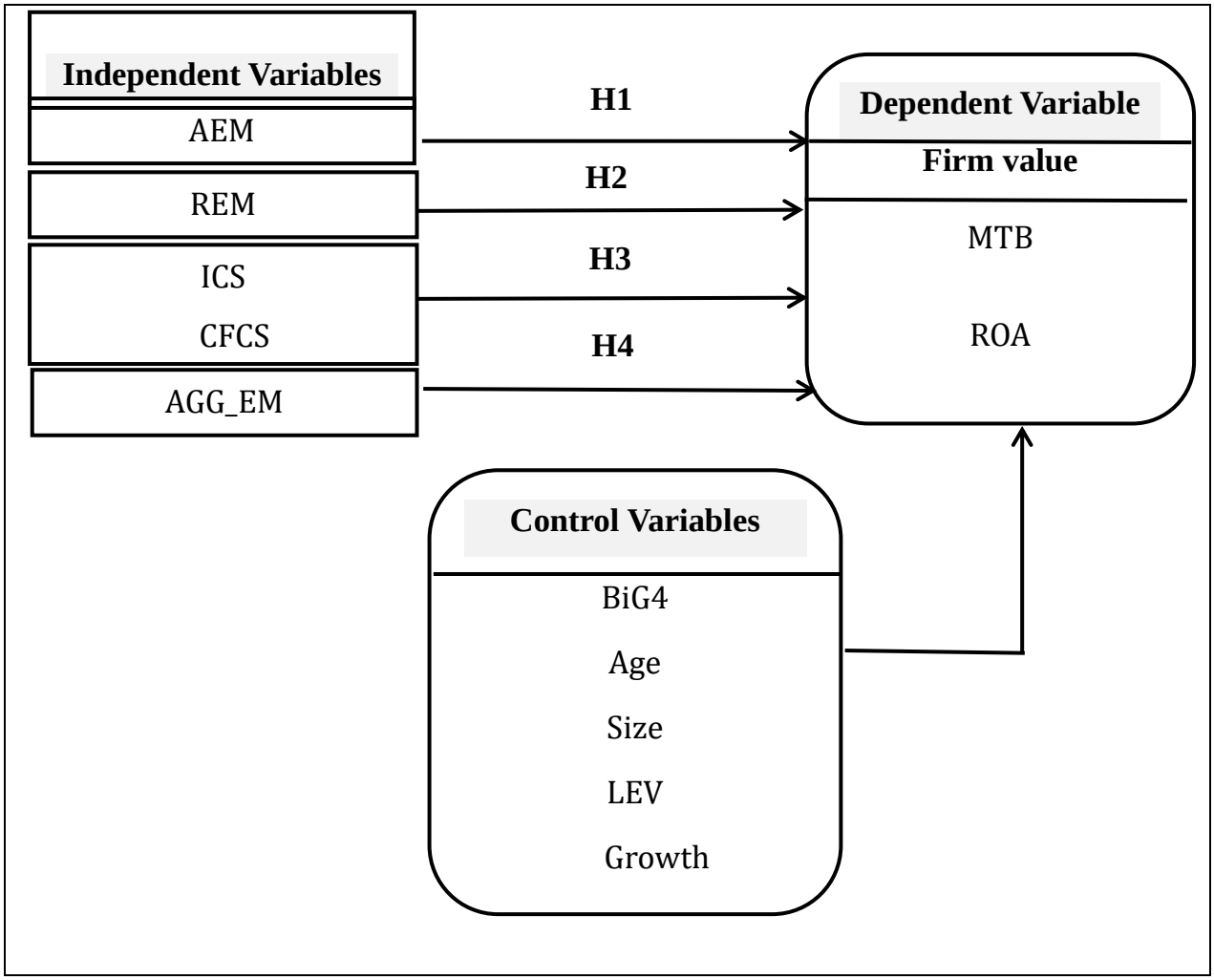


Figure (4.1): The Relationship among the Study Variables
Source: prepared by the researcher

4.2.4 Model Specification

The current study depends on several models to achieve its objectives. Two alternative measures of firm value which are MTB and ROA were used to investigate the impact of EM practices (AEM, REM, ICS, CFCS, and IEM) on firm value. The study proposed the following models:

A. Investigating the Impact of Accrual-based Earnings Management on Firm Value

Model (1): $MTB_{it} = \alpha + B_1\, AEM_{it} + B_2\, BIG4_{it} + B_3\, Age_{it} + B_4\, Size_{it} + B_5\, LEV_{it} + B_6\, Growth_{it} + \varepsilon_{it}$.

Model (2): $ROA_{it} = \alpha + B_1\, AEM_{it} + B_2\, BIG4_{it} + B_3\, Age_{it} + B_4\, Size_{it} + B_5\, LEV_{it} + B_6\, Growth_{it} + \varepsilon_{it}$.

B. Investigating the Impact of Real Earnings Management on Firm Value

Model (3): $MTB_{it} = \alpha + B_1\ REM_{it} + B_2\ BIG4_{it} + B_3\ Age_{it} + B_4\ Size_{it} + B_5\ LEV_{it} + B_6\ Growth_{it} + \varepsilon_{it}$.

Model (4): $ROA_{it} = \alpha + B_1\ REM_{it} + B_2\ BIG4_{it} + B_3\ Age_{it} + B_4\ Size_{it} + B_5\ LEV_{it} + B_6\ Growth_{it} + \varepsilon_{it}$.

C. Investigating the Impact of Income Classification Shifting, and Cash Flow Classification Shifting on Firm Value

Model (5): $MTB_{it} = \alpha + B_1\ ICS_{it} + B_2\ CFCS_{it} + B_3\ BIG4 + B_4\ Age_{it} + B_5\ Size_{it} + B_6\ LEV_{it} + B_7\ Growth_{it} + \varepsilon_{it.}$

Model (6): $ROA_{it} = \alpha + B_1\ ICS_{it} + B_2 CFCS_{it} + B_3\ BIG4 + B_4\ Age_{it} + B_5\ Size_{it} + B_6\ LEV_{it} + B_7\ Growth_{it} + \varepsilon_{it}$.

D. Investigating the Impact of Aggregate Earnings Management on Firm Value

Model (7): $MTB_{it} = \alpha + B_1\ AGG_EM + B_2\ BIG4_{it} + B_3\ Age_{it} + B_4\ Size_{it} + B_5\ LEV_{it} + B_6\ Growth_{it} + \varepsilon_{it}$.

Model (8): $ROA_{it} = \alpha + B_1\ AGG_EM + B_2\ BIG4_{it} + B_3\ Age_{it} + B_4\ Size_{it} + B_5\ LEV_{it} + B_6\ Growth_{it} + \varepsilon_{it}$.

4.2.5 Statistical Techniques

The present study uses the following statistical techniques to test hypotheses and obtain results:

- **The Normal Distribution Tests**

Used to test whether the study variables follow the normal distribution, in addition to determining the parametric and non- parametric techniques which are compatible for analyzing the study data. The study used Kolmogrov – Simirnov and Shapiro – Wilk tests.

- **The Normality Test**

Used to test the normality of residuals, this test determines whether sample data have skewness and kurtosis that match a normal distribution. The study used the Jarque-Bera test.

- **The Multicollinearity Test**

Used to detect the existence of the multicollinearity problem in the study models. The study relied on the collinearity diagnostics by determining the value of variance inflation factor (VIF) and tolerance.

- **The Autocorrelation Test**

The study used the Wooldridge test to detect the existence of autocorrelation problem in the study models.

- **The Heteroskedasticity Test**

The study relied on Whites' tests to check for heteroscedastic errors in regression analysis. It examines if the independent variable affects error variance non–linearly.

- **Descriptive Statistics**

Descriptive statistics are used to describe the basic features of the study data using the mean, standard deviation, maximum and minimum values, in addition to the frequency of the dummy variables.

- **Regression Analysis**

A Generalized Least Square (GLS) test on the Stata is employed to increase the models' efficiency. FGLS is used for solving the normality of residuals, autocorrelation and heteroskedasticity problems in the panel data.

- **Kruskal Wallis Test**

Used to examine the significant differences among sectors in terms of the study variables.

4.3 Empirical Results and Testing Hypotheses

4.3.1 The Normal Distribution Test

For testing whether the study variables follow the normal distribution, the study depended on the Kolmogorov-Smirnov and Shapiro-Wilk tests, where the variables follow the normal distribution if the p-value (sig.) is more than (.05). This can be illustrated by the following table (4.3):

Table (4.3): Results of the Normal Distribution Test

Variable	Kolmogorov-Smirnov		Shapiro-Wilk	
	Statistic	Sig.	Statistic	Sig.
MTB	.323	.000	.467	.000
ROA	.200	.000	.481	.000
REM	.120	.000	.904	.000
AGG_EM	.076	.002	.976	.000
Age	.115	.000	.967	.000
Size	.068	.009	.979	.001
LEV	.055	.078	.959	.000
Growth	.137	.000	.806	.000

Results in the table (4.3) indicate that the significance values for Kolmogorov-Smirnov and Shapiro-Wilk tests were less than (.05) which reflects that the study variables are not normally distributed except for LEV which reveals a significance value more than (.05) and so follows the normal distribution. However, there is no effect for the non-normality of the research variables on the accuracy of the research model as the sample size was more than 30.

In accordance with AEM, ICS, CFCS, and BIG4 variables, they have not been included in the normal distribution test because they are dummy variables with binary values not subject to the conditions of the normal distribution.

4.3.2 Descriptive Statistics

Table (4.4) shows descriptive statistics by dividing the study variables into continuous variables and interval variables through panel (A) and panel (B) as follows:

Table (4.4): Descriptive Statistics Results

Panel A: Continuous Variables				
Variable	**Mean**	**Std. Dev.**	**Min.**	**Max.**
MTB	.969	2.424	-27.644	9.281
ROA	.0323	.13481	-1.68	.53
REM	-.0001	.189	-1.152	.733
Age	3.344	.436	2.079	4.736
Size	20.898	1.92	16.414	24.938
Lev	.482	.243	.009	1.831
Growth	.102	.251	-.51	1.531
Panel B: Interval Variables				
Variable	**Value (1)**		**Value (0)**	
	Frequency	**Percent**	**Frequency**	**Percent**
AEM	81	33.75	159	66.25
ICS	108	45	132	55
CFCS	118	49.17	122	50.83
BIG4	80	33.33	160	66.67

As shown in table (4.4**)**, the mean value of MTB was (.969) with a minimum and maximum (-27.644, 9.281) respectively; a negative minimum value is because that some firms in the study sample have a negative equity. The findings are slightly below average values revealed by Zaher (2019); Elsayed (2021); Osman (2022) which reached (1.114, 1.955, 1.33) respectively. Regarding the second measure of firm value ROA, it showed a mean value of (.0323) and a minimum and maximum value (-1.68, .53) respectively; the negative value of ROA means that some firms suffer from high losses, implying a quite poor performance in some of the Egyptian firms in the study sample. This result goes

in the same vein as those of Majeed and Yan (2021); Hussien (2023) which showed mean values of (.0348, .0420) respectively; the negative value of ROA Additionally, the mean value of REM ranged from (-1.152 to .733) with a mean value of (-.0001). This value agrees with the result of Almaleeh et al. (2023) where the REM mean value was (-.000012).

Turning to control variables, (Age) has a mean value of (3.344) with a minimum and maximum value of (2.079, 4.736). This result is consistent with the values reported by Elsayed et al. (2023); Elhabashy et al. (2023) which reached mean values of (3.406, 2.745) respectively. Moreover, (Size) ranged from (16.414 to 24.938), with a mean value of (20.898). This finding goes in the same vein as these reported by Elsayed (2021); Osman (2022); Almaleeh et al. (2023) which revealed mean values of (20.494, 20.50. 20.711) respectively. For LEV, the mean value was (.482) with a minimum and maximum value of (.009, 1.831) respectively; a high value of LEV more than one is because the assets of some firms are highly below their liabilities and so have a negative equity which indicates that these firms have a high level of indebtedness. This result comes in line with the findings of Zalata and Robert (2017); Zaher (2019); Elhabasy and Elkelety (2023) where the mean values were (.48, .425, .553) respectively. As for Growth, its minimum and maximum value reached (-.51, 1.531) respectively, with a mean value of (.102) this value comes in line with the average value of Hussien (2023) which reached (.133).

Regarding the interval variables represented in AEM, ICS, CFCS, and BIG4, the study found that the percentage of firms engaged in AEM during the study period reached (81) observations at (33.75%). While the percentage of firms that engaged in ICS reached (108) observations at (45%). Additionally, the number of firms observations that engaged in CFCS reached (118) at (49.17%) Finally, the results indicated that around 33% of the study sample is being audited by BIG 4 firm with (80) observations.

4.3.3 Validation Tests of the OLS Models

The degree of study models' validity and acceptability for statistical analysis can be ensured by performing three preliminary tests. These tests are normality, multicollinearity, autocorrelation, and heteroscedasticity. This can be explained as follows:

4.3.3.1 The Normality Test

For testing the normality of residuals, the Jarque-Bera test was conducted. That is, if the P-value of the Jarque-Bera test is more than 0.05, this means that

the residuals are normally distributed (Thadewald and Büning, 2007). This can be exhibited in table (4.5) as follows:

Table (4.5): The Normality Test Results

Models	MTB		ROA	
	Test statistic	P-value	Test statistic	P-value
1&2	29	.000	120	.000
3&4	29	.000	140	.000
5&6	29	.000	130	.000
7&8	68	.000	120	.000
Result	Normality of residuals Problem			

The results as illustrated in table (4.5), reveal that the P-value of the Jarque-Bera normality test for all study models were lower than 0.05, indicating that the residuals are not normally distributed.

4.3.3.2 The Multicollinearity Test

To verify the quality and validity of the study models, the multicollinearity test was conducted to detect the existence of the multicollinearity problem. As such problem may result in imprecise regression coefficients, failure to reach statistical significance, change in the predicted signs of coefficients, and substantial changes in the estimated coefficients when adding or removing a few observations (Asteriou and Hall, 2007). The study uses the collinearity diagnostic measure to identify two values of Variance Inflation Factor (VIF) and Tolerance. If the VIF value is less than (10) and the tolerance is greater than (.05), this indicates that there is no multicollinearity problem. (Shrestha, 2020). This can be illustrated in the table (4.6) as follows:

Table (4.6): The Multicollinearity Test Results

Variables	Collinearity Statistics	
	VIF	TOL
AEM	1.301	.769
REM	1.373	.729
ICS	1.277	.783
CFCS	1.289	.776
AGG_EM	1.533	.652
BIG4	1.385	.722
Age	1.119	.894
Size	1.65	.606
LEV	1.513	.661
Growth	1.022	.978
Result	No Multicollinearity Problem	

As shown in table (4.6), the findings confirm the inexistence of multicollinearity problem as all study variables show VIF values less than (10) and tolerance values greater than (.05).

4.3.3.3 The Autocorrelation Test

The study conducted an autocorrelation test to ensure that the residuals are not serially correlated because this problem may cause estimated variances of the regression coefficients to be biased and inconsistent and therefore, hypotheses testing is no longer valid (Asteriou and Hall, 2007). To identify the problem of serial autocorrelation, Wooldridge test was performed. That is, if the Wooldridge p-value is more than (.05), this infers no autocorrelation problem (Drukker, 2003). This can be illustrated in the table (4.7) as follows:

Table (4.7): Wooldridge Test Results

Models	MTB		ROA	
	F test	P-value	F test	P-value
1&2	140.920	.000	30.058	.000
3&4	164.863	.000	30.471	.000
5&6	186.500	.000	28.107	.000
7&8	145.063	.000	32.239	.000
Result	Autocorrelation Problem			

The results reported in table (4.7) show that the p-values of Wooldridge test for all study models were less than (.05), implying the presence of serial autocorrelation problem among the study variables.

4.3.3.4 The Heteroscedasticity Test

To check for heteroscedasticity, or differences in variance of the errors in regression model, the study performed the White's test. A heteroskedasticity problem would be present if the p-value of White's test is less than (.05) (Asteriou and Hall, 2007). This can be explained in the table (4.8) as follows:

Table (4.8): The Heteroscedasticity Test Results

Models	MTB		ROA	
	Chi^2	P-value	Chi^2	P-value
1&2	39.35	.0340	229.81	.000
3&4	37.78	.0635	229.39	.000
5&6	46.89	.0434	231.70	.000
7&8	39.22	.0465	229.03	.000
Result	Heteroskedasticity Problem			

As table (4.8) shows, the p-values of White's test were less than (.05) which indicates that the study models suffer from heteroskedasticity problem (having non constant variance of errors).

Accordingly, due to the presence of normality of residuals, autocorrelation, and heteroskedasticity problems in the study models; linear regression is no longer the best linear unbiased estimator, and the generalized least square regression (GLS) is to be employed because of its proper estimation for solving normality of residuals, autocorrelation, and heteroskedasticity problems and it has been adopted by previous studies such as (Al-Absy et al. 2018; Chakroun et al. 2021; Elhabasy and Elkelety, 2023; Musa et al. 2023; ; Almaleh et al., 2023; Mahmoud et al., 2023).

4.3.4 Statistical Analysis for Testing Hypotheses

This section presents the testing of hypotheses and illustrates the empirical results found using the statistical analysis techniques including Generalized Least Square (GLS) test and Kruskal Wallis Test.

4.3.4.1 Investigating the Impact of Accrual-based Earnings Management on Firm Value

For investigating the impact of AEM on firm value, the study used two alternative measures of firm value which are MTB and ROA. GLS regression analysis was employed to test the first hypothesis. This can be illustrated by table (4.9) as follows:

Table (4.9): GLS Regression Analysis Results for the Impact of AEM on Firm Value

Variables	Model (1): MTB		Model (2): ROA	
	B	**P -value**	**B**	**P -value**
AEM	.172	.001***	-.007	.024**
BIG4	.345	.000***	.027	.02**
Age	.16	.155	-.009	.325
Size	.313	.000***	.036	.000***
LEV	-4.039	.000***	-.46	.000***
Growth	.913	.000***	.124	.000***
Constant	-4.446	.000***	-.474	.000***
Model Summary				
Wald Chi^2	367.956		381.197	
p-value	.000		.000	

*** p<.01, ** p<.05, * p<.1

The results in table (4.9) show a positive impact of AEM on MTB, as regression coefficient was (.172) and the p-value was (.001), Such positive impact implying that managers' involvement in EM practices could mislead the market about the firm's economic performance, leading to an increase in the value of firms, concluding that managers are more likely to engage in EM practices to meet the expectations of analyst forecast and stakeholders, and to show how their firms are more stable in the market in order to enhance its reputation and help investors better predict future performance which as a result reflects on the share price positively and increase the value of the firm in the short term accordingly. To conclude, the impact of EM practices on firm value mainly depends on how managers employ them i.e., informative perspective or opportunistic perspective. Taken together, their impact on firm value could be positive in the short term but such positive impact might be changed in the long term (Abo Taleb, 2022). This result goes in the same vein as those of Susanto (2017); Abbas and Ayub (2019); Olatungji and Juwa (2020) which reported that AEM practices increase firm value. However, this outcome is inconsistent with the findings of Challen and Siregar (2012); Wibowo (2018); Darmawan et al. (2019) which provided evidence of no impact of AEM on firm value.

For control variables, the results show a positive impact of BIG4 on firm value (coefficient = .345, P-value= .000), this result supports that BIG4 auditing firms can enhance the firm's value and their reputation in capital market. This is because investors perceive that firms who are audited by BIG4 firms provide proper, reliable, and authentic financial information which in turn improves investors' confidence and increase firm value accordingly. This result is in accordance with research conducted by (Afza and Nazir, 2013; Wijaya, 2020) However, this finding is contrary to results reported by Yolandita, and Cahyonowati (2022) which showed a negative impact of BIG4 on firm value.

Furthermore, the findings show that firm value is positively influenced by Size, (coefficient = .0313, p-value = .000). Such result indicate that investors tend to be more interested in firms with large scale as large firms are in a more stable financial condition, more stable operations, relatively more able to generate profits, and more flexible in obtaining funds due to the accessibility to the capital market. Therefore, investors interpret this as a positive signal concerning the firm's ability to provide a greater rate of return, which as a result increases stock prices and increases the firm's value accordingly. This result is consistent with the results of (Mohamed, 2017; Atiningsih and Izzaty, 2021; Marpaung et al., 2022). In contrast, (Susanti and Restiana, 2018; Hirdinis, 2019)

confirmed that Size has a negative impact on firm value. (Antro et al., 2020) proved that Size has no effect on firm value.

Regarding leverage, the results reveal a negative impact of leverage on firm value with a coefficient value of (- 4.039) and p-value = (.000). These results confirm that low leveraged firms are more likely to create value for shareholders. This is consistent with the findings of (Vo and Ellis, 2017; Mohamed, 2017; Gharaibeh and Qader, 2017) who revealed that the higher the leverage, the firm value is lower. However, these findings contradict the findings of Yuliyanti et al. (2021) which revealed a positive and significant effect of leverage on firm value.

Moreover, the results indicate the existence of a positive impact of Growth on firm value a with a coefficient value of (.913) and p-value = (.000). The reason is that the existence of asset growth opportunities provides a positive signal about the firm's growth in the future and indicates that the firm's performance is improving. i.e., firms with rapid growth opportunities are expected to have better value as these firms are favorable to investors who have higher prospects for recovering their investment. These results are in line with the research results of (Data et al., 2017; Badruzaman et al., 2019; Marpaung et al., 2022). In contrast the findings of Nurdiansari et al (2022) showed no significant impact of Growth on firm value.

On the other hand, (Age) has an insignificant impact on firm value with a p-value greater than (.05). Although this result contradicts with research conducted by (Leite and Carvalhal, 2016; Susanti and Restiana, 2018) who found a positive impact of firm age on firm value. Whereas research conducted by (Chay et al., 2015) showed a negative impact of firm age on firm value.

Referring to the second measure of firm value (ROA), findings reveal a negative impact of AEM on firm value (coefficient = -.007, p-value= .024), this evidence indicates that the greater intense of EM practices adversely affect share prices and consequently the value of the firm. Such interpretation is consistent with the findings of Gill et al. (2013); Yaniarti (2016); Almari et al. (2021). For control variables (BIG4, Size, and Growth) the results confirm that they have a positive impact on ROA as their regression coefficients were (.027, .036, .124) respectively, and p-values were (.02, .000, .000) respectively, but LEV show a negative impact (coefficient = -.46, p-value = .000) Age illustrated no significant impact on ROA.

For the first hypothesis, both GLS models have a high fit because the Wald Chi^2 values reached (367.956, 381.197) respectively and the overall significance of the models have p- values of (.000). Accordingly, the regression equations for investigating the impact of AEM on firm value can be presented as follows:

MTB= -4.446 +**.172 (AEM)** + **.345 (BIG4)** + .16 (Age)+ **.313 (Size)** - **4.039 (LEV)** + **(.913) Growth.**

ROA= -.474 -**.007 (AEM**)+ **.027 (BIG4)** -.009 (Age)+ **.036 (Size)** -**.46 (LEV)**+ **.124 (Growth).**

Thus, as a result of the study findings, there is a strong indication to reject the first hypothesis ***(H1)*** related to **"AEM has an insignificant impact on the value of firms listed in the Egyptian Exchange".**

4.3.4.2 Investigating the Impact of Real Earnings Management on Firm Value

To test the second hypothesis, the study depended on GLS regression analysis for investigating the impact of REM on firm value as measured by MTB and ROA. This can be illustrated by the following table:

Table (4.10): GLS Regression Analysis Results for the Impact of REM on Firm Value

Variable	Model (3): MTB		Model (4): ROA	
	B	**P -value**	**B**	**P -value**
REM	-.195	.204	-.06	.000***
BIG4	.199	.034**	.032	.002***
Age	.021	.817	-.011	.129
Size	.328	.000***	.04	.000***
LEV	-3.93	.000***	-.46	.000***
Growth	1.04	.000***	.15	.000***
Constant	-4.24	.000***	-.564	.000***
Model Summary				
Wald Chi^2	270.911		425.925	
p-value	.000		.000	

**** p<.01, ** p<.05, * p<.1*

The results reported in table (4.10) reveal an insignificant impact of REM on MTB, this result is inconsistent with the findings of Challen and Siregar (2021); Susanto (2017); Mellado-Cid et al. (2017), Darmawan et al. (2019); Olatunji and Juwon(2020); Tulcanaza-Prieto (2022) which provided evidence of a significant negative impact of REM on firm value, while Abbas and Ayub (2019); Alsoboa (2020) reported that REM practices have a significant positive impact on firm

value. For control variables, the results confirm that MTB is positively influenced by BIG4, Size, and Growth with (coefficient = .199, .328, 1.04) and (p-value =.034, .000, .000) respectively but negatively influenced by LEV with coefficient value of (-3.93) and p-value of (.000), therefore, it is concluded that these variables play an important role in determining firm value. In contrast, the control variable (Age) is not proven to have a significant impact on MTB.

Relating to the ROA measure as shown in panel (B), the findings reveal a negative impact of REM on firm value (coefficient= -.06, p-value = .000). This finding is consistent with Susanto (2017); Mellado-Cid et al. (2017); Abbas and Ayub (2019); Darmawan et al. (2019); Tulcanaza-Prieto (2022); Hussein (2023) who concluded that REM practices have an adverse impact on the value of the firm. Referring to control variables, it is proven that BIG4, Size, and Growth have a positive impact on ROA as their regression coefficients were (.032, .04, .15) and p-values were (.002, .000, .0000) respectively. Whilst the coefficient of LEV is negatively significant (coefficient= -.46, p-value= .000). On the other hand, Age shows a p-value greater than (.05), indicating an insignificant impact on ROA.

Table (4.10) also confirms that the proposed models have a high fit as the overall significance of both models have p-values of (.000); besides, the Wald Chi^2 were (270.911, 425.925) respectively. Thus, the regression equations for investigating the impact of REM on firm value can be presented as follows:

MTB= **-4.24** - .195 (REM) + **.199 (BIG4)** + .021(Age)+ **.328 (Size) -3.93(LEV)+ 1.04 (Growth)**.

ROA = **-.564** -**.06 (REM)** + **.032 (BIG4)** -.011(Age)+ **.04 (Size) + -.46 (LEV) + .15 (Growth).**

Accordingly, depending on the previous regression models, the research rejects the second hypothesis ***(H2)*** expressed as **"REM has an insignificant impact on the value of firms listed in the Egyptian Exchange".**

4.3.4.3 Investigating the Impact of Income Classification Shifting and Cash Flow Classification Shifting on Firm Value

For Investigating the impact of ICS and CFCS on firm value, two alternative measures of firm value were utilized including MTB and ROA. To test the third hypothesis, the study employed GLS regression analysis. This can be explained by table (4.11) as follows:

Table (4.11): GLS Regression Analysis Results for the Impact of ICS and CFCS on Firm Value

Variable	Model (5): MTB		Model (6): ROA	
	B	**P -value**	**B**	**P -value**
ICS	-.175	.003***	.016	.000***
CFCS	.175	.01***	.013	.000***
BIG4	.287	.036**	.027	.021**
Age	.185	.13	-.003	.771
Size	.251	.000***	.036	.000***
LEV	-3.33	.000***	-.428	.000***
Growth	.854	.000***	.128	.000***
Constant	-3.431	.000***	-.538	.000***
Model Summary				
Wald Chi^2	114.996		363.421	
p-value	.000		.000	

**** p<.01, ** p<.05, * p<.1*

The results of the GLS regression analysis in the previous table (4.11) show a negative impact of ICS on MTB (coefficient = -.175, P-value= .003). Whereas the analysis reveals that the coefficient of CFCS is positive with a p-value (.01), this result is inconsistent with Osman et al. (2022), who showed that firm engaging in CFCS suffer from lower firm value. Regarding control variables, the results reveal a positive influence of BIG4, Size, and Growth on MTB with coefficient values of (.287, .251, .854) and p-values (.036, .000, .000) respectively. In contrast, (LEV) has a negative impact on MTB (coefficient=-3.33, P-value= .000). On the other side, firm age (Age) has an insignificant impact on MTB with a p-value greater than (.05).

Turning to the second measure of firm value (ROA), the results report a positive impact of ICS and CFCS practices on ROA with regression coefficients of (.016, .013) and p-values less than (.01). With regard to the control variables, evidence shows that only (Age) has an insignificant impact on ROA (p-value >.05). While the coefficient values of other control variables (BIG4, Size, and Growth) are positively significant (coefficients =.027, .036, .128) and (p-values = .021, .000, .000). But (LEV) had a coefficient value of (-.428) and significant value of (.000), implying that there is a negative impact of LEV on firm value.

Table (4.11) also indicates that the proposed models have a high fit because the overall significance of both models had p-values of (.000) and Wald Chi^2 values of (114.996, 363.421) respectively. Thus, the regression equations for

investigating the impact of ICS shifting and CFCS on firm value can be presented as follows:

MTB= **-3.431 - .175 (ICS) +.175 (CFCS)** + **.287 (BIG4)** + .185 (Age)+ **.251 (Size) - 3.33 (LEV) + (.854) Growth.**

ROA= **-.538** +**.016 (ICS)** +**.013 (CFCS)** + **.027 (BIG4)** -.003 (Age)+ **.036 (Size) -.428 (LEV)**+ **.128 (Growth).**

As a result, the third hypothesis ***(H3)*** related to **"ICS and CFCS have an insignificant impact on the value of firms listed in the Egyptian Exchange"** is rejected.

4.3.4.4 Investigating the Impact of Aggregate Earnings Management on Firm Value

To test the fourth hypothesis, the study relied on GLS regression analysis for investigating the impact of AGG_EM on firm value as measured by MTB and ROA. This can be illustrated by the following table:

Table (4.12): GLS Regression Analysis Results for the Impact of AGG_EM on Firm Value

Variable	Model (7): MTB		Model (8): ROA	
	B	**P -value**	**B**	**P -value**
AGG_EM	.011	.047**	.001	.009***
BIG4	.253	.002***	.027	.024**
Age	.078	.343	-.009	.307
Size	.335	.000***	.035	.000***
LEV	-4.089	.000***	-.435	.000***
Growth	1.108	.000***	.119	.000***
Constant	-4.704	.000***	-.49	.000***
Model Summary				
Wald Chi^2	319.788		304.776	
p-value	.000		.000	

**** p<.01, ** p<.05, * p<.1*

Along with the evidence extracted from table (4.12) it is noted that AGG_EM practices positively influence MTB, showing a positive coefficient = (.011) and p-value = (.047). For control variables, the results show a positive impact of BIG4, Size, and Growth on MTB as their coefficient values were (.253, .335, 1.108) respectively and p-values were less than (.01).while LEV had a negative impact coefficient of (- 4.089) with a p-value of (.000) For Age, the results show a p-value greater than (.05), indicating an insignificant impact on MTB.

Likewise, ROA as a second measure of firm value is proven to be positively influenced by AGG_EM as regression coefficient was (.001) and the significance p-value was (.009). Relating to control variables (BIG4, Size, and Growth), the results indicate that they are positive with coefficient values of (.027, .035, .119) and p- values of (.024, .000, .000) respectively. While the coefficient of LEV is negative (coefficient = -.435, p-value= .000). On the contrary, the regression coefficient of Age is not statistically significant for ROA

Table (4.12) also reveals that the Wald Chi^2 are significantly higher for both models as Wald Chi^2 values were (319.788, 304.776) respectively and p-value = (.000), showing that the proposed models have a high fit. Thus, the regression equations for investigating the impact of AGG_EM on firm value can be presented as follows:

MTB= **-4.704**+**.011 (AGG_EM)** + **.253 (BIG4)** + .078 (Age)+ **.335 (Size)** - **4.089 (LEV)** + **1.108 (Growth).**

ROA= **-.49** +**.001 (AGG_EM**) + **.027 (BIG4)** -.009 (Age)+ **.035 (Size)** -**.435 (LEV)**+ **.119 (Growth).**

Accordingly, depending on the previous regression models, the research rejects the fourth hypothesis ***(H4)*** expressed as **"AGG_EM has an insignificant impact on the value of firms listed in the Egyptian Exchange".**

4.3.5 Investigating the differences among Sectors regarding the Study Variables

The study relied on the Kruskal-Wallis Test for comparing MTB, ROA, AEM, REM, ICS, CFCS, and AGG_EM among sectors in the Egyptian Exchange. Where a significant level less than (.05) indicates the existence of significant differences among sectors in the Egyptian Exchange regarding the study variables. This can be illustrated through the following tables:

Table (4.13): Kruskal-Wallis Test Results for investigating the differences among Sectors regarding MTB

MTB				
Sectors	**Mean rank**	**Sig.**	**Chi^2**	**Ranking up**
Contracting and Construction Engineering	89.50	**.000**	**41.197**	6
Health care, pharmaceuticals	**176.60**			1
Basic resources	133.86			2
Travel & Leisure	**88.77**			7
Building materials	96.87			5
Industrial goods, services, and Automobiles	124.87			3
Real estate	124.67			4

Table (4.13) shows that the test significance level for MTB was (.000) which is lower than (.05), indicating the existence of significant differences among the Egyptian Exchange sectors regarding MTB for a chi^2 value of (41.197). This is apparent from the expansion in the mean rank of MTB among the Egyptian Exchange which ranges from (88.77 to 176.60). The mean rank of the sectors (Contracting and Construction Engineering, Health care, pharmaceuticals, Basic resources, Travel & Leisure, Building materials, Industrial goods, services, and Automobiles, Real estate) amounted to (89.50, 176.60, 133.86, 88.77, 96.87, 124.87, 124.67) respectively.

Table (4.14): Kruskal-Wallis Test Results for investigating the differences among Sectors regarding ROA

ROA				
Sectors	**Mean rank**	**Sig.**	**Chi^2**	**Ranking up**
Contracting and Construction Engineering	121.27	**.000**	**51.695**	4
Health care, pharmaceuticals	**168.34**			1
Basic resources	117.80			5
Travel & Leisure	95.06			6
Building materials	**58.97**			7
Industrial goods, services, and Automobiles	153.17			2
Real estate	123.91			3

Along with the results from table (4.14), it is noted that ROA significant level is (.000) which is less than (.05) with a chi^2 value (51.695), this reveals that there are significant differences among the Egyptian Exchange sectors. The mean rank was between the minimum and maximum range of (58.97, 168.34). The mean rank of the sectors (Contracting and Construction Engineering, Health care, pharmaceuticals, Basic resources, Travel & Leisure, Building materials, Industrial goods, services, and Automobiles, Real estate) amounted to (121.27, 168.34, 117.80, 95.06, 58.97, 153.17, 123.91) respectively. The increase in the mean rank is the reason for the significant difference among the Egyptian Exchange Sectors concerning ROA.

Table (4.15): Kruskal-Wallis Test Results for investigating the differences among Sectors regarding AEM

AEM				
Sectors	**Mean rank**	**Sig.**	**Chi^2**	**Ranking up**
Contracting and Construction Engineering	112.00	**.409**	**6.125**	2
Health care, pharmaceuticals	131.43			6
Basic resources	**107.43**			1
Travel & Leisure	128.00			5
Building materials	120.00			4
Industrial goods, services, and Automobiles	**132.00**			7
Real estate	114.67			3

Results illustrated in Table (4.15) indicate that the test significance level for AEM (.409) is higher than (.05), which means the inexistence of significant differences among the Egyptian Exchange sectors regarding AEM for a chi^2 value of (6.125). This is evident from the decrease in the mean rank of AEM among the Egyptian Exchange that ranges from (107.43 to 132.00), as the mean rank of the sectors (Contracting and Construction Engineering, Health care, pharmaceuticals, Basic resources, Travel & Leisure, Building materials, Industrial goods, services, and Automobiles, Real estate) amounted to (112.00, 131.43, 107.43, 128.00, 120.00, 132.00, 114.67) respectively.

Table (4.16): Kruskal-Wallis Test Results for investigating the differences among Sectors regarding REM

REM				
Sectors	**Mean rank**	**Sig.**	**Chi^2**	**Ranking up**
Contracting and Construction Engineering	121.27	**.978**	**1.185**	4
Health care, pharmaceuticals	123.14			3
Basic resources	116.46			6
Travel & Leisure	118.94			5
Building materials	**110.97**			7
Industrial goods, services, and Automobiles	123.03			2
Real estate	**126.96**			1

The results of table (4.16) reveal that the test significant level for REM was (.978) which is higher than (.05) for a Chi^2 value of 1.185, and with a minimum and maximum mean rank (110.97, 126.96) respectively. The mean rank of the sectors (Contracting and Construction Engineering, Health care, pharmaceuticals, Basic resources, Travel & Leisure, Building materials, Industrial goods, services, and Automobiles, Real estate) amounted to (121.27, 123.14, 116.46, 118.94, 110.97, 123.03, 126.96) respectively. The decrease in

the mean rank among the Egyptian Exchange Sectors indicates that there are no significant differences among the Egyptian Exchange Sectors regarding REM.

Table (4.17): Kruskal-Wallis Test Results for investigating the differences among Sectors regarding ICS

ICS				
Sectors	**Mean rank**	**Sig.**	**Chi^2**	**Ranking up**
Contracting and Construction Engineering	122.50	**.602**	**4.554**	2
Health care, pharmaceuticals	117.93			5
Basic resources	121.36			4
Travel & Leisure	**138.50**			1
Building materials	118.50			3
Industrial goods, services, and Automobiles	114.50			6
Real estate	**111.83**			7

It can be seen from the results in the table (4.17) that the significant level for ICS was (.602) which is more than (.05), indicating no significant differences among the Egyptian Exchange Sectors concerning ICS for a chi^2 value of (4.554) with a minimum and maximum mean rank (111.83, 138.50) respectively. The mean rank of the sectors (Contracting and Construction Engineering, Health care, pharmaceuticals, Basic resources, Travel & Leisure, Building materials, Industrial goods, services, and Automobiles, Real estate) amounted to (122.50, 117.93, 121.36, 138.50, 118.50, 114.50, 111.83) respectively. The decrease in the mean rank among the Egyptian Exchange Sectors indicates that there are no significant differences among the Egyptian Exchange Sectors regarding ICS.

Table (4.18): Kruskal-Wallis Test Results for investigating the differences among Sectors regarding CFCS

CFCS				
Sectors	**Mean rank**	**Sig.**	**Chi^2**	**Ranking up**
Contracting and Construction Engineering	117.50	**.357**	**6.625**	4
Health care, pharmaceuticals	130.07			6
Basic resources	112.93			3
Travel & Leisure	**109.50**			1
Building materials	129.50			5
Industrial goods, services, and Automobiles	**137.50**			7
Real estate	112.17			2

Referring to CFCS, it is noted that its significant level is (.357) which is more than (.05) with a chi^2 value (6.625. The mean rank is between the minimum and maximum range of (109.50, 137.50). The mean rank of the sectors (Contracting and Construction Engineering, Health care, pharmaceuticals, Basic resources, Travel & Leisure, Building materials, Industrial goods, services, and

Automobiles, Real estate) amounted to (117.50, 130.07, 112.93, 109.50, 129.50, 137.50, 112.17) respectively. The decrease in the mean rank among the Egyptian Exchange Sectors indicates that there are no significant differences among the Egyptian Exchange Sectors.

Table (4.19): Kruskal-Wallis Test Results for investigating the differences among Sectors regarding AGG_EM

AGG_EM				
Sectors	**Mean rank**	**Sig.**	**Chi^2**	**Ranking up**
Contracting and Construction Engineering	79.55	**.000**	**74.173**	6
Health care, pharmaceuticals	122.53			4
Basic resources	126.89			3
Travel & Leisure	128.06			2
Building materials	82.03			5
Industrial goods, services, and Automobiles	**79.28**			7
Real estate	**188.50**			1

Table (4.19) shows that the test significance level for AGG_EM was (.000) which is lower than (.05), indicating the existence of significant differences among the Egyptian Exchange Sectors regarding AGG_EM for a chi^2 value of (74.173). This is evident from the expansion in the mean rank of AGG_EM among the Egyptian Exchange Sectors that ranges from (79.28 to 188.50). The mean rank of the sectors (Contracting and Construction Engineering, Health care, pharmaceuticals, Basic resources, Travel & Leisure, Building materials, Industrial goods, services, and Automobiles, Real estate) amounted to (79.55, 122.53, 126.89, 128.06, 82.03, 79.28, 188.50) respectively.

Based on the above, the fifth hypothesis ***(H5)*** which indicated that "**There are insignificant differences among sectors in the Egyptian Exchange regarding the study variables"** is rejected.

4.4 Chapter Summary:

This chapter aims to empirically investigate the impact of EM practices (AEM, REM, ICS, CFCS, and AGG_EM) on the value of firms listed in the Egyptian Exchange, therefore, this chapter included two main parts. Firstly, it discussed the study methodology by indicating the sample selection, data collection method, variables description and measurement, model specification, and statistical techniques. Secondly, it illustrates the empirical results; it shows the findings of the normal distribution test, descriptive statistics results, validation tests, and the statistical analysis for testing hypotheses.

Fifth hypotheses are proposed, and the statistical analysis revealed the following results:

- AEM has a significant positive impact on MTB while results showed a significant negative impact on ROA, therefore the first hypothesis *(H1)* related to "AEM has an insignificant impact on the value of firms listed in the Egyptian Exchange" is rejected.
- The findings found an insignificant impact of REM on MTB, but it was negatively significant on ROA. This result provides a strong indication to reject the second hypothesis *(H2)* related to **"**REM has an insignificant impact on the value of firms listed in the Egyptian Exchange**"**.
- The results found a significant negative impact of ICS on MTB whilst CFCS reported a significant positive impact on MTB. Moreover, evidence reported a significant positive impact of ICS and CFCS on ROA, such result supports rejecting the third hypothesis ***(H3)*** related to "ICS and CFCS have an insignificant impact on the value of firms listed in the Egyptian Exchange".
- A significant positive impact of IEM was shown on each of MTB and ROA, this finding provides evidence to reject the fourth hypothesis ***(H4)*** which states that "AGG_EM has an insignificant impact on the value of firms listed in the Egyptian Exchange".
- The results reported the existence of significant differences among sectors in the Egyptian Exchange regarding the study variables including MTB, ROA, AEM, REM, ICS, CFCS, and AGG_EM. Hence the fifth ***(H5)*** hypothesis related to "There are insignificant differences among sectors in the Egyptian Exchange regarding the study variables" is rejected.

Chapter 5

Summary, Conclusion, and Recommendations

5.1 Summary:

The current study consists of five chapters. The first chapter provides the general framework of the study which includes an introduction, the main research problem, research objectives, research importance, research limitations, literature review and study hypotheses and finally the research outline.

The second chapter is divided into eight sections for addressing firm value and the EM phenomenon. This chapter discussed the concept of firm value and argues that there is no unique definition of firm value, but it could be described depending on the perception of each stakeholder and expected interests that will be realized. Moreover, the main determinants of firm value are explained including firm size, firm age, profitability, liquidity, leverage, dividend policy, investment opportunity, capital structure, and firm growth. Four main approaches of firm value are presented which are asset approach, income approach, market approach, and economic approach. Furthermore, this chapter presents the different definitions of EM and argues that EM practices occur when managers intentionally intervene in financial reporting to manipulate reported earnings through accounting practices and/or real earnings activities to achieve specific targets. Such manipulation could be either firm value maximizing or opportunistic. Also, this chapter demonstrates the main motivations behind EM practices including capital market expectations motivations, contractual motivations, and regulatory motivations. This chapter presents the main determinants of EM which include audit quality, internal corporate governance, auditing standards, firm life cycle, firm size, financial leverage, and human factors such as gender diversity, and managerial ability. Finally, two main approaches of EM are presented which are the accounting EM approach and REM approach.

In the third chapter, two sections are presented: the relationship among EM tools and the measurement of such tools. This chapter indicates that the relationship among EM tools could be complementary, simultaneously, or substitutional based on time, cost, and constraints of each strategy. Further, this chapter illustrates several models to detect and measure EM practices through AEM, CS, and REM.

The fourth chapter is divided into two sections. The first section provides the study methodology including sample selection, data collection method, variables description and measurement, model specification, and statistical techniques. In the second section the empirical results are presented. This section illustrates the

normal distribution test, descriptive statistics, and the statistical analysis for testing hypotheses.

The fifth chapter includes the summary, conclusions, recommendations of the present study, and suggests some avenues for future research.

5.2 Conclusions

This study aimed to propose a multidimensional model for measuring EM practices and its impact on firm value by utilizing a sample of 48 firms listed in the Egyptian Exchange from 7 sectors with 240 observations during the period from 2017 to 2021.

The current study empirically investigates the impact of EM practices namely, AEM, REM, ICS, CFCS, and AGG_EM on firm value. By using MTB and ROA as proxies for firm value, the study reported the following findings:

- The results found a significant positive impact of AEM on MTB while evidence showed a significant negative impact of AEM on ROA.
- The findings reported that REM has no significant impact on MTB, but it had a significant negative impact on ROA.
- For CS practices, it was proven that ICS had a significant negative impact on MTB whilst CFCS revealed a significant positive influence on MTB, moreover, evidence showed a significant positive impact of ICS and CFCS practices on ROA.
- The results revealed the existence of significant positive impact of AGG_EM practices on the two proxies of firm value (MTB and ROA).
- Referring to control variables, the results confirmed that the firm value measured by MTB and ROA was positively and significantly influenced by BIG4, Size, and Growth but negatively and significantly affected by LEV, indicating that these variables are more influential and play a crucial role in determining firm value. On the other hand, Age was shown to be insignificant for MTB and ROA.
- The results reported the existence of significant differences among sectors in the Egyptian Exchange regarding the study variables including MTB, ROA, and AGG_EM but insignificant differences regarding AEM, REM, ICS, and CFCS.
- The study indicated that Egyptian firms engage in EM tools jointly, complementary, this means that they don’t prefer one strategy over the other to attain their desired earnings targets.

- When investigating the existence of ICS and CFCS practices in Egyptian firms the study concluded that Egyptian firms are more prone to engage in ICS practices by shifting COGS to operating expenses in order to inflate gross profit rather than shifting other operating income to total revenues, showing a positive and significant impact of operating expenses on unexpected gross profit but other operating income had an insignificant impact on unexpected gross profit. Moreover, the results illustrated that Egyptian firms are more inclined in CFCS practices by shifting operating cash flows as investing and financing cash flows, as a negative and significant impact of investing and financing cash flows was found on unexpected cash flow.

5.3 Recommendations

The study findings should be of interest to standard setter, regulators, investors, and auditors, as results highlight the importance of an awareness of EM issue and its impact on the value of firms. Therefore, some recommendations can be given as follows:

- For standard setter the findings emphasize the necessity to set enough standards and regulations that mitigate the managers' opportunistic behavior. Disclosure requirements should be enhanced as they offer more opportunities for managers to manipulate earnings.
- Regulators should refine monitoring systems; more aggressive penalties must be imposed on firms involved in EM practices.
- Investors should make a comprehensive review of financial statements items before making investment decisions, as well they should be cautious and aware of the items disclosed individually within the financial statement instead of depending mainly on aggregate figures for portfolio evaluation.
- Auditors should be more vigilant about the seriousness of earnings manipulation to search for more relevant techniques that better detect and limit such practices.
- Finally, future research should pay great attention to CS to get a deeper understanding of EM in order to suggest more remedies for such practices.

• 5.4 Future Research

The current study has some limitations that can be considered as suggestions for future research. The sample employed in this study is limited to nonfinancial

Egyptian firms, using just five years as a study period. As such, future research can extend the study sample to include banks and financial firms, moreover, using a longer period may lead to different results.

The study depends mainly on the manipulation of gross profit rather than core earnings to investigate the existence of income classification shifting practices. However, there are many additional forms of income classification shifting for inflating core earnings (operating profit) that future studies might investigate such as misclassifying core expenses as income decreasing special items or non-operating expense and misclassifying non-operating revenues as total revenues.

Considering the potential impact of adopting IFRS on the earnings management practices can be addressed by future research to investigate the changes in Egyptian firms' behavior pre and post IFRS implementation.

Another avenue for future research would be to empirically explore the factors that could affect managers' choice of an EM technique such as audit quality, corporate governance, accounting standards, firm life cycle, gender diversity, and managerial ability. More future studies are needed to explore this topic in other countries and economies.

References

Abba, H.I., Lawal, L.O., ACA, P. & Sadah, A.A. (2020). Financial determinants of firm value of listed deposit money banks in Nigeria. ***Al-Hikmah Management Review,*** *5*(1), 87-102.

Abbas, A., & Ayub, U. (2019). Role of earnings management in determining firm value: An emerging economy perspective. ***International Journal of Advanced and Applied Sciences***, *6*(6), 103-116.

Abdelwahed, G.M.M. (2018). ***Earnings managements constraints in the UK*** **and** ***the US***. Doctoral dissertation, Portsouth University.

Abdl Kader, A. M. A. (2020). ***A proposed framework for constraining earnings*** management ***based on factors affecting the choice of earnings management approach: An Empirical study on Egyptian listed firms***. Doctoral dissertation, Alexandria University. (*In Arabic*)

Abdou, H. A., Ellelly, N. N., Elamer, A. A., Hussainey, K., & Yazdifar, H. (2021). Corporate governance and earnings management nexus: Evidence from the UK and Egypt using neural networks. ***International Journal of Finance & Economics***, *26*(4), 6281-6311.

Abdullah, S. N., & Ismail, K. N. I. K. (2016). Women directors, family ownership and earnings management in Malaysia. ***Asian Review of Accounting***, *24*(4), 525-550.

Abernathy, J. L., Beyer, B., & Rapley, E. T. (2014). Earnings management constraints and classification shifting. ***Journal of Business Finance & Accounting***, 41(5-6), 600-626.

Abiahu, M. F. C., Egbunike, A., Udeh, D. F., Francis, E., & Amahalu, N. (2020). Corporate life cycle and classification shifting in financial statements: Evidence from quoted manufacturing firms in Nigeria. ***Amity Business Review***, *20*(2), 75-91.

Abo Taleb, K. A. M (2022). The impact of earnings management practices on the stock market return with an empirical study on firms listed in Egyptian stock exchange. ***Journal of Accounting Research,*** 9(2), 614-655. (*In Arabic*)

Adel, O.M.S. (2019). ***The effect of earnings management on the relationship*** between ***financing decisions and stock liquidity in the Egyptian stock exchange: An applied study***. Mater thesis, Ain Shams University.

Afridi, F. A., Khan, Y., Zafar, S., & Ayaz. B. (2022). The effect of firm size, investment opportunity set, and capital structure on firm value. ***International Journal of Social Sciences and Entrepreneurship,*** *2*(2), 32-46.

Afza, T., & Nazir, M. S. (2014). Audit quality and firm value: A case of Pakistan. ***Journal of Applied Sciences, Engineering and Technology***, 7(9), 1803-1810.

Aggarwal, D., & Padhan, P. C. (2017). Impact of capital structure on firm value: Evidence from Indian hospitality industry. ***Theoretical Economics Letters***, 7(4), 982-1000.

Aghab, E. (2020). ***The impact of corporate governance on the trade-off between accrual-based and real-based earnings management in the UK and India.*** Doctoral dissertation, Central Lancashire University.

Agung, G., Hasnawati, S., & Huzaimah, R. F. (2021). The effect of investment decision, financing decision, dividend policy on firm value. ***Jurnal Bisnis dan Manajemen***, 17(1), 1-12.

Ahmad, L., Suhara, E., & Ilyas, Y. (2016). The effect of audit quality on earning management within manufacturing companies listed on Indonesian stock exchange. ***Research Journal of Finance and Accounting***, 7(8), 132-138.

Ahmad, M., Anjum, T., & Azeem, M. (2014). Investigating the impact of corporate governance on earning management in the presence of firm size; evidence from Pakistan. ***International Interdisciplinary Research Journal***, 3(2), 84-90.

Ahmed, M., A. (2013). ***Evaluating the impact of the human capital on the value of the firm: An Empirical study on family firms***. Doctoral dissertation, Menoufiya University. (In Arabic)

Al Madbouly, D. M. K. (2021). The effect of financial leverage and other firm's characteristics on real earnings management and accrual earnings management: An empirical study on listed firms in the Egyptian Stock Exchange. ***Scientific Journal for Financial and Commercial Studies and Research***, 2(1), 54-126.

References

Al-Absy, M. S. M., Ismail, K. N. I. K., & Chandren, S. (2018). Board chairmen's involvement in the nomination and remuneration committees and earnings management. ***Australasian Accounting, Business and Finance Journal***, 12(4), 60-76.

Al-Absy, M., Ismail, K. N. I. K., & Chandren, S. (2021). The association between real activities and accruals earnings management in Malaysian listed companies. ***Contaduría y Administración***, 66(3), 1-31.

Alfadhael, S., & Jarraya, B. (2021). Earnings management and its applications in Saudi Arabia context: Conceptual framework and literature review. ***Accounting***, 7(5), 993-1008.

Alfonso, E., Cheng, C. A., & Pan, S. (2015). Income classification shifting and mispricing of core earnings. ***Journal of Accounting, Auditing & Finance***, 1-32.

Alghemary, M., Al-Najjar, B., & Polovina, N. (2024). What do we know about real earnings management in the GCC?. ***Journal of Accounting in Emerging Economies***, 14(3), 659-691.

Alhabsji, T., Rahayu, S. M., & Handayani, S. R. (2017). Effect of growth, liquidity, business risk and asset usage activity, toward capital structure, financial performance and corporate value (Study at Manufacturing Companies Listed in Indonesia Stock Exchange in 2010-2015). ***European Journal of Business and Management***, 9(24), 9-25.

Alhadab, M. M. S. (2012). ***Real and accrual earnings management, regulatory environments, audit quality and IPO failure risk***. Doctoral dissertation, Leeds University.

Al-Haddad, L., & Whittington, M. (2019). The impact of corporate governance mechanisms on real and accrual earnings management practices: evidence from Jordan. ***Corporate Governance: The International Journal of Business in Society***, 19(6), 1167-1186.

Ali, B., & Kamardin, H. (2018). Real earnings management: a review of literature and future research. ***Asian Journal of Finance & Accounting***, 10(1), 440-456.

Ali, N. K. A. M. (2021). ***The impact of board characteristics on the relationship between real earnings management and cost of equity; An applied study***. Master thesis, Ain Shams University.

References

Ali, U., Noor, M., Khurshid, M. K., & Mahmood, A. (2015). Impact of firm size on earnings management: A study of textile sector of Pakistan. European ***Journal of Business and Management***, 7(28), 47-56.

Almaleeh, N. M. S. (2019). Motivations for classification shifting: A systematic review. ***Asian Journal of Finance & Accounting***, 11(1), 29-40.

Almaleeh, N. M. S., Elkholy, O. A. A., & Al-Shafei, M. M. R. (2023). The Tradeoff between Accruals and Real Activities Earnings Management by More Capable Managers: Evidence from Egypt. ***Scientific Journal for Commerce and Finance***, 43(3), 74-110.

Almand, A., Cantrell, B., & Dickinson, V. (2023). Accruals and firm life cycle: Improving regulatory earnings management detection. ***Advances in Accounting***, 60, 100642.

Almari, M. O. S., Weshah, S. R. S., Saleh, M. M. A., Aldboush, H. H. H., & Ali, B. J. (2021). Earnings management, ownership structure and the firm value: An empirical analysis. ***Journal of Management Information and Decision Sciences***, 24(7), 1-14.

Al-Nsour. O. J. (2020). Earning management and its impact on firms value (An Empirical Study on the Jordanian Industrial Public Shareholding Companies listed in Amman Stock Exchange). ***Al-Mithqal for Economic& Administrative Science***. 6(1), 41-72. (In Arabic)

Alquhaif, A. S., Latif, R. A., & Chandren, S. (2017). Women in board of directors and real earnings management: Accretive share buyback in Malaysia. ***Asian Journal of Finance & Accounting***, 9(2), 48-59.

Al-Sayed, M.S.H. (2021). ***The Impact of sticky cost behavior on earnings quality of the listed companies in the Egyptian exchange: An Empirical Study***. Doctoral dissertation, Menoufia University.

Alsharairi, M., Khamis, R., & Alkhalaileh, M. (2020). How does real earnings management affect firms' future profitability? Evidence from Jordan. ***Corporate Ownership & Control***, 18(1), 47-55.

Al-Shattarat, B. (2017). ***Real earnings management activities, meeting earnings benchmarks and future performance: UK evidence***, Doctoral dissertation, Plymouth University.

References

Alsoboa, S. S. (2019). Investigating the relationships between the firm value and real earnings management in Jordanian industrial listed firms. ***16th International Conference on Accounting***, 18.

Aly, D. A. M., & Nguyen, N. B. (2013). The appropriateness of international financial reporting standards (IFRS) to Vietnam. ***Scientific Journal for Economic & Commerce***, 2, 1-48.

Alzoubi, E. S. S. (2016). Audit quality and earnings management: evidence from Jordan. ***Journal of Applied Accounting Research***, 17(2), 170-189.

Alzoubi, E. S. S. (2018). Audit quality, debt financing, and earnings management: Evidence from Jordan. ***Journal of International Accounting, Auditing and Taxation***, 30, 69-84.

Ameria, H. E. S. A. (2020). ***Examining the relationship between political connections, related party transactions and firm value in Egyptian listed companies: Empirical study***. Master Thesis, Cairo University.

Anagnostopoulou, S. C., & Tsekrekos, A. E. (2017). The effect of financial leverage on real and accrual-based earnings management. ***Accounting and Business Research***, 47(2), 191-236.

Anagnostopoulou, S. C., Gounopoulos, D., Malikov, K., & Pham, H. (2021). Earnings management by classification shifting and IPO survival. ***Journal of Corporate Finance***, 66, 101796.

Anthonius & Murwaningsari, E. (2018). The analysis of earnings management with classification shifting by using discontinued operations in Indonesia. ***Journal of Business and Retail Management Research***, 12(4), 108-116.

Anton, S. G., & Carp, M. (2020). The effect of discretionary accruals on firm growth. Empirical evidence for SMEs from emerging Europe. ***Journal of Business Economics and Management***, 21(4), 1128-1148.

Antoro, W., Sanusi, A., & Asih, P. (2020). The effect of profitability, company size, company growth on firm value through capital structure in food and beverage companies on the Indonesia stock exchange 2014-2018 period. ***International Journal of Advances in Scientific Research and Engineering***, 6(9), 36-43.

Arun, T. G., Almahrog, Y. E., & Aribi, Z. A. (2015). Female directors and earnings management: Evidence from UK companies. ***International Review of Financial Analysis***, 39, 137-146.

References

Asteriou, D., & Hall, S. G. (2007). ***Applied econometrics***. Palgrave Macmillan.

Atiningsih, S., & Izzaty, K. N. (2021). The effect firm size on company value with profitability as intervening variable and dividend policy as moderating variable. ***International Journal of Economics, Business and Accounting Research***, 5(4), 378-388.

Attia, E. F. A. (2016). ***Evaluating the relationship between ownership structure as corporate governance mechanism and accounting earnings management tools on the financial performance: A case of Egypt***. Doctoral dissertation, Cairo University.

Awuye, I. S., & Aubert, F. (2022). The impact of leverage on earnings management and the trade-off between discretionary accruals and real earnings management. ***Journal of Accounting and Taxation***, 14(1), 89-101.

Azzam, M., AlQudah, A., Al-Haddad, L., & Haija, A. A. (2021). The trade-off between accrual-based and real earnings management: evidence from Jordan. ***International Journal of Accounting, Auditing and Performance Evaluation***, 17(1-2), 73-92.

Badruzaman, J., Ridho, A., & Saputra, J. (2019). Analysis of asset growth and profit growth through supply chain management toward company value. ***International Journal of Supply Chain Management***, 8(5), 336-348.

Baik, B., Cho, H., Choi, W., & Lee, K. (2016). Who classifies interest payments as financing activities? An analysis of classification shifting in the statement of cash flows at the adoption of IFRS. ***Journal of Accounting and Public Policy***, 35(4), 331-351.

Bandanuji, A., & Khoiruddin, M. (2020). The effect of business risk and firm size on firm value with debt policy as intervening variable. ***Management Analysis Journal***, 9(2), 200-210.

Bansal, M. (2021). Impact of corporate life cycle on misclassification practices: evidence from IFRS adoption in India. ***Journal of Applied Accounting Research***, 23(3), 628-649.

Bansal, M. (2024). Do shifting practices vary across the firm life cycle?. ***Australian Journal of Management***, 49(2), 142-169.

References

Bansal, M., Kumar, A., & Badhani, K. N. (2021). Do Indian firms engage in classification shifting to report inflated core earnings?. ***Managerial Finance***, 47(11), 1533-1552.

Bansal, M., Kumar, A., Bhattacharyya, A., & Bashir, H. A. (2023). Predictors of revenue shifting and expense shifting: Evidence from an emerging economy. ***Journal of Contemporary Accounting & Economics***, 19(1), 100339.

Bartov, E., Gul, F. A., & Tsui, J. S. (2000). Discretionary-accruals models and audit qualifications. ***Journal of Accounting and Economics***, 30(3), 421-452.

Barua, A., Lin, S., & Sbaraglia, A. M. (2010). Earnings management using discontinued operations. ***The Accounting Review***, 85(5), 1485-1509.

Barus, I. S. L., Sarumpaet, T. L., Edison, A., Maisyarah, R., & Pulungan, E. (2019). Relationship between Leverage and Firm Size Toward to Real Earning Management (Unit Analysis of Mining Company Indonesia Exchange Stock Period 2012 Until 2015). ***Journal of Reviews on Global Economics***, 8, 672-687.

Bassiouny, S. W. (2016). The impact of firm characteristics on earnings management: an empirical study on the listed firms in Egypt. ***Journal of Business and Retail Management Research***, 10(3), 91-101.

Behery, A. H., Abulezz, M. E., Boghdady, A. B. A. (2019). Corporate governance and overproduction for state owned and privately owned companies .***Journal for Commercial Research***, 41(3), 3-26.

Birk, M. (2018). ***Empirical research on financial notes to the accounts and earnings management***. Doctoral dissertation, Gloucestershire University.

Boghdady, A. B. A. (2019). ***The impact of corporate governance mechanisms on real and accrual earnings management in the state-owned versus privately-owned companies: Empirical evidence from Egypt***. Doctoral dissertation, Zagazig University.

Burnett, B. M., Cripe, B. M., Martin, G. W., & McAllister, B. P. (2012). Audit quality and the trade-off between accretive stock repurchases and accrual-based earnings management. ***The accounting review***, 87(6), 1861-1884.

Chae, S. J., & Nakano, M. (2015). The effect of classification shifting on analyst forecast accuracy: Evidence from Japan. ***Hitotsubashi Journal of Commerce and Management***, 49, 25-35.

Chakroun, S., Ben Amar, A., & Ben Amar, A. (2022). Earnings management, financial performance and the moderating effect of corporate social responsibility: evidence from France. ***Management Research Review***, 45(3), 331-362.

Challen, A. E., & Siregar, S. V. (2012). Audit quality on earnings management and firm value. ***Jurnal Keuangan dan Perbankan***, 14(1), 30-43.

Chang, C. C., Kao, L. H., & Chen, H. Y. (2018). How does real earnings management affect the value of cash holdings? Comparisons between information and agency perspectives. ***Pacific-Basin Finance Journal***, 51, 47-64.

Chang, L. L., Guan, Y. D., & Hsiao, D. F. (2017). An exploratory study of female directors' constraint on real earnings management. ***Corporate Management Review***, 37(2), 31-85.

Charfeddine, C., Riahi, R., & Omri, A. (2013). The determinants of earnings management in developing countries: A study in the Tunisian context. ***The IUP Journal of Corporate Governance***, 12(1), 35-49.

Chay, J. B., Kim, H., & Suh, J. (2015). Firm age and valuation: Evidence from Korea. ***Asia-Pacific Journal of Financial Studies***, 44(5), 721-761.

Chen, C. L., Huang, S. H., & Fan, H. S. (2012). Complementary association between real activities and accruals-based manipulation in earnings reporting. ***Journal of Economic Policy Reform***, 15(2), 93-108.

Chen, L. (2020). ***The Effect of Investor Sentiment on Earnings Management***. Doctoral dissertation, Texas University.

Chi, W., Lisic, L. L., & Pevzner, M. (2011). Is enhanced audit quality associated with greater real earnings management?. ***Accounting Horizons***, 25(2), 315-335.

Choi, A., Choi, J. H., & Sohn, B. C. (2018). The joint effect of audit quality and legal regimes on the use of real earnings management: International evidence. ***Contemporary Accounting Research***, 35(4), 2225-2257.

References

Choi, J. J., Mao, C. X., & Upadhyay, A. D. (2015). Earnings management and derivative hedging with fair valuation: Evidence from the effects of FAS 133. ***The Accounting Review***, 90(4), 1437-1467.

Chowdhury, S. N., & Eliwa, Y. (2021). The impact of audit quality on real earnings management in the UK context. ***International Journal of Accounting & Information Management***, 29(3), 368-391.

Chung, H., Kim, B. J., Lee, E. Y., & Sunwoo, H. Y. (2021). Debt financing and classification shifting of private firms. ***Managerial Auditing Journal***, 36(7), 921-950.

Cohen, D. A., & Zarowin, P. (2010). Accrual-based and real earnings management activities around seasoned equity offerings. ***Journal of Accounting and Economics***, 50(1), 2-19.

Cohen, D. A., Dey, A., & Lys, T. Z. (2008). Real and accrual-based earnings management in the pre-and post-Sarbanes-Oxley periods. ***The Accounting Review***, 83(3), 757-787.

Cohen, D., Pandit, S., Wasley, C. E., & Zach, T. (2020). Measuring real activity management. ***Contemporary Accounting Research***, 37(2), 1172-1198.

Cordiaz, E. F., & Erlina, C. S. (2021). The effect of profitability, liquidity, and capital structure on firm value with dividend policy as a moderating variable in the company mining listed on the Indonesia stock exchange. ***International Journal of Research and Review***, 8(12), 328-337.

Darmawan, I. P. E., Sutrisno, T., & Mardiati, E. (2019). Accrual Earnings Management and Real Earnings Management: Increase or Destroy Firm Value?. ***International Journal of Multicultural and Multireligious Understanding***, 6(2), 8-19.

Darrough, M., Lee, Y. G., & Oh, H. I. (2019). Classification shifting within non-recurring items. ***Asia-Pacific Journal of Accounting & Economics***, 26(3), 185-206.

DeAngelo, L. E. (1986). Accounting numbers as market valuation substitutes: A study of management buyouts of public stockholders. ***The Accounting Review***, 61(3), 400-420.

Dechow, P. M., & Dichev, I. D. (2002). The quality of accruals and earnings: The role of accrual estimation errors. ***The Accounting Review***, 77, 35-59.

Dechow, P. M., Sloan, R. G., & Sweeney, A. P. (1995). Detecting earnings management. ***The Accounting Review***, 70(2), 193-225.

Deef, A. E. T. E. (2022). The impact of the relationship between managerial ability and earnings management by classification shifting on transparency of financial reporting and firm value: An empirical evidence from the Saudi capital market. ***Journal of Accounting Research***, 9(2), 1-31.

Desoky, A. M. (2016). The impact of cash dividends and firm factors on earnings management–Empirical evidence from Bahrain. ***Journal of Accounting Research***, 3(2), 1-26.

Dewi, N. W. S. K., Ratnadi, N. M. D., Yadnyana, I. K., & Suaryana, I. G. N. A. (2022). Selection of profit management strategy: Testing at the company life cycle stage. ***Linguistics and Culture Review***, 6(1), 530-549.

Dewiningrat, A. I., & Baskara, I. G. K. (2020). Does dividend policy moderate the relationship between profitability, IOS, and liquidity toward firm value. ***American Journal of Humanities and Social Sciences Research***, 4(7), 49-52.

Doukakis, L. C. (2014). The effect of mandatory IFRS adoption on real and accrual-based earnings management activities. ***Journal of Accounting and Public Policy***, 33(6), 551-572.

Drukker, D. M. (2003). Testing for serial correlation in linear panel-data models. ***The stata journal***, 3(2), 168-177.

Duong, L., & Evans, J. (2016). Gender differences in compensation and earnings management: Evidence from Australian CFOs. ***Pacific-Basin Finance Journal***, 40, 17-35.

Durana, P., Michalkova, L., Privara, A., Marousek, J., & Tumpach, M. (2021). Does the life cycle affect earnings management and bankruptcy?. ***Oeconomia Copernicana***, 12(2), 425-461.

Ebaid, I. E. S. (2016). International accounting standards and accounting quality in code-law countries: The case of Egypt. ***Journal of Financial Regulation and Compliance***, 24(1), 41-59.

Eilifsen, A., & Knivsflå, K. H. (2021). Core earnings management: How do audit firms interact with classification shifting and accruals management?. ***International Journal of Auditing***, 25(1), 142-165.

References

EL Deeb, M. S., & Ramadan, M. S. (2020). The impact of financial distress, firm size, and audit quality on earnings' management evidence from companies listed in the Egyptian Stock Exchange. ***Alexandria Journal of Accounting Research***, 4(1), 1-46.

El Diri, M. (2018). ***Introduction to earnings management***. Springer. Available at: http://dx.doi.org/10.1007/978-3-319-62686-4

El Diri, M., Lambrinoudakis, C., & Alhadab, M. (2020). Corporate governance and earnings management in concentrated markets. ***Journal of Business Research***, 108, 291-306

El Garf, Y. A. E. M. The impact of the Egyptian accounting standards developments on earnings management: An empirical study on listed companies In Egyptian stock exchange. ***Journal of Accounting Research***, 2(4), 62-136. (In Arabic)

El Madbouly, D. M. (2022). The impact of corporate governance, firm characteristics on voluntary electronic social media disclosure during the COVID 19 global crisis: An empirical study on listed firms in the Egyptian stock of exchange. ***Alexandria Journal of Accounting Research***, 6(1), 1-57.

El Shafey, S. F. M. A. (2020). ***Earnings management under information asymmetry at initial public offering in Egypt: An empirical study***. Master thesis, Tanta University.

Elhabashy, H. A., & El-Kelety, I. A. (2023). The impact of institutional ownership and controlling shareholder on the relationship between CEO power and earnings quality in Egypt. ***Scientific Journal for Commercial Research***, 4(1), 9-54.

Elhabashy, H. A., Elkholy, O. A. A., & Eid, A. L. (2023). The impact of managerial ability and business strategy on real earnings management: Evidence from Egypt. **Journal of Accounting Research**, 10(2), 55-107.

ElHawary, E., & Hassouna, D. (2021). Earnings management determinants: a study of Egyptian listed firm characteristics post the Egyptian revolution. ***Corporate Governance and Organizational Behavior Review***, 5(2), 165-183.

Elkalla, T. (2021). Do firms substitute earnings management methods? The case of the MENA region. **Journal of Research in Emerging Markets**, 3(1), 32-43.

ElSayed, M. S. H. (2019). The impact of company characteristics on real earnings management practices and their reflection on future operational performance. ***Scientific Journal for Commercial Research***, 32(1), 9-84. (In Arabic)

Elsayed, M. S. H. (2021). The Impact of debt structure on future stock price crash risk: Evidence from Egypt. ***Alexandria Journal of Accounting Research***, 5(1), 1-46.

Elsayed, M. S. H., Elkholy, O. A. A. A. A., & Marzouk, M. (2023). The impact of financial statement comparability on income classification shifting: Evidence from Egypt. ***Journal of Accounting Research***, 10(3),1-29.

Evans, M. E., Houston, R. W., Peters, M. F., & Pratt, J. H. (2015). Reporting regulatory environments and earnings management: US and non-US firms using US GAAP or IFRS. ***The Accounting Review***, 90(5), 1969-1994.

Ezat, A. N. M. (2015). Financial statements timeliness and real earnings management in the Egyptian context. ***Scientific Journal for Commerce and Finance***, 2(35), 2-56.

Ezat, A. N. M. (2021). Gross profit manipulation through classification shifting: Evidence from the Egyptian environment. ***Scientific Journal for Commercial and Environmental Studies***, 12(4), 185-240.

Ezat, A. N. M. (2021). The Relationship between earnings benchmarks and earnings management across the stages of a listed Egyptian firm's life cycle. ***Scientific Journal for Accounting Studies***, 3(4), 1-87.

Fajaria, A. Z., & Isnalita, N. I. D. N. (2018). The effect of profitability, liquidity, leverage and firm growth of firm value with its dividend policy as a moderating variable. ***International Journal of Managerial Studies and Research***, 6(10), 55-69.

Fan, Y., & Liu, X. (2017). Misclassifying core expenses as special items: Cost of goods sold or selling, general, and administrative expenses?. ***Contemporary Accounting Research***, 34(1), 400-426.

Fan, Y., Barua, A., Cready, W. M., & Thomas, W. B. (2010). Managing earnings using classification shifting: Evidence from quarterly special items. ***The Accounting Review***, 85(4), 1303-1323.

Fan, Y., Thomas, W. B., & Yu, X. (2019). The impact of financial covenants in private loan contracts on classification shifting. ***Management Science***, 65(8), 3637-3653.

Ferentinou, A. C., & Anagnostopoulou, S. C. (2016). Accrual-based and real earnings management before and after IFRS adoption: The case of Greece. ***Journal of Applied Accounting Research***, 17(1), 2-23.

Fields, T. D., Lys, T. Z., & Vincent, L. (2001). Empirical research on accounting choice. ***Journal of Accounting and Economics***, 31(1-3), 255-307.

Francis, J., LaFond, R., Olsson, P., & Schipper, K. (2005). The market pricing of accruals quality. ***Journal of Accounting and Economics***, 39(2), 295-327.

Gao, J., Gao, B., & Wang, X. (2017). Trade-off between real activities earnings management and accrual-based manipulation-evidence from China. ***Journal of International Accounting, Auditing and Taxation***, 29, 66-80.

Garanina, T. (2024). CSR disclosure and state ownership: implications for earnings management and market value. ***Journal of Accounting in Emerging Economies***, 14(3), 513-547.

Ge, W., & Kim, J. B. (2014). Boards, takeover protection, and real earnings management. ***Review of Quantitative Finance and Accounting***, 43, 651-682.

Gharaibeh, A. M. O., & Qader, A. A. A. A. (2017). Factors influencing firm value as measured by the Tobin's Q: Empirical evidence from the Saudi Stock Exchange (TADAWUL). ***International Journal of Applied Business and Economic Research***, 15(6), 333-358.

Gill, A., Biger, N., Mand, H. S., & Mathur, N. (2013). Earnings management, firm performance, and the value of Indian manufacturing firms. ***International Research Journal of Finance and Economics***, 116(1), 121-131.

Graham, J. R., Harvey, C. R., & Rajgopal, S. (2005). The economic implications of corporate financial reporting. ***Journal of Accounting and Economics***, 40(1-3), 3-73.

Gunny, K. A. (2010). The relation between earnings management using real activities manipulation and future performance: Evidence from meeting earnings benchmarks. ***Contemporary Accounting Research***, 27(3), 855-888.

Ha, K., & Thomas, W. B. (2019). Classification shifting and earnings predictability. Working paper, Available at https://dx.doi.org/10.2139/ssrn.3299152

Ha, N. T. L., & Minh, B. T. (2020). Determinants of firm value in Vietnam: A research framework. ***International Journal of Science and Research***, 9(1), 626-631.

Habbash, M., & Alghamdi, S. (2015). The perception of earnings management motivations in Saudi public firms. ***Journal of Accounting in Emerging Economies***, 5(1), 122-147.

Habib, A. (2004). Impact of earnings management on value-relevance of accounting information: empirical evidence from Japan. ***Managerial Finance***, *30*(11), 1-15.

Habib, A., & Hasan, M. M. (2019). Corporate life cycle research in accounting, finance and corporate governance: A survey, and directions for future research. ***International Review of Financial Analysis***, 61, 188-201.

Habib, A., Ranasinghe, D., Wu, J. Y., Biswas, P. K., & Ahmad, F. (2022). Real earnings management: A review of the international literature. ***Accounting & Finance***, 62(4), 4279-4344.

Hammad, M. A. M. A. (2017). The effect of using real earnings management (REM) activities on the subsequent operating performance of Egyptian companies: An applied study. ***Accounting Thoughts***, 21(1), 10-62. (In Arabic)

Handayani, Y. D., & Ibrani, E. Y. (2020). The effect of audit committee characteristics on earnings management and its impact on firm value. ***International Journal of Commerce and Finance***, 6(2), 104-116.

Hasan, M. M., Hossain, M. S., & Gotti, G. (2022). Product market competition and earnings management: the role of managerial ability. ***Review of Accounting and Finance***, 21(5), 486-511.

Hasan, M. T., & Rahman, A. A. (2017). IFRS Adoption and earnings management: A review and justification of earnings management model

for developing countries. ***ELK Asia Pacific Journal of Finance and Risk Management***, 8(3), 43-60.

Hasan, M. T., Hossain, M. K., Rekabder, M. S., Molla, M. S., & Ashif, A. S. M. (2022). IFRS adoption and real earnings management in Bangladesh: The role of board characteristics. ***Cogent Business & Management***, 9(1), 1-20.

Hassaan, M. H. M., & Abdallah, W. S. M. (2024). Management Inducements for Cash Flows Classification Shifting in Egyptian Companies: Analysis of Core Operating Cash Flows Inflation Strategies—An applied study. ***Journal of Accounting Research,*** 11(2), 1-43.

Hassanein, A. (2015). ***Informativeness of unaudited forward-looking financial disclosure: evidence from UK narrative reporting***. Doctoral Dissertation, Plymouth University.

Healy, P. M. (1985). The effect of bonus schemes on accounting decisions. ***Journal of Accounting and Economics***, 7(1-3), 85-107.

Healy, P. M., & Wahlen, J. M. (1999). A review of the earnings management literature and its implications for standard setting. ***Accounting Horizons***, 13(4), 365-383.

Hendawy, A. M. A. (2020). ***The relationship between corporate social responsibility and information asymmetry and its impact on firm value***. Master Thesis, Kafr El Sheikh University. (In Arabic)

Hirdinis, M. (2019). Capital structure and firm size on firm value moderated by profitability. ***International Journal of Economics and Business Administration***, 2(1), 174-191.

Ho, L. C. J., Liao, Q., & Taylor, M. (2015). Real and accrual-based earnings management in the pre-and post-IFRS periods: Evidence from China. ***Journal of International Financial Management & Accounting***, *26*(3), 294-335.

Hoang, K. M. T., & Phung, T. A. (2019). The effect of financial leverage on real and accrual-based earnings management in Vietnamese firms. ***Economics & Sociology***, 12(4), 299-333.

Huang, X. S., & Sun, L. (2017). Managerial ability and real earnings management. ***Advances in Accounting***, 39, 91-104.

Hussain, A., Akbar, M., Kaleem Khan, M., Akbar, A., Panait, M., & Catalin Voica, M. (2020). When does earnings management matter? Evidence across the corporate life cycle for non-financial Chinese listed companies. ***Journal of Risk and Financial Management***, 13(12), 313.

Hussien, M. S. A. (2023). The impact of managerial ability and managerial overconfidence on the relationship between real earnings management and firm value. An experimental study on industrial firms listed in the Egyptian Stock Exchange. ***Journal of Accounting Research***, 10(3), 75-155. (In Arabic)

Hwang, J., Choi, S., Choi, S., & Lee, Y. G. (2022). Corporate social responsibility and classification shifting. ***Journal of Accounting and Public Policy***, 41(2), 106918.

Ibrahim, A. E. A. (2012). ***Investigating impact of historical costing on real earnings management: An Empirical study***. Master thesis, Cairo University.

Idris, M. I. (2012). ***The impact of ownership structure and external audit on accruals and real activities earnings management in Jordan***. Doctoral dissertation, Gloucestershire University.

Imeni, M., Fallah, M., & Edalatpanah, S. A. (2021). The effect of managerial ability on earnings classification shifting and agency cost of Iranian listed companies. ***Discrete Dynamics in Nature and Society***, 2021, 1-10.

Inoue, S. (2021). Classification shifting using discontinued operations and impact on core earnings: evidence from Japan. ***Journal of Financial Reporting and Accounting***, 19(2), 211-233.

International Accounting Standards Board (IASB). (2018). ***Conceptual framework for financial reporting, London***. Available at: http :// www. ifrs.org.

Ipino, E., & Parbonetti, A. (2017). Mandatory IFRS adoption: the trade-off between accrual-based and real earnings management. ***Accounting and Business Research***, 47(1), 91-121.

Ismail, M. A. (2017). Real earnings management versus accrual-based and its effect on firm performance: Evidence from Egypt. ***Alexandria Journal of Accounting Research***, 1(2), 1-38.

Ismail, M. A. A. (2020). Effect of audit quality on earnings quality given firms operational characteristics: An empirical study on companies listed on the Egyptian Stock Exchange. ***Alexandria Journal of Accounting Research***, 4(1), 1-44.

Iswajuni, I., Manasikana, A., & Soetedjo, S. (2018). The effect of enterprise risk management (ERM) on firm value in manufacturing companies listed on Indonesian Stock Exchange year 2010-2013. ***Asian Journal of Accounting Research***, 3(2), 224-235.

Jaafar, H. (2011). ***Accounting for intangible assets, firm life cycle and the value relevance of intangible assets***. Doctoral dissertation, Tasmania University.

Jackson, A. B. (2018). Discretionary accruals: earnings management... or not?. Journal of Accounting, ***Finance and Business Studies***, 54(2), 136-153.

Jevasuwan, S. (2018). ***Essays on earnings management in response to natural disasters***. Doctoral dissertation, Exeter University.

Jiraporn, P., Miller, G. A., Yoon, S. S., & Kim, Y. S. (2008). Is earnings management opportunistic or beneficial? An agency theory perspective. ***International Review of Financial Analysis***, 17(3), 622-634.

Joo, J. H., & Chamberlain, S. L. (2017). The effects of governance on classification shifting and compensation shielding. ***Contemporary Accounting Research***, 34(4), 1779-1811.

Kamarudin, K. A., & Ismail, W. A. W. (2014). The risk of earnings quality impairment**.** ***Procedia-Social and Behavioral Sciences***, 145, 226-236.

Kang, S. A. & Kim, T. (2019). A research on the relationship between accrual-based earnings management and real earnings management in the retail industry. ***Journal of Distribution Science***, *17*(12), 5-12.

Keung, E., & Shih, M. S. (2014). Measuring discretionary accruals: Are ROA-matched models better than the original Jones-type models?. ***Review of Accounting Studies***, 19, 736-768.

Khaled, B. K. M. (2018). ***The Relationship between CSR Disclosure Quality and Accrual and Real Earnings Management: Large-Scale Evidence from India***. Doctoral dissertation, Central Lancashire University.

Khalil, M. M. M. (2010). ***Earnings management, agency costs and corporate governance: Evidence from Egypt***. Doctoral dissertation, Hull University.

Khanh, H. T. M., & Nguyen, V. K. (2018). Audit quality, firm characteristics and real earnings management: The case of listed Vietnamese firms. ***International Journal of Economics and Financial Issues***, 8(4), 243-249.

Khuong, N. V., Anh, L. H. T., & Van, N. T. H. (2022). Firm life cycle and earnings management: The moderating role of state ownership. ***Cogent Economics & Finance***, 10(1), 1-28.

Kolsi, M. C., & Matoussi, H. (2011). Securitisation transactions, real manipulation and stock market valuation. ***International Journal of Accounting and Finance***, 3(2), 147-164.

Kothari, S. P., Leone, A. J., & Wasley, C. E. (2005). Performance matched discretionary accrual measures. ***Journal of Accounting and Economics***, 39(1), 163-197.

Kourdoumpalou, S. (2017). Detecting earnings management: a review of the proxies. ***International Journal of Critical Accounting***, 9(2), 103-118.

Kuo, J. M., Ning, L., & Song, X. (2014). The real and accrual-based earnings management behaviors: Evidence from the split share structure reform in China. ***The International Journal of Accounting***, 49(1), 101-136.

Lakhal, F., Aguir, A., Lakhal, N., & Malek, A. (2015). Do women on boards and in top management reduce earnings management? Evidence in France. ***Journal of Applied Business Research***, 31(3), 1107-1118.

Lanouar, C., Riahi, R., & Omri, A. (2013). The determinants of earnings management in developing countries: A study in the Tunisian context. ***The IUP Journal of Corporate Governance***, 12(1), 35-49.

Lattanzio, G., & Thomas, W. B. (2020). The persistence and valuation effects of classification shifting. Available at: https://dx.doi.org/10.2139/ssrn.3384962

Lee, K. W., & Yeo, G. H. H. (2016). The association between integrated reporting and firm valuation. ***Review of Quantitative Finance and Accounting***, 47, 1221-1250.

Lee, L. F. (2012). Incentives to inflate reported cash from operations using classification and timing. ***The Accounting Review***, 87(1), 1-33.

References

Leite, R. D. O., & Carvalhal, A. L. C. D. (2016). Firm age, value, performance and corporate governance in Brazil. ***Corporate, Ownership & Control***, 13(4), 8-12.

Leuz, C., Nanda, D., & Wysocki, P. D. (2003). Earnings management and investor protection: an international comparison. ***Journal of financial economics***, 69(3), 505-527.

Li, V. (2019). The effect of real earnings management on the persistence and informativeness of earnings. ***The British Accounting Review***, 51(4), 402-423.

Li, X. D., & Guo, Y. (2018). Abnormal audit fees and earnings management using classification shifting. ***Journal of Accounting and Finance***, 18(5), 113-139.

Lindenberg, E. B., & Ross, S. A. (1981). Tobin's q ratio and industrial organization. ***Journal of Business***, 54(1), 1-32.

Liu, Y., Wei, Z., & Xie, F. (2016). CFO gender and earnings management: Evidence from China. ***Review of Quantitative Finance and Accounting***, 46, 881-905.

Lumapow, L. S. (2019). The effect of investment and non-monotonic of managerial ownership on corporate value. ***International Journal of Applied Business and International Management***, 4(1), 71-82.

Mahmoud, A. E. (2020). Classification shifting of income statement items on the accounting profits quality with An Empirical study on Egyptian Corporation. ***Alexandria Journal of Accounting Research***, 4(1), 345-417. (In Arabic)

Mahmoud, A. E. (2023). The impact of earning management practices through classification shifting of cash flow statement items on analyst forecasts accuracy with an empirical study on Egyptian corporation. Alexandria ***Journal of Accounting Research***, 7(2), 47-114. (In Arabic)

Mahmoud, E. E., Abdel-Kader, M. G., & Mahmoud, K. A. (2023). The relationship between earnings management and credit ratings and their impact on firm performance: Evidence from Egypt. ***Scientific Journal for Commercial and Environmental Studies***, 14(1), 586-640.

Mai, D. D. T. V. T., & Ngoc, T. (2021). Too big to cheat? An empirical study of the impact of firm size on earnings management in Vietnam. ***Journal of Contemporary Issues in Business and Government***, 27(5), 234-247.

Majeed, M. A., & Yan, C. (2021). Financial statement comparability, state ownership, and the cost of debt: Evidence from China. ***International Business and Finance***, *58*, 101497.

Makhaiel, N. K. B. (2019). The influence of Egyptian context on the trade-off between earnings management approaches: accounting, operational and investment. ***Journal of Financial Reporting and Accounting***, 17(1), 133-168.

Malikov, K., Coakley, J., & Manson, S. (2019). The effect of the interest coverage covenants on classification shifting of revenues. ***The European Journal of Finance***, 25(16), 1572-1590.

Malikov, K., Manson, S., & Coakley, J. (2018). Earnings management using classification shifting of revenues. ***The British Accounting Review***, 50(3), 291-305.

Marisetty, V. B., & Moturi, S. (2023). Boundaries of the firm and real earnings management. ***Journal of International Accounting, Auditing and Taxation***, 52, 100567.

Marpaung, N, Yahya, I., & Sadalia, I. (2022). The effect of liquidity, profitability, capital structure, asset growth, and firm size on the firm value with dividend policy as a moderating variable in food and beverage sub-sector of manufacturing companies listed on the Indonesia Stock Exchange. ***International Journal of Research and Review***, 9(7), 268-282.

McNichols, M. F. (2002). Discussion of the quality of accruals and earnings: The role of accrual estimation errors. ***The Accounting Review***, 77, 61-69.

McVay, S. E. (2006). Earnings management using classification shifting: An examination of core earnings and special items. ***The Accounting review***, 81(3), 501-531.

Mellado-Cid, C., Jory, S. R., & Ngo, T. N. (2018). Real activities manipulation and firm valuation. ***Review of Quantitative Finance and Accounting***, 50, 1201-1226.

Miletic, S., & Vuckovic, M. S. (2020). An analysis of accrual-based earnings management in large Serbian companies. ***Ekonomika preduzeća***, *68*(7-8), 469-483.

Mnif, Y., & Ben Hamouda, A. (2021). Audit quality and the trade-off between real and accrual earnings management in the oil and gas industry: the GCC evidence. ***Journal of Applied Accounting Research***, 22(2), 223-251.

Mohamed, G. T. (2017). ***Impact of firm characteristics on firm value of listed Healthcare firms in Nigeria***. Master thesis, Ahmadu Bello University.

Mohamed, H. A. H. (2017). ***Analytical study of the variable that affect the choice between earnings management approaches and its impact on the firms' future financial performance measures in light of accounting standers***. Doctoral dissertation, Cairo University. (In Arabic)

Mohamed, M. G. M. (2014). ***The impact of cost behavior and ownership structure on earnings management.*** Master thesis, Cairo University.

Morovvati Siboni, Z., & Pourali, M. R. (2015). The relationship between investment opportunity, dividend policy and firm value in companies listed in TSE: Evidence from IRAN. ***European Online Journal of Natural and Social Sciences***, 4(1), 263-272.

Mugera, B. W. (2013). ***The effect of foreign exchange risk management on the value of firms listed at The Nairobi Securities Exchange***. Doctoral dissertation, University of Nairobi.

Mulchandani, K., Mulchandani, K., & Jain, M. (2024). Firm's life cycle and cash flow classification: evidence from Indian firms. ***Asian Review of Accounting***, 32(3), 443-462.

Musa, A., Abdul Latif, R., & Abdul Majid, J. (2023). CEO attributes, board independence, and real earnings management: Evidence from Nigeria. ***Cogent Business & Management***, 10(1), 1-21.

Muthén, B., & Kaplan, D. (1985). A comparison of some methodologies for the factor analysis of non-normal Likert variables. ***British Journal of Mathematical and Statistical Psychology***, 38(2), 171-189.

Nagar, N., & Radhakrishnan, S. (2017). Firm life cycle and real-activity based earnings management. Working paper. Available at https://dx.doi.org/10.2139/ssrn.2701680

Nagar, N., & Raithatha, M. (2016). Does good corporate governance constrain cash flow manipulation? Evidence from India. ***Managerial Finance***, 42(11), 1034-1053.

Nagar, N., & Sen, K. (2014). Classification shifting in the cash flow statement: Evidence from India. ***Pace University Accounting Research Paper***. Available at: https://dx.doi.org/10.2139/ssrn.2710508

Nagar, N., & Sen, K. (2016). How does regulation affect the relation between family control and reported cash flows? Comparative evidence from India and the United States. ***Corporate Governance: An International Review***, 24(5), 490-508.

Nagar, N., Desai, N., & Jacob, J. (2021). Do big 4 auditors limit classification shifting? Evidence from India. ***Journal of International Accounting***, Auditing and Taxation, 42, 100376.

Naz, I., Bhatti, K., Ghafoor, A., & Khan, H. H. (2011). Impact of firm size and capital structure on earnings management: Evidence from Pakistan. ***International Journal of Contemporary Business studies***, 2(12), 22-31.

Nazir, M. S., & Afza, T. (2018). Does managerial behavior of managing earnings mitigate the relationship between corporate governance and firm value? Evidence from an emerging market. ***Future Business Journal***, 4(1), 139-156.

Noagy, H. M. (2017). The impact of audit characteristics on firm value: An applied study on nonfinancial firms listed in the stock exchange. ***Scientific Journal for Commerce and Finance***, 37(4), 134-172. (In Arabic)

Noh, M., Moon, D., & Parte, L. (2017). Earnings management using revenue classification shifting–evidence from the IFRS adoption period. ***International Journal of Accounting & Information Management***, 25(3), 333-355.

Novatiani, R. A. (2021). Analysis of financial performance affecting corporate value. ***Turkish Journal of Computer and Mathematics Education***, 12(8), 1006-1009.

Nurdiansari, R., Sriwahyuni, A., Apriani, R., & Fadhilah, N. H. K. (2022,). The effect of dividend policy, debt policy, and asset growth on firm value with

managerial ownership as moderating variables. ***International Conference on Economics, Management and Accounting***, 100-108.

Nurmadi, R., & Novietta, L. (2022). The effect of profitability and liquidity on firm value with dividend policy as moderating variable. ***Accounting and Business Journal***, 4(1), 62-74.

Olatunji, O. C., & Juwon, A. M. (2020). Accrual earnings management, real earnings management and firm's value of quoted manufacturing companies in Nigeria. ***Euro Economica***, 39(3); 119-140.

Orieby, D.A.M. (2017). ***Exploring earnings management using structured transactions under IFRS with an empirical study***. Master thesis, Tanta University.

Osman, S. S., El Awam, A. M., & Hessian, M. (2022). Determinants of classification shifting of cash flows items under IFRS and its effect on the firm value: An applied study on the listed companies on Egyptian Stock Exchange. ***Accounting Thought***, 4(26), 85-113.

Ouyang, W., Ngo, T., & Wang, H. (2023). Revenue classification shifting: Does CEO gender matter?. ***Review of Pacific Basin Financial Markets and Policies***, 26(1), 1-42.

Pan, S. (2013). ***Income classification shifting and financial analysts' forecasts***. Doctoral dissertation, Louisiana State University.

Pan, S., Lacina, M., & Shin, H. (2019). Income classification shifting and financial analysts' forecasts. ***Review of Pacific Basin Financial Markets and Policies***, 22(2), 1-48.

Poonawala, S. H., & Nagar, N. (2019). Gross profit manipulation through classification shifting. ***Journal of Business Research***, 94, 81-88.

Potharla, S., Bhattacharjee, K., & Samontaray, D. P. (2021). Real earnings management and persistence of firm value: Evidence from India. ***The Journal of Asian Finance, Economics and Business***, 8(12), 323-336.

Pratama, A. (2018). Do related party transactions and tax avoidance affect firm value?. ***Review of Integrative Business and Economics Research***, 7, 106-116.

Purwani, T., & Santoso, R. B. (2023). Analysis of firm value determination on the IDX sharia growth index. ***International Business and Accounting Research Journal***, 7(1), 90-100.

Putra, A. A., Mela, N. F., & Putra, F. (2021). Managerial ability and real earnings management in family firms. Corporate Governance: ***The International Journal of Business in Society***, 21(7), 1475-1494.

Putri, I. G. A. P. T., & Rahyuda, H. (2020). Effect of capital structure and sales growth on firm value with profitability as mediation. ***International Research Journal of Management, IT and Social Sciences***, 7(1), 145-155.

Ra, S. S. E., & Embong, Z. (2014). IFRS adoption and information quality: Evidence from emerging market. ***Asian Journal of Accounting & Governance***, 5, 37-45.

Radja, F. L., & Artini, L. G. S. (2020). The effect of firm size, profitability and leverage on firm value (Study on manufacturing companies sector consumer goods industry listed in Indonesian stock exchange period 2017-2019). ***International Journal of Economics and Management Studies***, 7(11), 18-24.

Rahayu, F. P., Wahyuni, S., Pramono, H., & Inayati, N. I. (2023). The influence of profitability, firm size, and capital structure on firm value with managerial ownership as moderation variables (Empirical study of basic material sector companies on the IDX in 2019-2021). ***Jurnal Ekonomi dan Bisnis Digital***, 2(1), 157-174.

Reschiwati, R., Budiantini, A., & Gusmiarni, G. (2021). Examining the effect of firm-specific factors on the automotive industry in Indonesia. ***International Journal of Research in Business and Social Science***, 10(3), 36-45.

Ronen, J., & Yaari, V. (2008). ***Earnings management, emerging insights in theory, practice, and research***. Springer, New York.

Roychowdhury, S. (2006). Earnings management through real activities manipulation. ***Journal of Accounting and Economics***, 42(3), 335-370.

Saadah, T. I. S. (2021). A multi-dimensional vision for modeling the mechanisms for measuring the quality of accounting earnings according to the conceptualization methodology: An extended standard study.

Scientific Journal for Financial and Commercial Studies and Research, 2(2), 1229-1313. (In Arabic)

Sajid, M., & Safdar, R. (2023). Supply chain finance, firm performance and industry competition: empirical evidence from Pakistan. ***Business Process Management Journal***, 29(7), 1989-2008.

Salah, W. (2020). The international financial reporting standards and firm performance: A systematic review. ***Applied Finance and Accounting***, 6(2), 1-10.

Santoso, A. (2023). The effect of free cash flow and leverage on earnings management: Moderating role of good corporate governance. ***Asian Management and Business Review***, 3(1), 14-23.

Schipper, K. (1989). Commentary on earnings management. ***Accounting Horizons***, 3 (4), 91-102.

Sellami, M., & Fakhfakh, H. (2013). Effect of the mandatory adoption of IFRS on real and accruals-based earnings management: Empirical evidence from France. ***International Journal of Accounting and Economics Studies***, 2(1), 22-33.

Shah, F. M., & Khalidi, M. A. (2020). Determinants of firm value in Shariah compliant companies. ***Market Forces***, 15(1), 86-100.

Shahwan, Y. (2021). The impact of audit quality, corporate governance, and company characteristics on earnings management. ***Global Journal of Economics & Business***, 11(1), 107-112.

Sherine, C., Wiyanto, H., & Budiono, H. (2022). The effect of investment decision, funding decision, and profitability on the firm value of consumer goods industry registered in Indonesia stock exchange during 2017-2020. ***Tenth International Conference on Entrepreneurship and Business Management***. Available at: https://doi.org/10.2991/aebmr.k.220501.084

Shrestha, N. (2020). Detecting multicollinearity in regression analysis. ***American Journal of Applied Mathematics and Statistics***, 8(2), 39-42.

Siboni, Z. M., & Pourali, M. R. (2015). The relationship between investment opportunity, dividend policy and firm value in companies listed in TSE: Evidence from IRAN. ***European Online Journal of Natural and Social Sciences: Proceedings,*** 4(1), 263-272.

Simamora, A. J. (2021). Managerial ability, real earnings management, and earnings quality. ***JFBA: Journal of Financial and Behavioural Accounting***, 1(1), 1-28.

Singh, K. (2021). ***Impact of Firm specific factors on firm value in Nepalese commercial banks***. Doctoral dissertation, Tribhuvan University.

Siregar, S. V., & Utama, S. (2008). Type of earnings management and the effect of ownership structure, firm size, and corporate-governance practices: Evidence from Indonesia. ***The International Journal of Accounting***, 43(1), 1-27.

Skousen, C., Sun, L., & Wu, K. (2019). The role of managerial ability in classification shifting using discontinued operations. ***Advances in Management Accounting***, 31, 113-131.

Srivastava, A. (2019). Improving the measures of real earnings management. ***Review of Accounting Studies***, 24(4), 1277-1316.

Suffian, M. T. M., Sanusi, Z. M., & Mastuki, N. A. (2015). Real earnings management and firm value: Empirical evidence from Malasyia. **Malaysian Accounting Review**, 14(1), 26-47.

Sukesti, F., Wibowo, E., & Prakasiwi, A. E. (2019). The factors that influence on firm value and company performance as mediation variables (Study manufacturing companies period 2015-2017). ***1st International Conference on Economics, Management, Accounting and Business***. Available at: http://dx.doi.org/10.4108/eai.8-10-2018.2288692

Sunardi (2018). The effect of earnings management on firm value before and when IFRS implementation, moderated life cycle company. ***Journal of Business and Economics***, 9(3), 275-285.

Susanti, N., & Restiana, N. G. (2018). What's the best factor to determining firm value. ***Jurnal Keuangan dan Perbankan***, 22(2), 301-309.

Susanto, Y. K. (2017). Accrual earnings management, real earnings management, firm value. ***International Journal of Business, Economics and Law***, 14(1), 1-6.

Suteja, J., Gunardi, A., Alghifari, E. S., Susiadi, A. A., Yulianti, A. S., & Lestari, A. (2023). Investment decision and firm value: moderating effects of corporate social responsibility and profitability of non-financial sector

companies on the Indonesia stock exchange. ***Journal of Risk and Financial Management***, 16(1), 40.

Tahu, G. P., & Susilo, D. D. B. (2017). Effect of liquidity, leverage and profitability to the firm value (dividend policy as moderating variable) in manufacturing company of indonesia stock exchange. ***Research Journal of Finance and Accounting***, 8(18), 89-98.

Thadewald, T., & Büning, H. (2007). Jarque–Bera test and its competitors for testing normality–a power comparison. ***Journal of applied statistics***, 34(1), 87-105.

The Egyptian Corporate Governance Code (ECCG). (2016), The Egyptian Institute of Directors, third Release.

The Egyptian standards for firms' evaluation. (2017), The Financial Regulatory Authority, (1). Available at: https://asa.gov.eg/Page.aspx?id=5_692

Trang, T. V., & Mai. T. N. (2021). Too big to cheat? An empirical study of the impact of firm size on earnings management in Vietnam. ***The journal of contemporary issues in business and government***, 27(5), 234-247.

Tree, D., Wang, D., & Yang, R. (2019). Balance sheet classification shifting and non-articulation. Available at: https://dx.doi.org/10.2139/ssrn.4102477

Triani, N., & Tarmidi, D. (2019). Firm value: impact of investment decisions, funding decisions and dividend policies. ***International Journal of Academic Research in Accounting, Finance and Management Sciences***, 9(2), 158-163.

Tulcanaza-Prieto, A. B., & Lee, Y. (2022). Real earnings management, firm value, and corporate governance: Evidence from the Korean market. ***International Journal of Financial Studies***, 10(1), 1-23.

Tulcanaza-Prieto, A. B., & Lee, Y. (2022). Real earnings management and firm value using quarterly financial data: Evidence from Korea. ***Global Business & Finance Review***, 27(1), 50-64.

Usman, M., Nwachukwu, J., Ezeani, E., Salem, R. I. A., Bilal, B., & Kwabi, F. O. (2024). Audit quality and classification shifting: evidence from UK and Germany. ***Journal of Applied Accounting Research***, 25(3), 448-475.

Uwuigbe, U., Uwuigbe, O. R., & Okorie, B. (2015). Assessment of the effects of firms characteristics on earnings management of listed firms in Nigeria. ***Asian Economic and Financial review***, 5(2), 218-228.

Vakilifard, H., & Mortazavi, M. S. (2016). The impact of financial leverage on accrual-based and real earnings management. ***International Journal of Academic Research in Accounting, Finance and Management Sciences***, 6(2), 53-60.

Veronica, A. (2015). The influence of leverage and its size on the earnings management. ***Research Journal of Finance and Accounting***, 6(8), 159-167.

Viana, D. B. C., Lourenço, I. M. E. C., & Paulo, E. (2022). The effect of IFRS adoption on accrual-based and real earnings management: emerging markets' perspective. ***Journal of Accounting in Emerging Economies***, 13(3), 485-508.

Vo, X. V., & Ellis, C. (2017). An empirical investigation of capital structure and firm value in Vietnam. ***Finance Research Letters***, 22, 90-94.

Wati, L. N. (2021). The Role of Political Connections and Family Ownership in Increasing Firm Value. ***Journal of Finance and Risk Perspectives***, 10, 40-53.

Wibowo, A. S., & Fuad, F. (2018). Growth illusion in the Indonesia stock exchange: relationship between earnings management and firm value. ***The Indonesian Journal of Accounting Research***, 21(1), 76-100.

Wijaya, A. L. (2020). The effect of audit quality on firm value: a case in Indonesian manufacturing firm. ***Journal of Accounting, Finance and Auditing Studies,*** 6(1), 1-15.

Wroblewski, D. R., Jarne Jarne, J. I., & Callao Gastón, S. (2016). ***The complex vision on the earnings management: the evidence from Eastern European countries***. Doctoral dissertation, Zaragoza University.

Wu, E. (2016). ***Classification shifting as an earnings management tool among European firms***. Masterarbeit, Erasmus Universiteit Rotterdam.

Xie, X., Chang, Y. S., & Shiue, M. J. (2022). Corporate life cycle, family firms, and earnings management: evidence from Taiwan. ***Advances in Accounting***, *56*, 100579.

Xu, R. Z., Taylor, G. K., & Dugan, M. T. (2007). Review of real earnings management literature. ***Journal of Accounting Literature***, 26, 195-228.

Yasser, S., & Soliman, M. (2018). The effect of audit quality on earnings management in developing countries: The case of Egypt. International ***Research Journal of Applied Finance***, 9(4), 216-231.

Yolandita, A. A., & Cahyonowati, N. (2022). The effects of audit quality on firm value of Indonesian financial service sector (FSS). ***Diponegoro Journal of Accounting***, 11(1), 1-8.

Yuliyanti, L., Waspada, I., Sari, M., & Nugraha, N. (2022). The effect of liquidity, leverage, and profitability on firm value with firm size as moderating variable. 6th Global Conference on Business, Management, and Entrepreneurship, ***Advances in Economics, Business and Management Research***, 220, 88-92.

Yuniarti, E., & Hanim, N. (2017). Effect on value earnings management company with good corporate governance practices as moderating variable. ***SHS-Web of Conferences***. Available at: https://doi.org/10.1051/shsconf/20173404007

Zaher, A. M. (2019). The effect of managerial overconfidence on accruals-based and real-activities earnings management: Evidence from Egypt. ***Academy of Accounting and Financial Studies Journal***, 23(4), 1-14.

Zalat, A. A. (2015). ***A proposed Framework for evaluating the effect of fair value accounting measurement of exploration assets on the firm value – A case study of public petroleum company***. Doctoral dissertation, Menoufiya University. (In Arabic)

Zalata, A. M., & Abdelfattah, T. (2021). Non-executive female directors and earnings management using classification shifting. ***Journal of Business Research***, 134, 301-315.

Zalata, A. M., & Roberts, C. (2017). Managing earnings using classification shifting: UK evidence. ***Journal of International Accounting, Auditing and Taxation***, 29, 52-65.

Zalata, A. M., Ntim, C. G., Alsohagy, M. H., & Malagila, J. (2022). Gender diversity and earnings management: the case of female directors with financial background. ***Review of Quantitative Finance and Accounting***, 58(1), 101-136.

Zalata, A., & Roberts, C. (2016). Internal corporate governance and classification shifting practices: An analysis of UK corporate behavior. ***Journal of Accounting, Auditing & Finance***, 31(1), 51-78.

Zamri, N., Rahman, R. A., & Isa, N. S. M. (2013). The impact of leverage on real earnings management. ***Procedia Economics and Finance***, 7, 86-95.

Zang, A. Y. (2012). Evidence on the trade-off between real activities manipulation and accrual-based earnings management. ***The Accounting Review***, 87(2), 675-703.

Zgarni, I., Hlioui, K., & Zehri, F. (2016). Effective audit committee, audit quality and earnings management: Evidence from Tunisia. ***Journal of Accounting in Emerging Economies***, 6(2), 138-155.

الملخص: تسعى هذه الدراسة إلى اقتراح نموذج متعدد الأبعاد لقياس ممارسات إدارة الأرباح وأثره على قيمة المنشأة. وقد قامت الدراسة التطبيقية باختبار أثر ممارسات إدارة الأرباح والتي تتمثل في إدارة الأرباح بالاستحقاقات، إدارة الأرباح بالأنشطة الحقيقية، إعادة تصنيف بنود قائمة الدخل، إعادة تصنيف بنود قائمة التدفقات النقدية، وإدارة الأرباح الإجمالية على قيمة المنشأة. وتم الاعتماد على مقياسين بديلين لقيمة المنشأة تمثل في نسبة القيمة السوقية إلى القيمة الدفترية ومعدل العائد على الأصول. وتم قياس إدارة الأرباح بالاستحقاقات عن طريق الاستحقاقات الاختيارية. بينما تم قياس إدارة الأرباح بالأنشطة الحقيقية من خلال إجمالي اداره الأرباح بالمبيعات، إدارة الأرباح بالإنتاج، واداره الأرباح بالنفقات الاختيارية. كما تم قياس ممارسات إعادة تصنيف بنود قائمة الدخل عن طريق مجمل الربح غير المتوقع، في حين تم قياس ممارسات إعادة تصنيف بنود قائمة التدفقات النقدية عن طريق التدفقات النقدية التشغيلية غير المتوقعة. أما إدارة الأرباح الإجمالية فقد تم قياسها عن طريق إيجاد متوسط الرتب لكل من إدارة الأرباح بالاستحقاقات، إدارة الأرباح بالأنشطة الحقيقية، إعادة تصنيف بنود قائمة الدخل، وإعادة تصنيف بنود قائمة التدفقات النقدية.

التصميم / المنهجية: أجريت الدراسة على عينة مكونة من (٤٨) شركة مقيدة في البورصة المصرية خلال الفترة من ٢٠١٧ إلى ٢٠٢١. وقد تم استخدام طريقة المربعات الصغرى المعممة (GLS) واختبار كروسكال واليس لاختبار فرضيات الدراسة.

النتائج: خلصت نتائج الدراسة إلى وجود أثر إيجابي معنوي لكل من إدارة الأرباح بالاستحقاقات، إعادة تصنيف بنود قائمة التدفقات النقدية، وإدارة الأرباح الإجمالية على نسبة القيمة السوقية إلى القيمة الدفترية، غير أنه كان لإعادة تصنيف بنود قائمة الدخل أثر سلبي. في حين لم يكن هناك أي أثر لإدارة الأرباح بالأنشطة الحقيقية على نسبة القيمة السوقية إلى القيمة الدفترية. وفيما يتعلق بالمقياس الثاني لقيمة المنشأة (معدل العائد على الأصول)، توصلت نتائج الدراسة إلى وجود أثر إيجابي معنوي لكل من إعادة تصنيف بنود قائمة الدخل، إعادة تصنيف بنود قائمة التدفقات النقدية وإدارة الأرباح الإجمالية على معدل العائد على الأصول. بينما كان لإدارة الأرباح بالاستحقاقات وإدارة الأرباح بالأنشطة الحقيقية أثر سلبي. علاوة على ذلك، توصلت نتائج الدراسة إلى وجود اختلاف بين قطاعات سوق الأوراق المالية المصري حول نسبة القيمة السوقية إلى القيمة الدفترية، معدل العائد على الأصول، وإدارة الأرباح الإجمالية فضلا عن عدم وجود اختلاف حول إدارة الأرباح بالاستحقاقات، إدارة الأرباح بالأنشطة الحقيقية، إعادة تصنيف بنود قائمة التدفقات النقدية، إعادة تصنيف بنود قائمة الدخل.

الاصالة / القيمة: تساهم هذه الدراسة في الادبيات الموجودة من خلال تقديم نموذج متعدد الأبعاد لقياس ممارسات إدارة الأرباح وأثره على قيمة المنشأة. ركزت غالبية الدراسات السابقة على ممارسات إدارة الأرباح بالاستحقاقات وإدارة الأرباح بالأنشطة الحقيقية، في حين تساهم هذه الدارسة في توفير أدلة تطبيقية على وجود ممارسات إعادة التصنيف في البيئة المصرية. وبالتالي، تقدم الدراسة تحليلا أكثر شمولا لممارسات إدارة الأرباح وتوفير أدلة تطبيقية حول أثر تلك الممارسات متجمعة على قيمة المنشأة. تعتبر نتائج الدراسة ذات قيمة بالنسبة إلى واضعي المعايير، والمستثمرين، والمحللين الماليين، والمراجعين.

الكلمات المفتاحية: إدارة الأرباح بالاستحقاقات، إدارة الأرباح بالأنشطة الحقيقية، إعادة تصنيف بنود قائمة الدخل، إعادة تصنيف بنود قائمة التدفقات النقدية، إدارة الأرباح الإجمالية.

لجنة المناقشة والحكم

أ.د / محمد زيدان إبراهيم **(مشرفاً ورئيسًا)**

أستاذ المحاسبة المالية

كلية التجارة - جامعة المنوفية

أ.د / فريد محرم فريد إبراهيم **(عضوًا)**

أستاذ المحاسبة المالية ووكيل كلية التجارة لشئون التعليم والطلاب

جامعة عين شمس

أ.د / محمد علي وهدان **(عضوًا)**

أستاذ المراجعة وعميد كلية التجارة – جامعة المنوفية

أ.د / محمد صابر حمودة السيد **(مشرفًا وعضوًا)**

أستاذ المحاسبة المالية ووكيل كلية التجارة لشئون خدمة

المجتمع وتنمية البيئة – جامعة المنوفية

أ.م.د / طارق إبراهيم صالح سعادة **(مشرفًا وعضوًا)**

أستاذ المحاسبة والمراجعة المساعد

كلية التجارة – جامعة المنوفية

كلية التجارة
قسم المحاسبة

نموذج مقترح متعدد الأبعاد لقياس ممارسات إدارة الأرباح وأثره على قيمة المنشأة: دراسة تطبيقية

رسالة مقدمة للحصول على درجة دكتور الفلسفة في العلوم التجارية (تخصص محاسبة)

اعداد

هبة الله جابر الصواف

المدرس المساعد بقسم المحاسبة

كلية التجارة – جامعة المنوفية

إشراف

الأستاذ الدكتور/ محمد زيدان إبراهيم

أستاذ المحاسبة المالية

كلية التجارة – جامعة المنوفية

الأستاذ الدكتور/ محمد صابر حمودة السيد

أستاذ المحاسبة المالية ووكيل الكلية لشئون خدمة المجتمع وتنمية البيئة

كلية التجارة – جامعة المنوفية

الدكتور/ طارق إبراهيم صالح سعادة

أستاذ المحاسبة المساعد

كلية التجارة – جامعة المنوفية

٢٠٢٤